ECOSPHERE

Droughts delay juvenile salmon migration and truncate diversity in habitat use

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4 Metropolitan Water District of Southern California, 1121 L Street, Suite 900, Sacramento, CA 95814, USA.

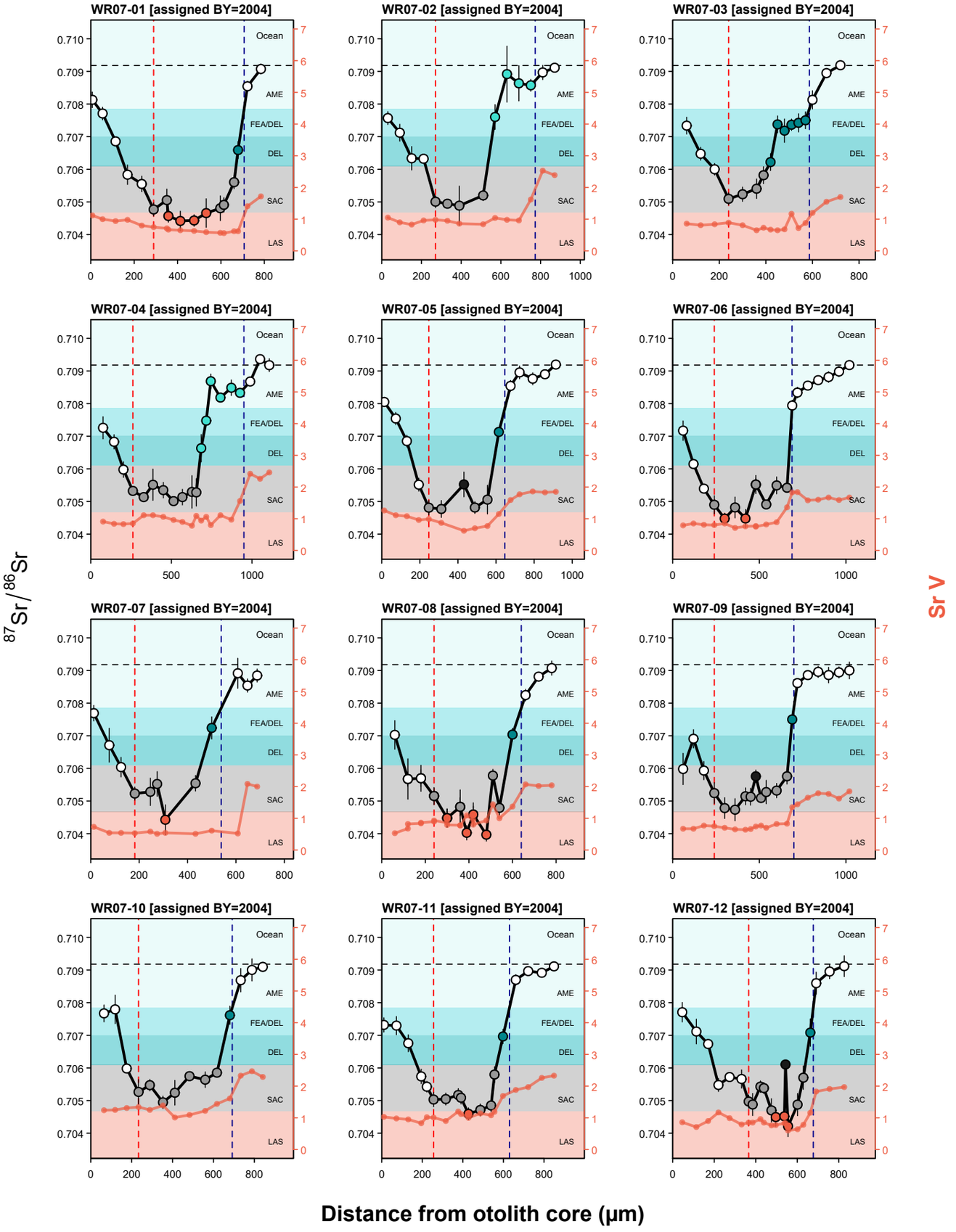
5 National Marine Fisheries Service, Southwest Fisheries Science Center, Santa Cruz, CA 95060, USA.

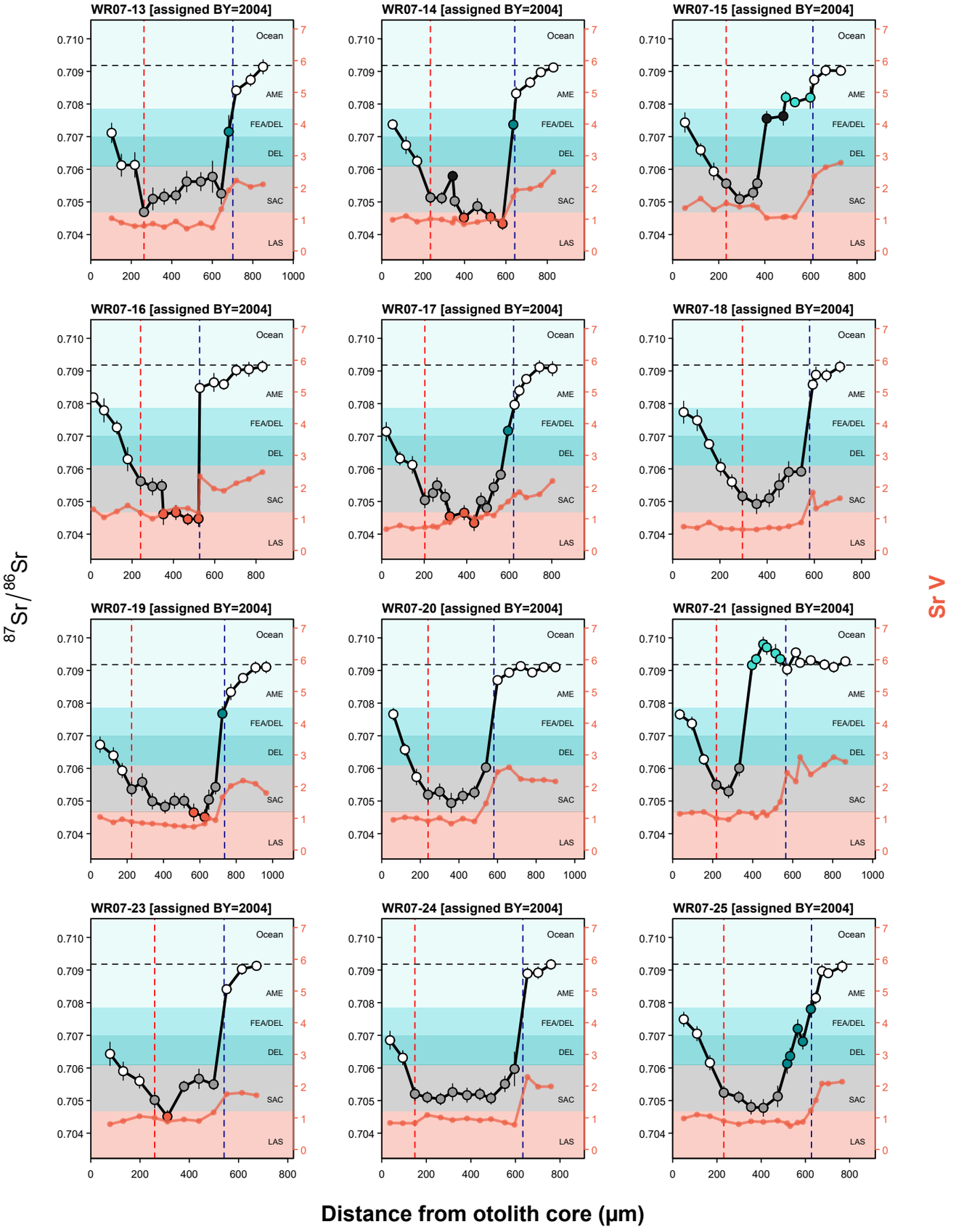
+ Equal contribution as the first author

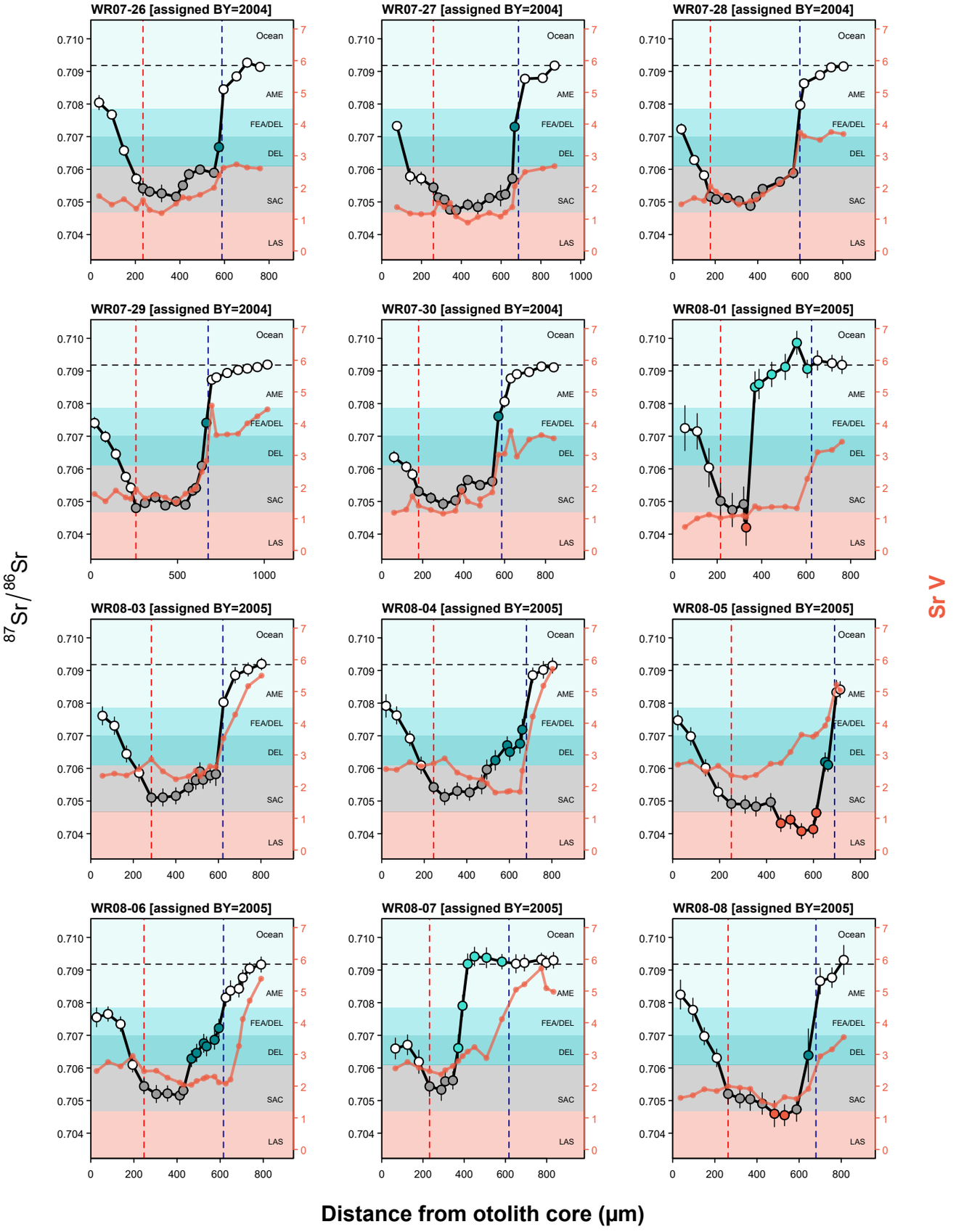
Correspondence: Anna M. Sturrock; Email: anna.sturrock@essex.ac.uk

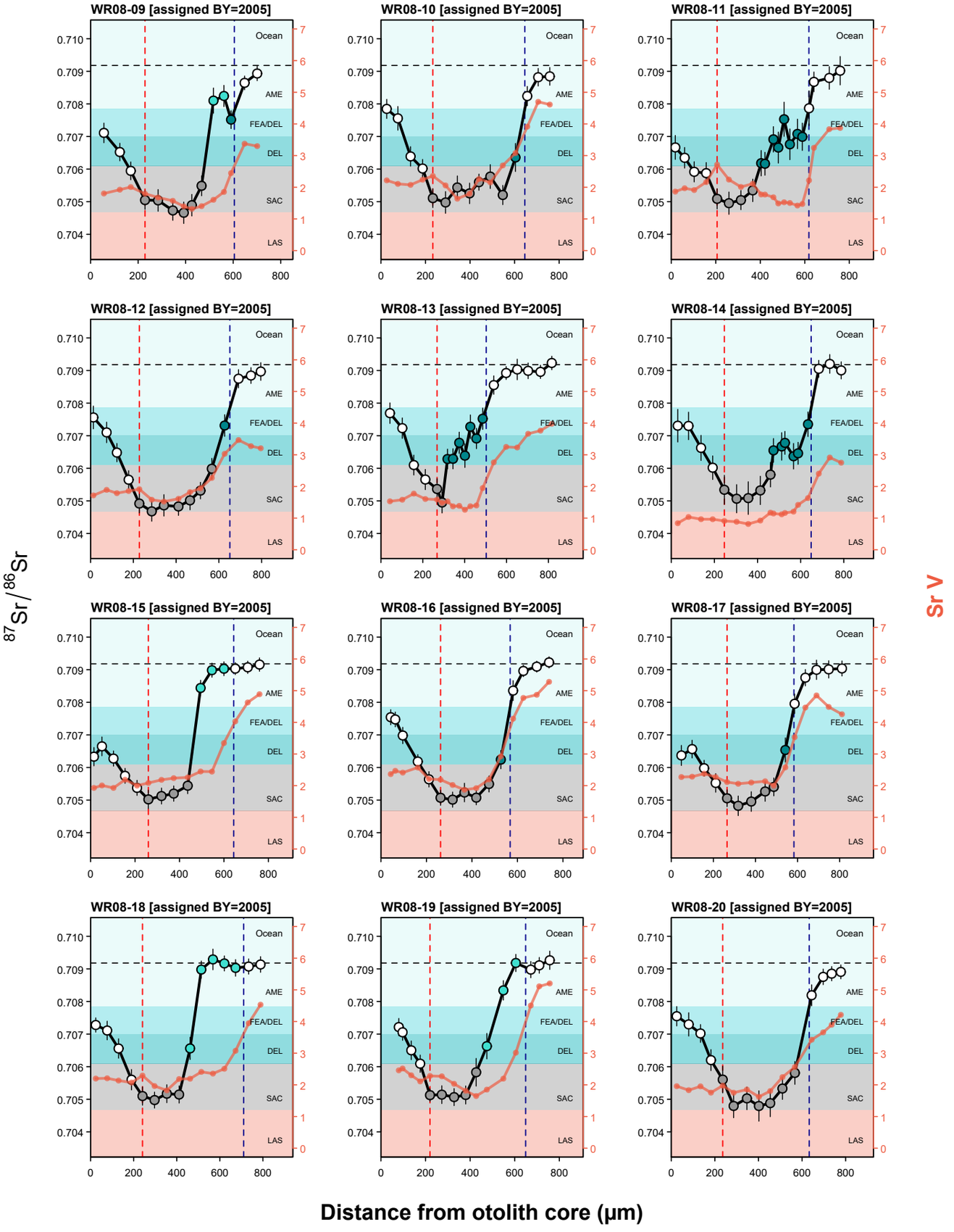
* Current address. University of Texas Marine Science Institute, 750 Channelview Dr., Port Aransas, TX 78373, USA.

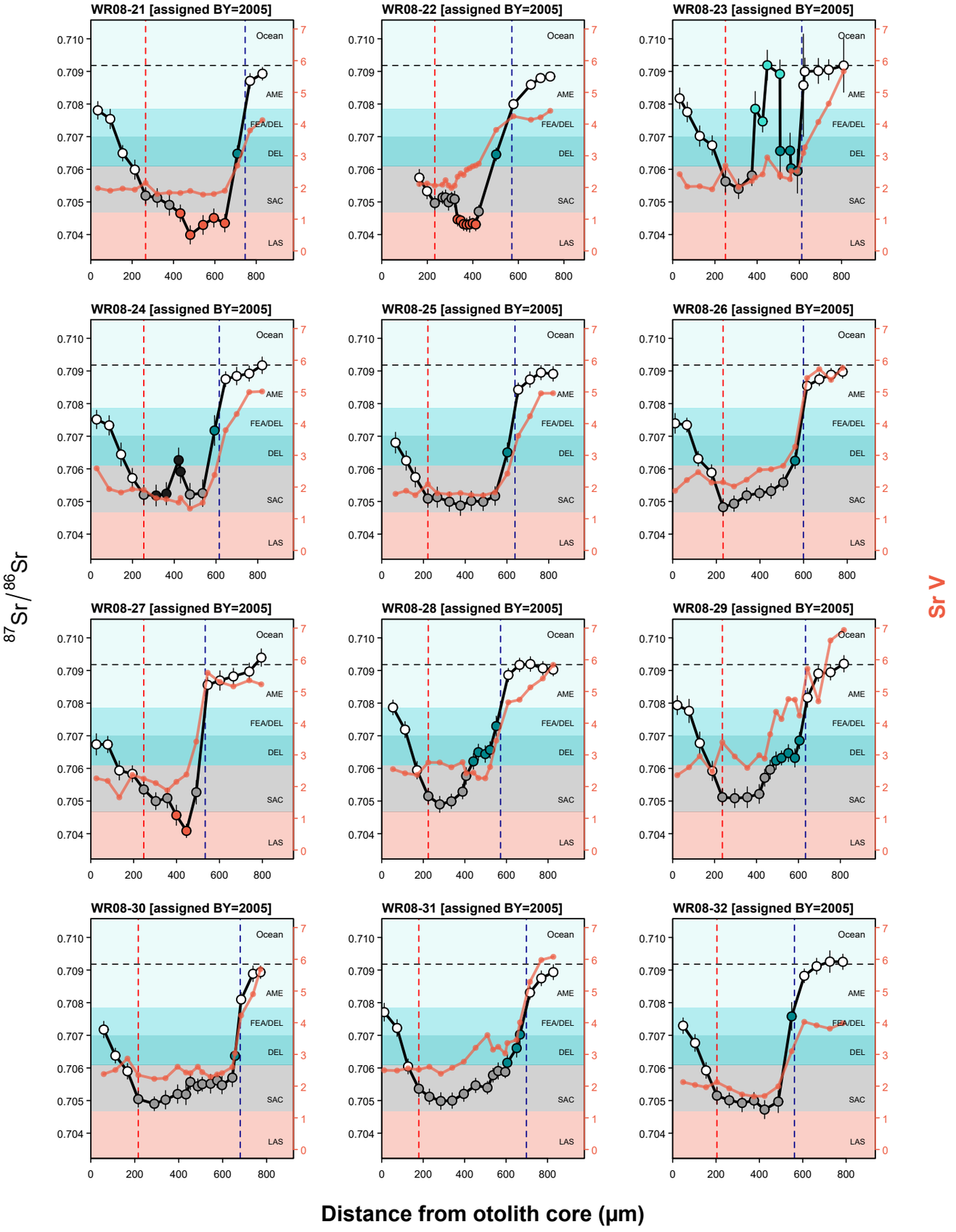
Appendix S2: Figure S1

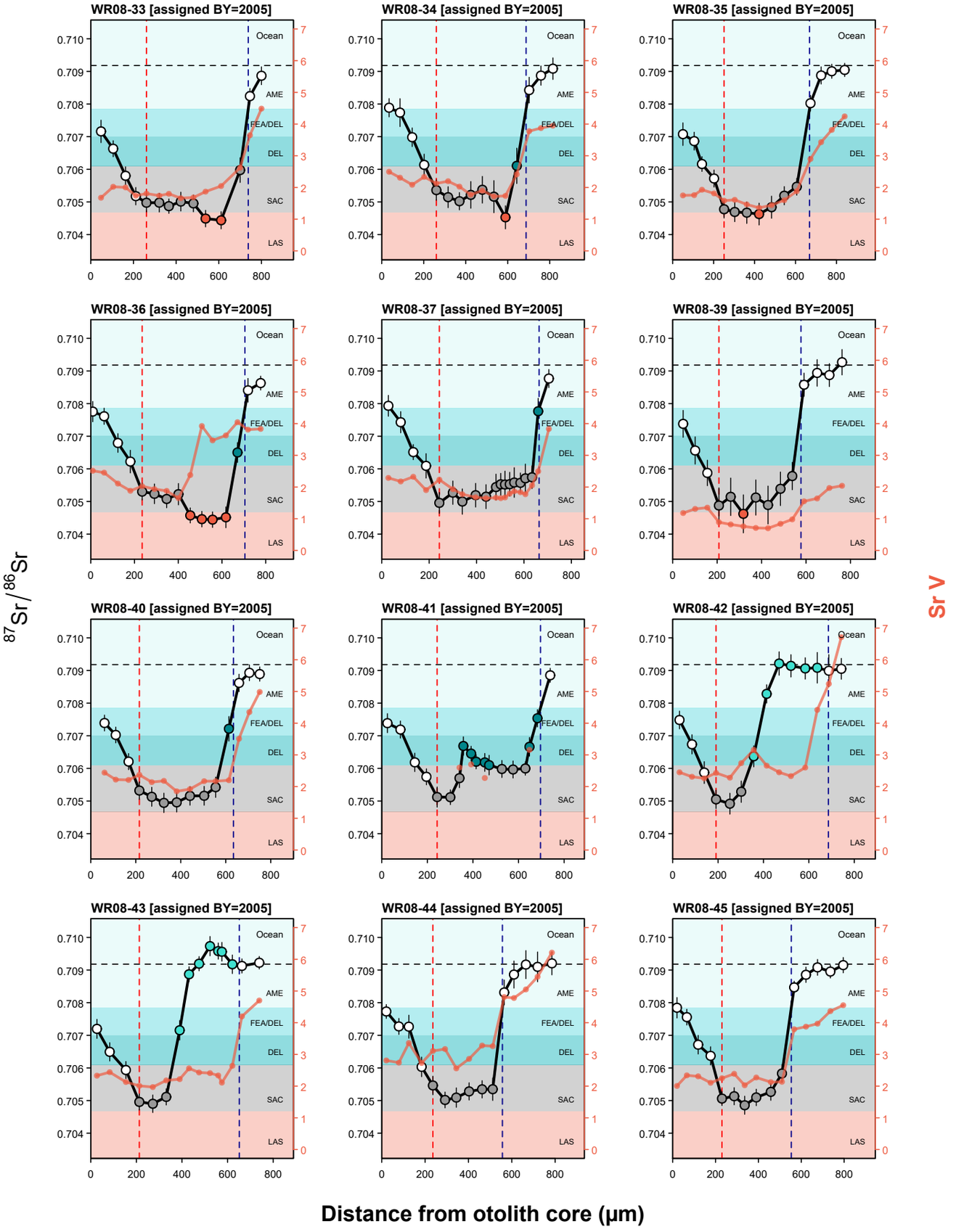

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Distance from otolith core (µm)

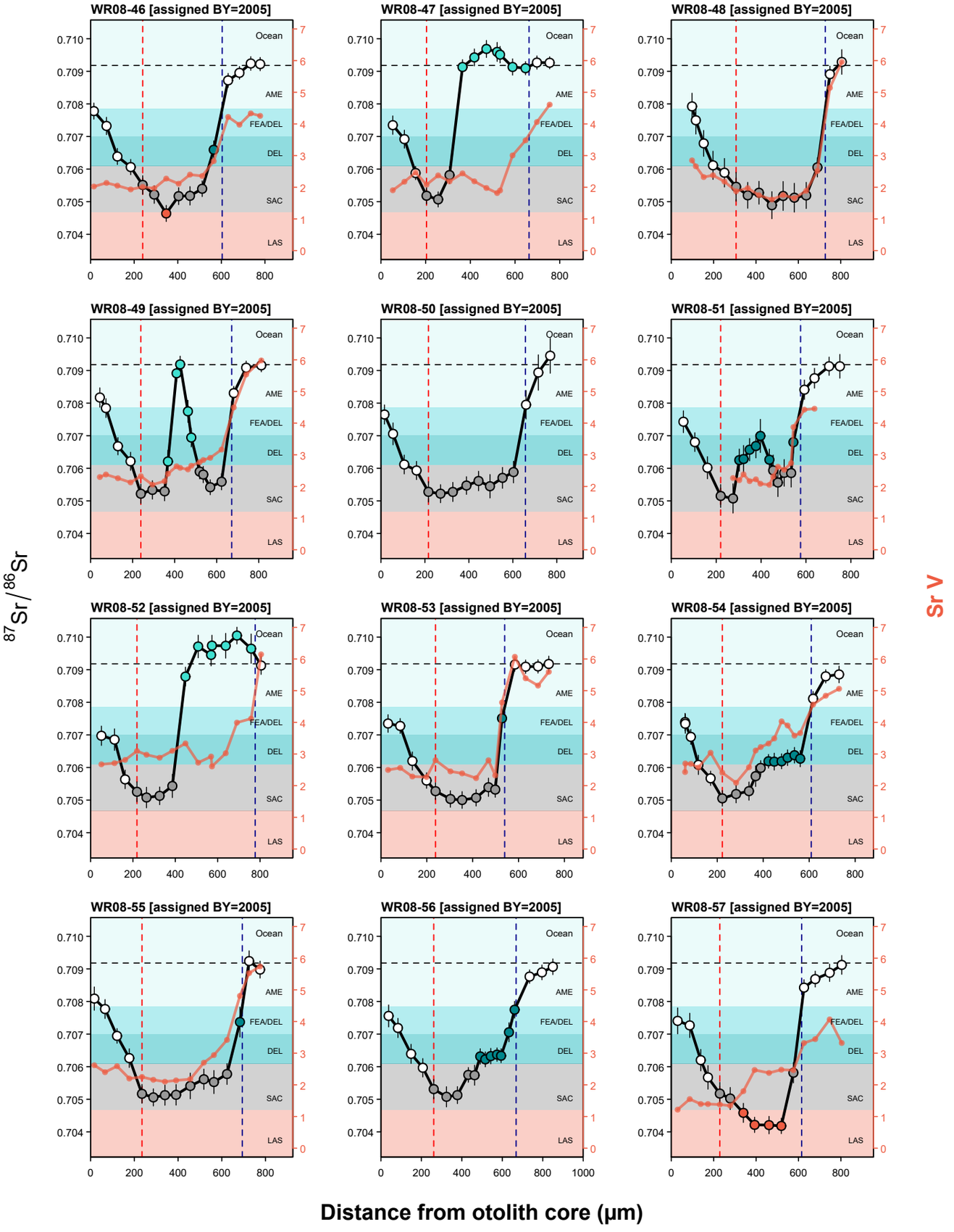

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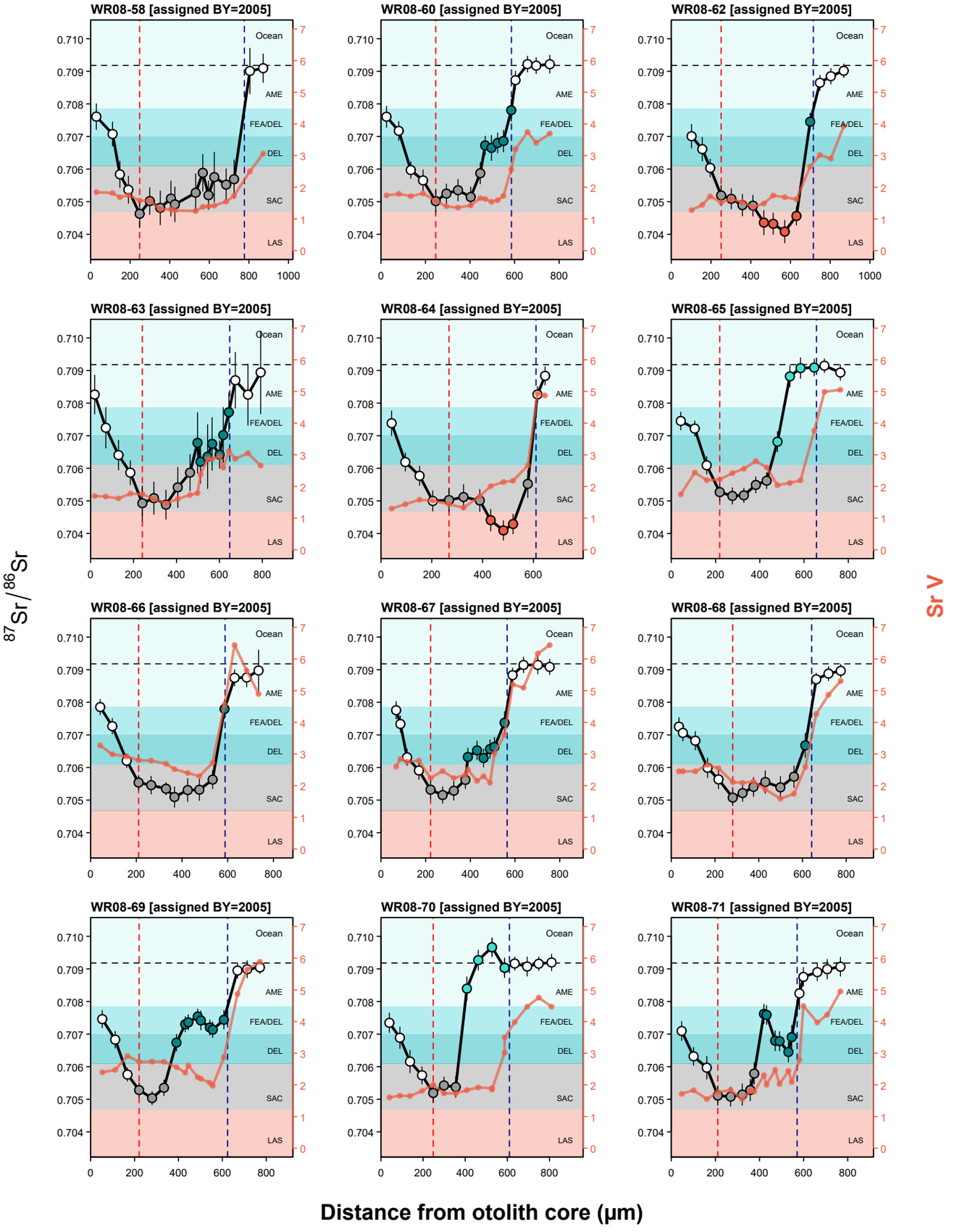

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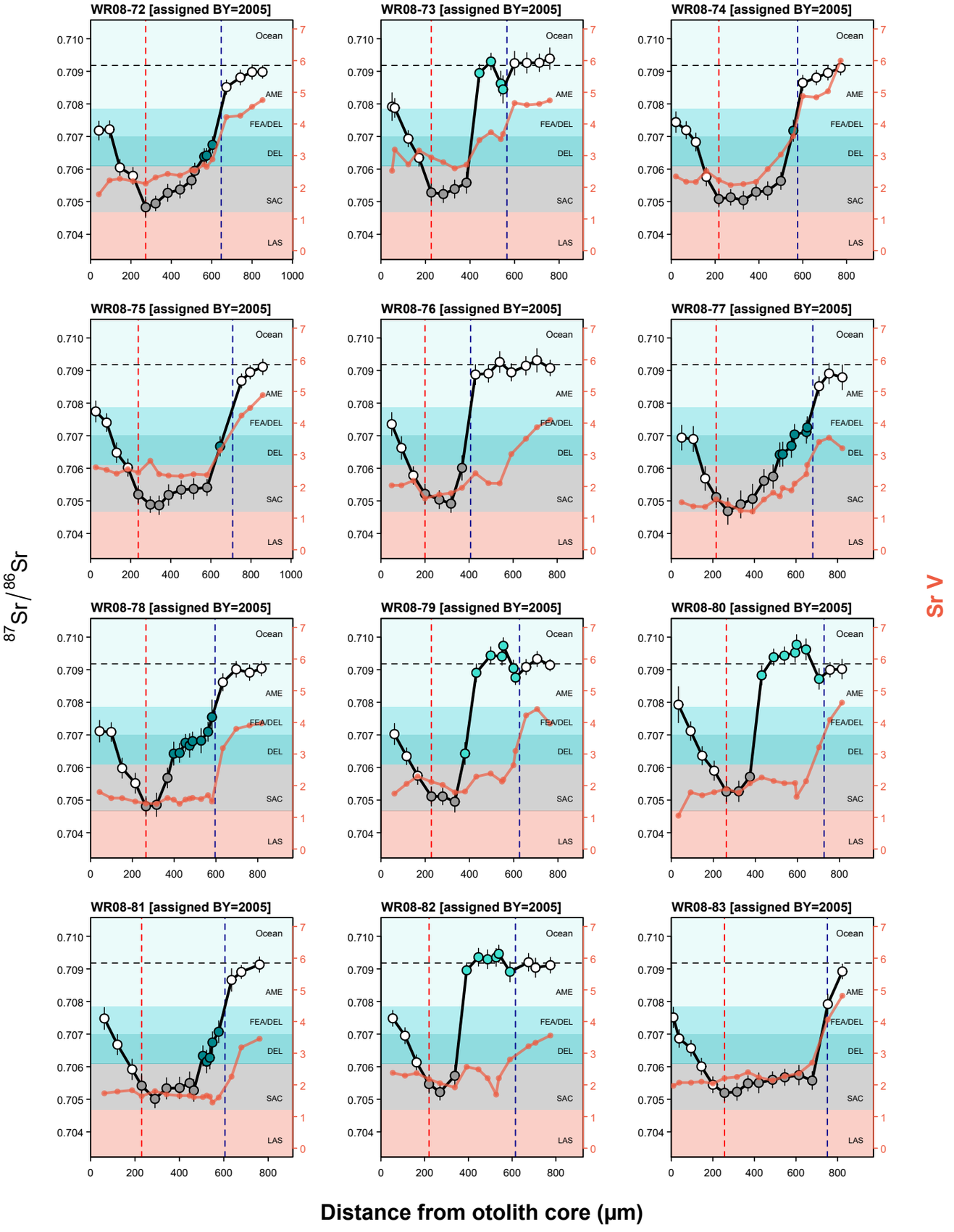

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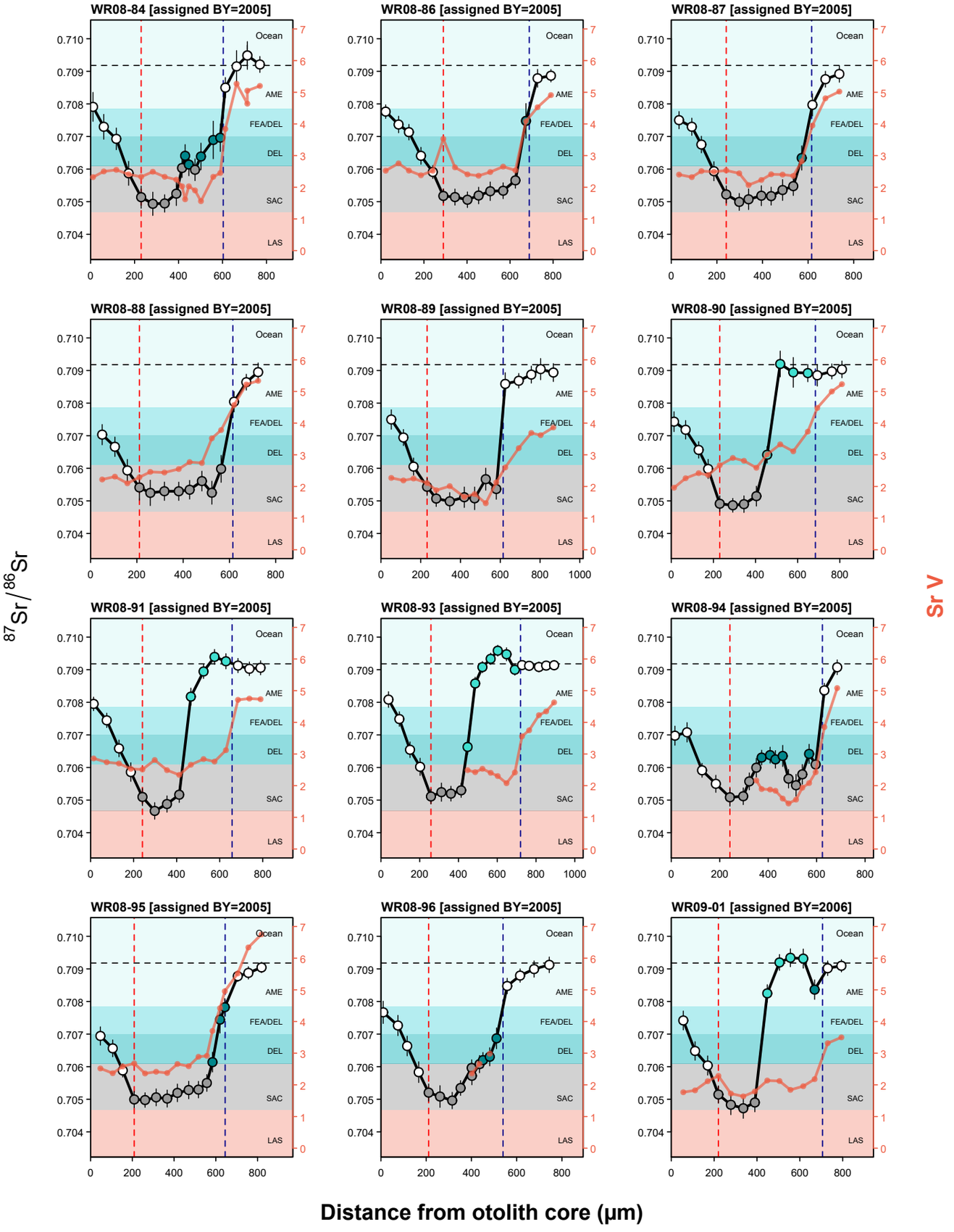

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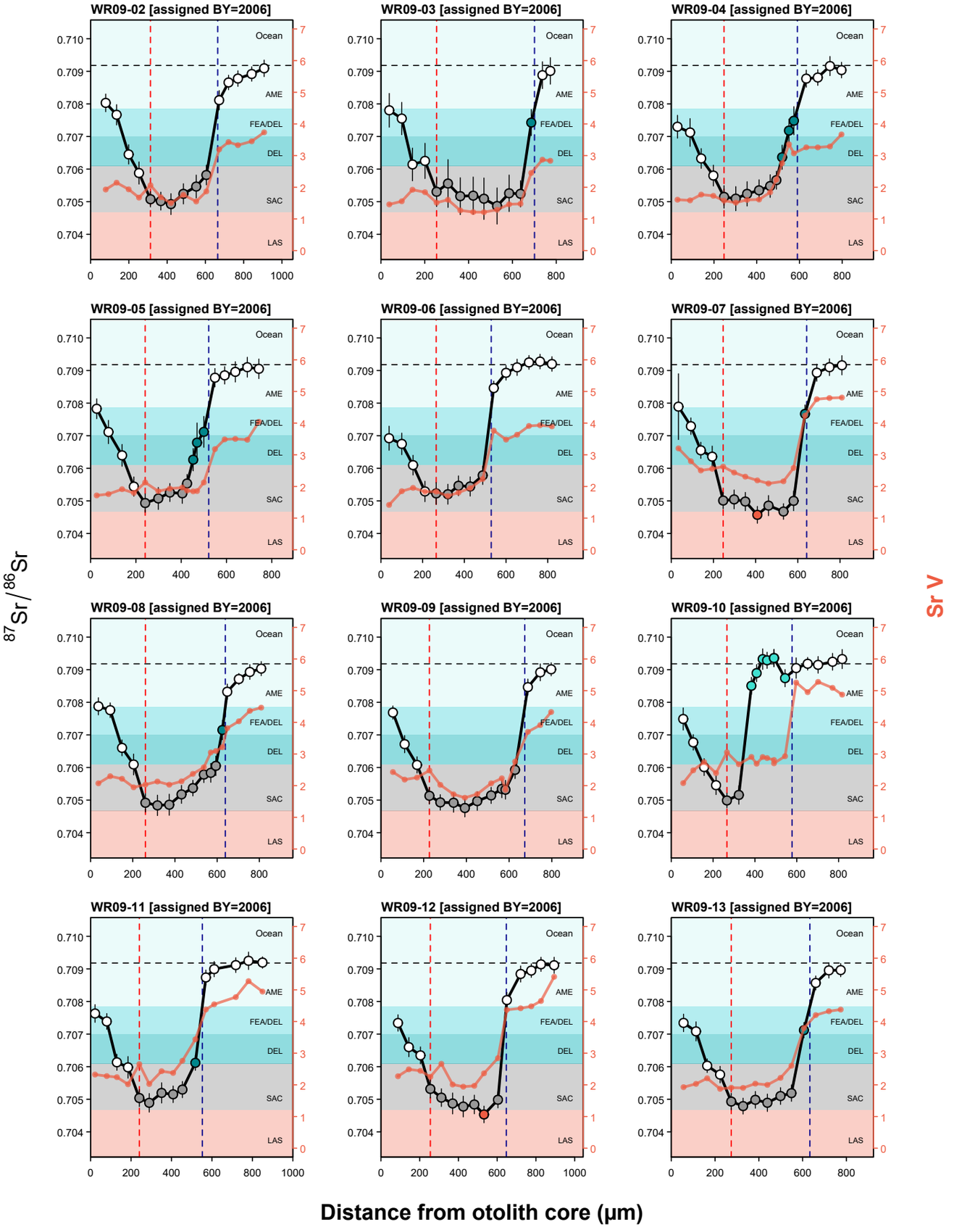

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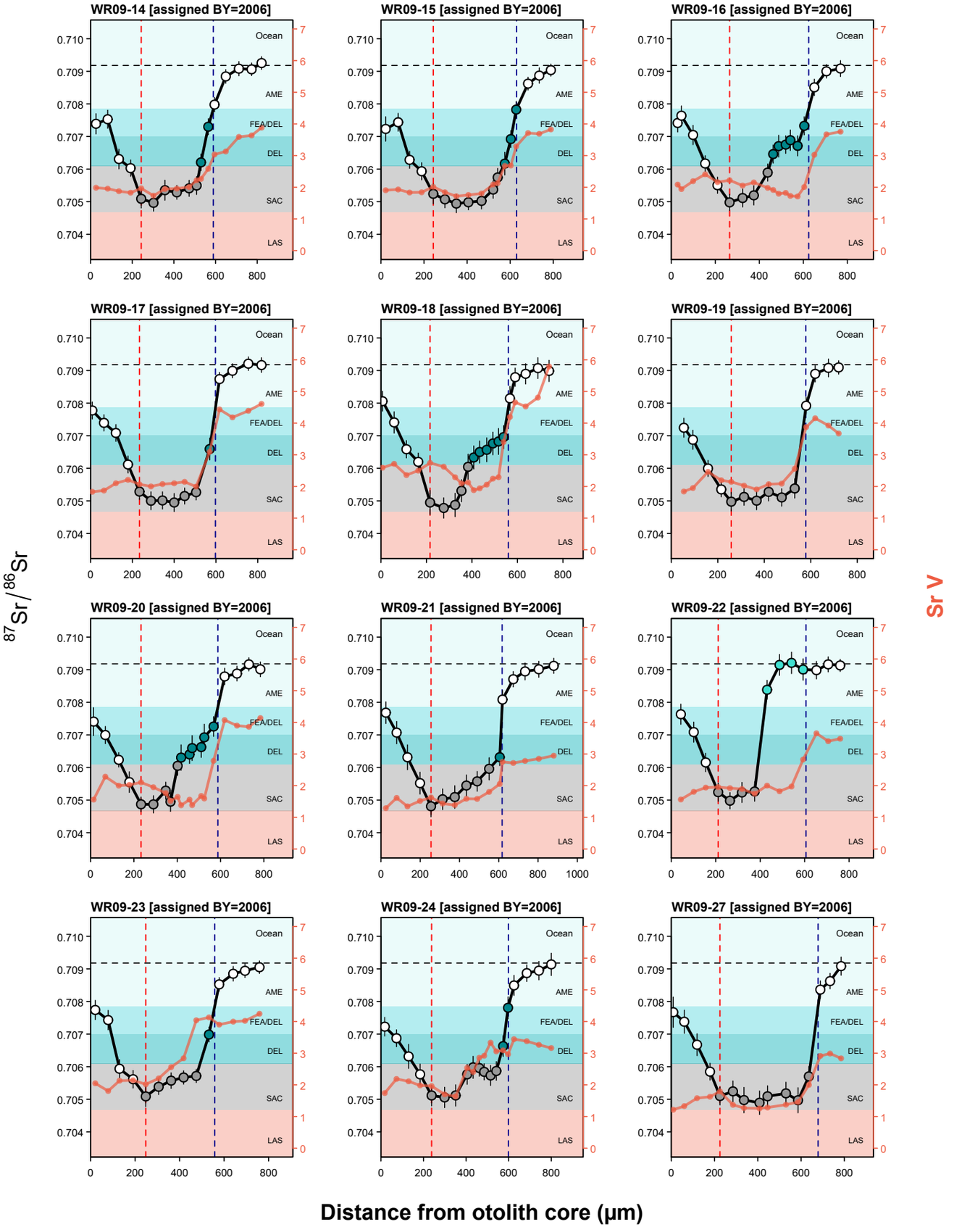

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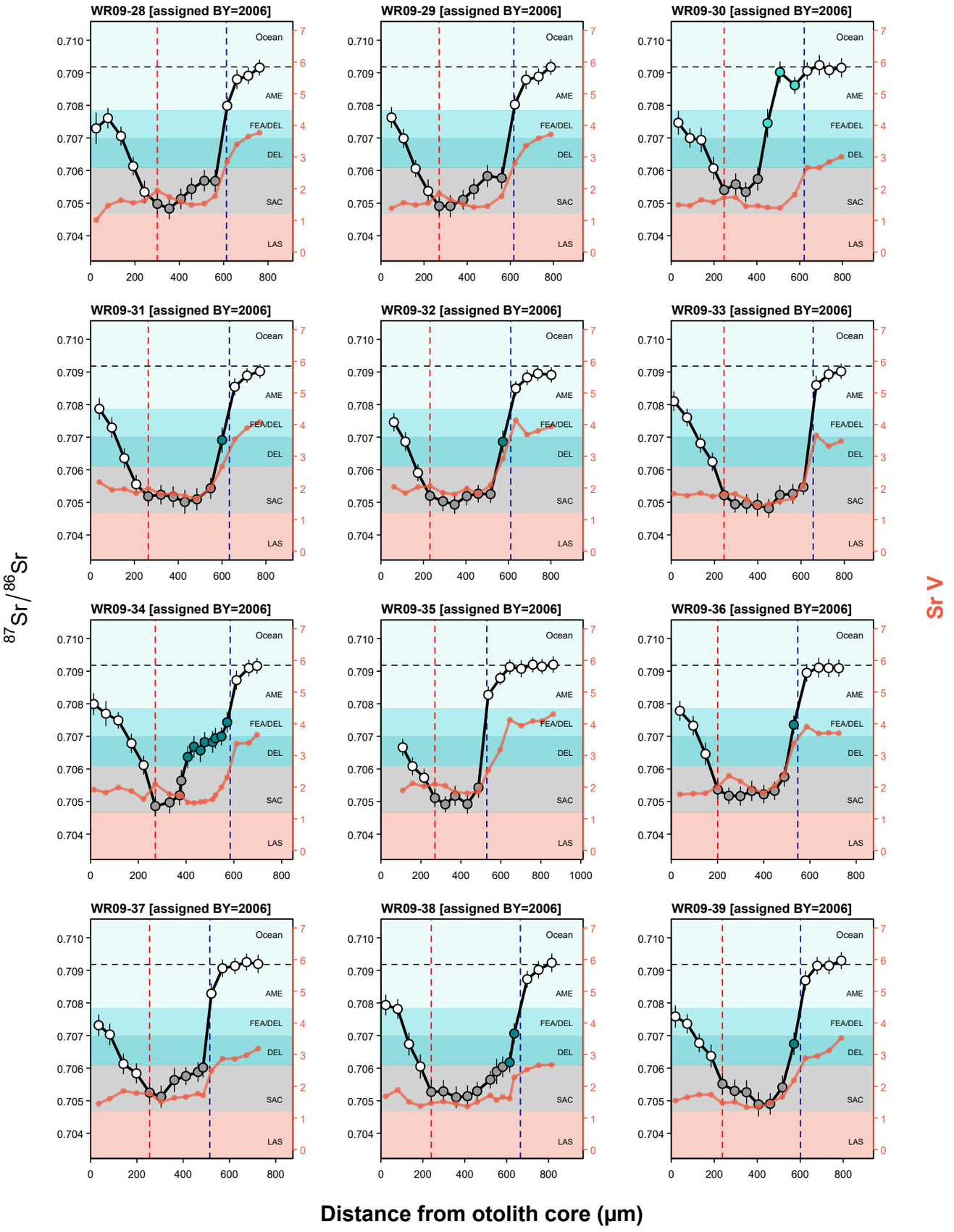

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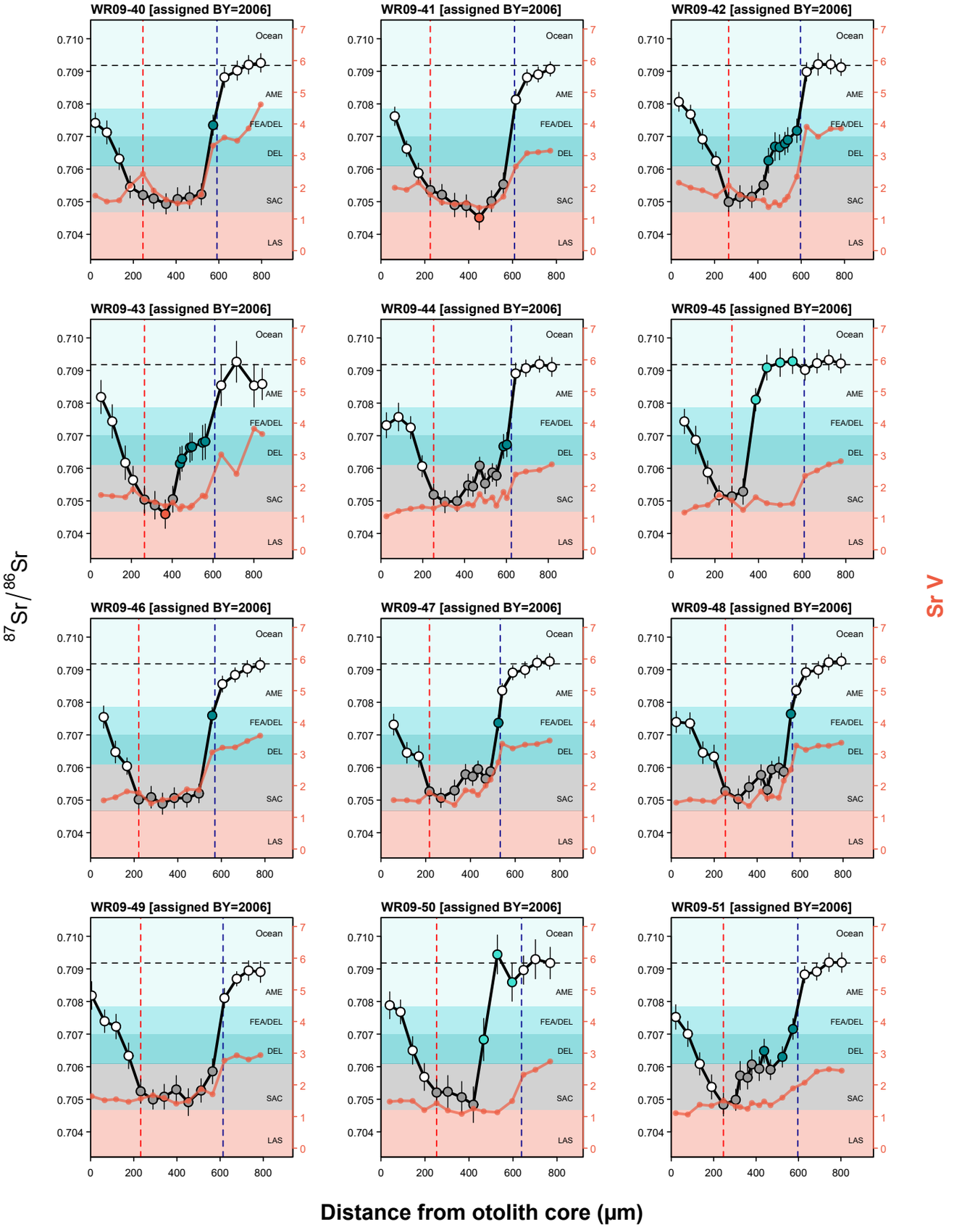

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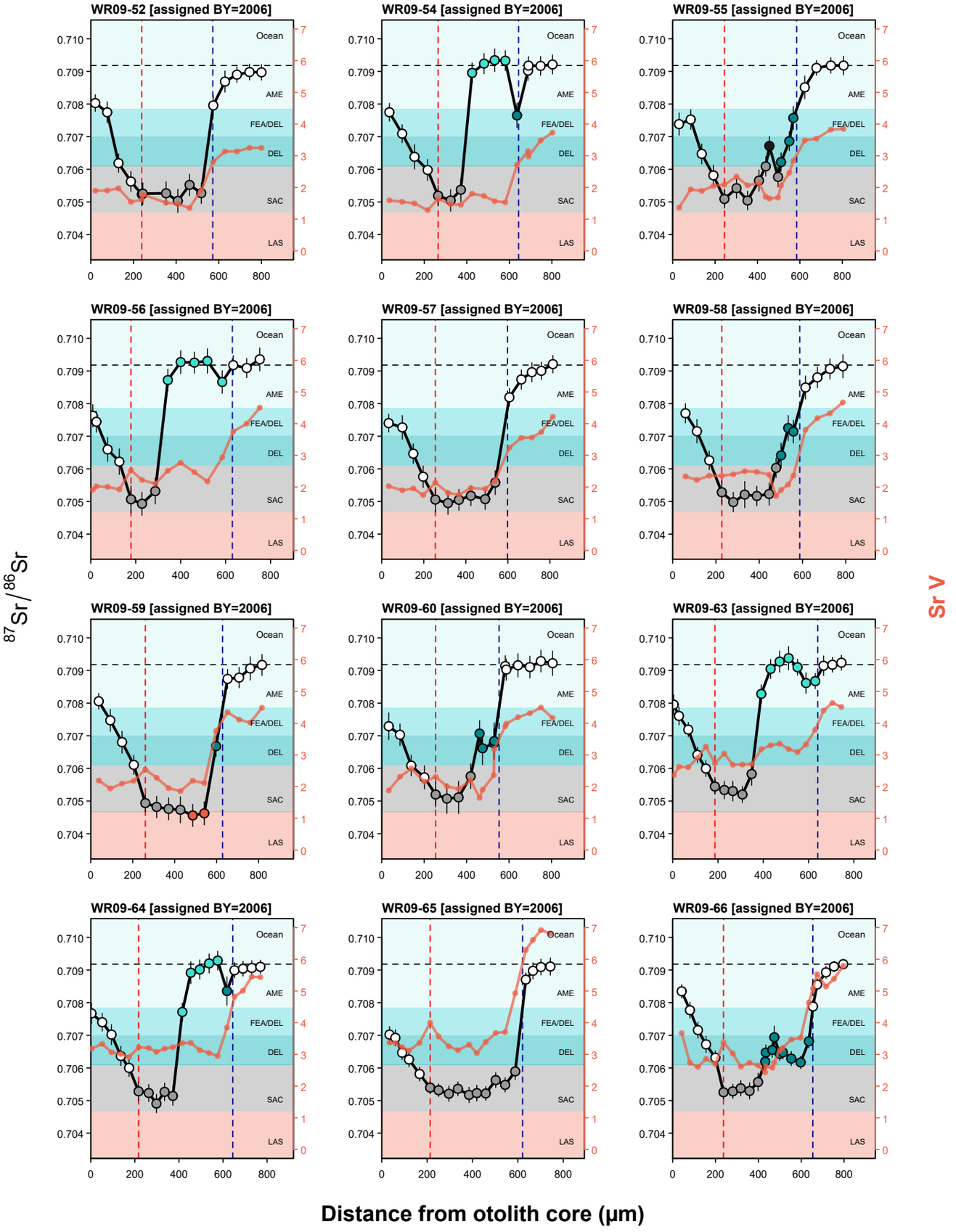

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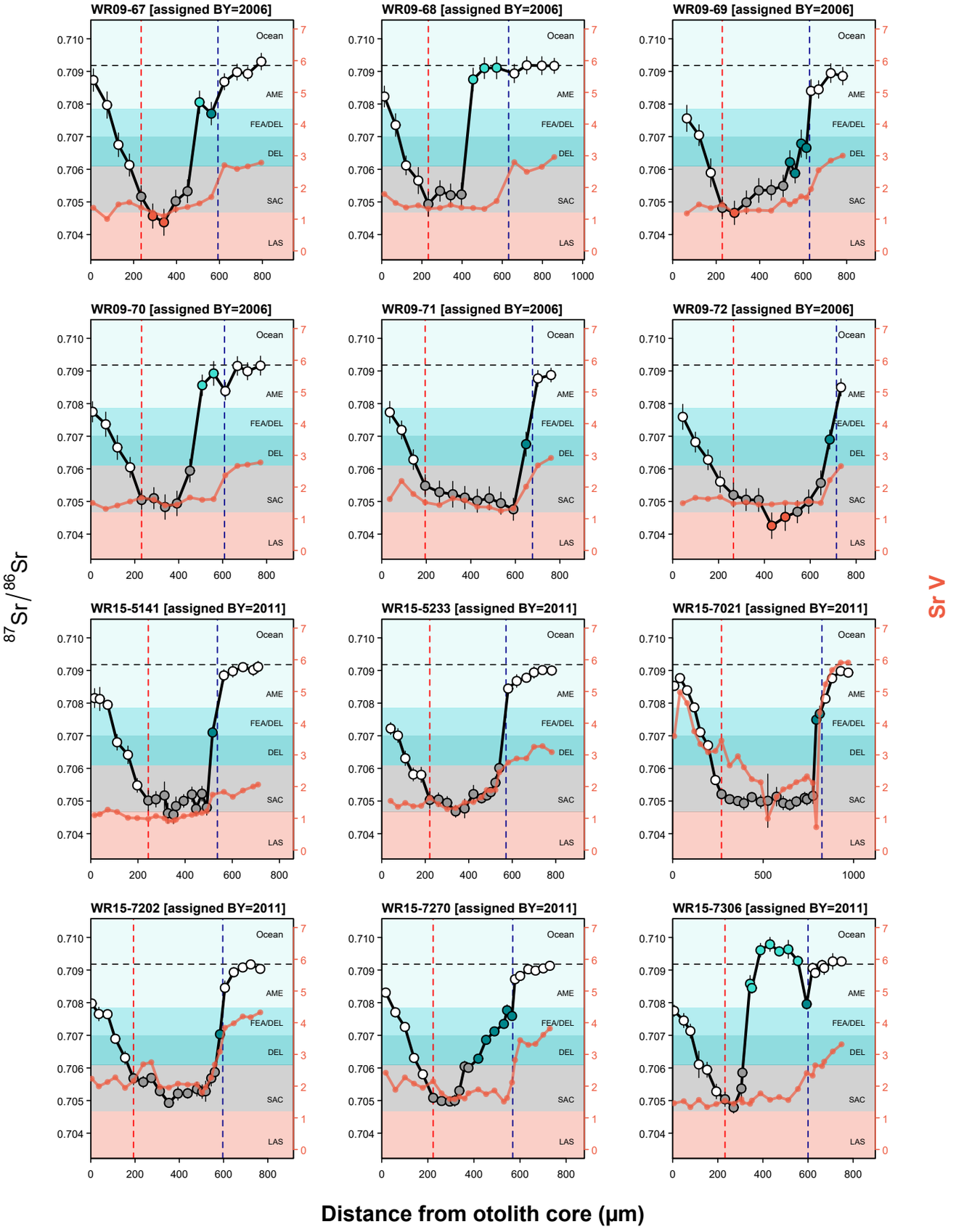

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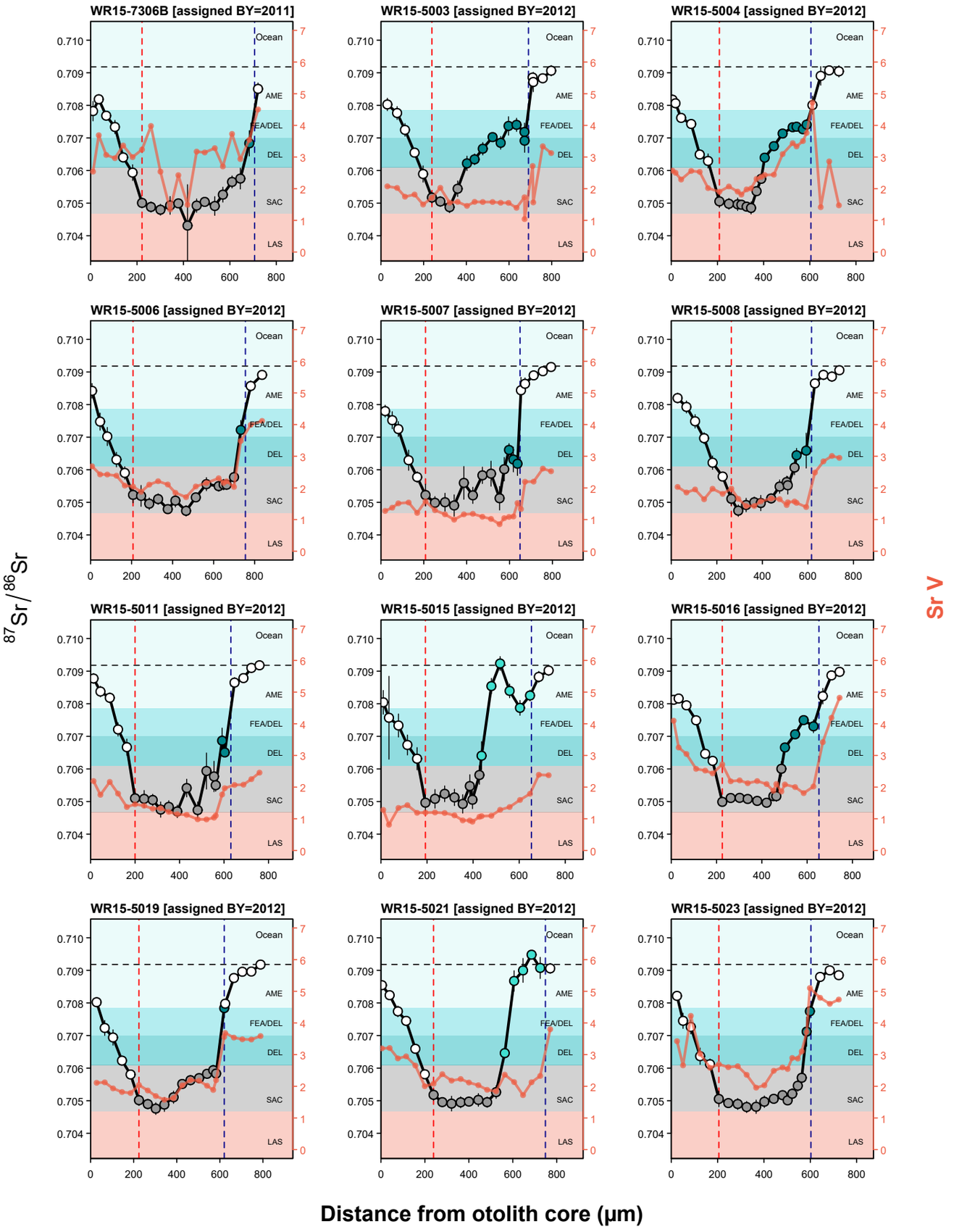

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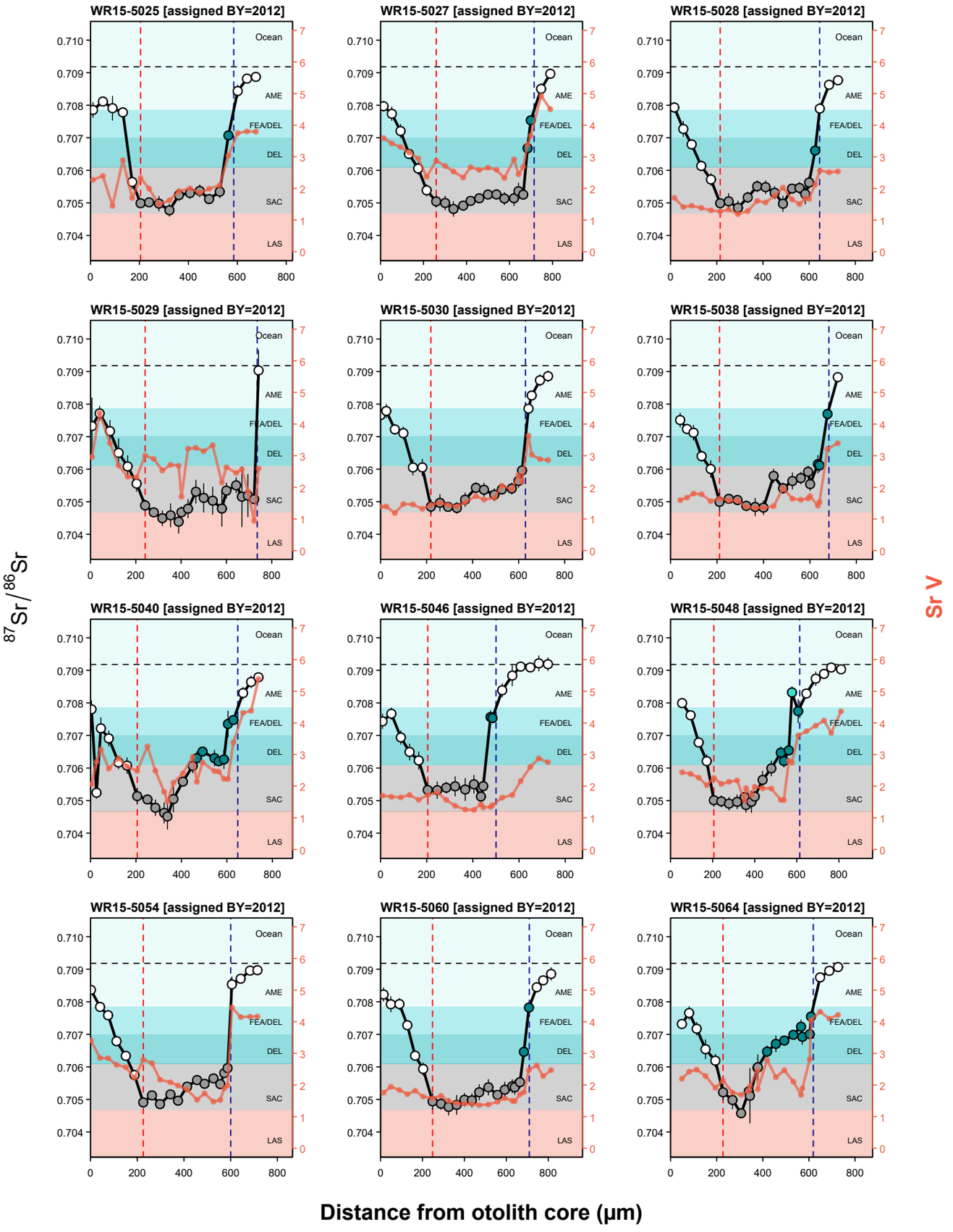

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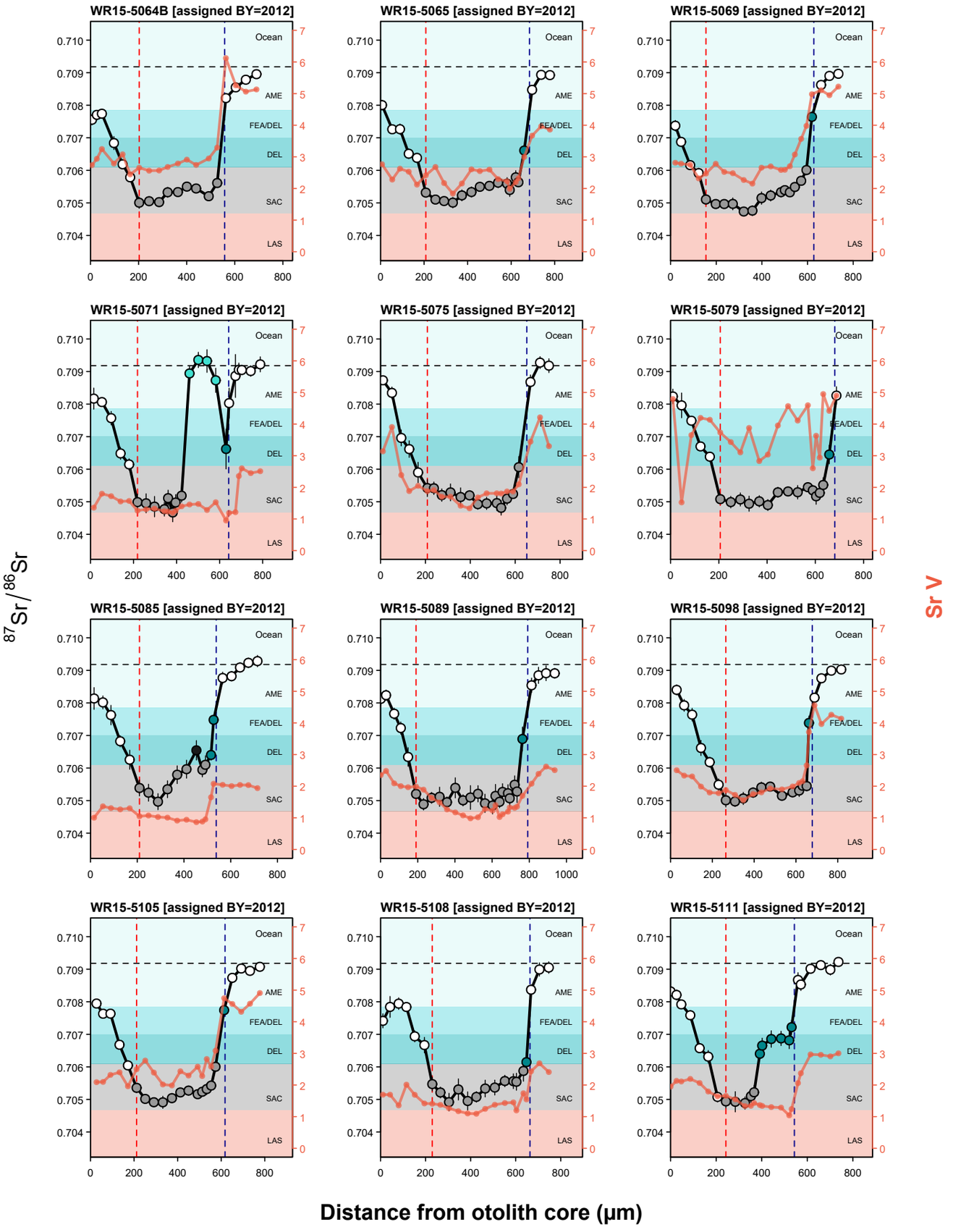

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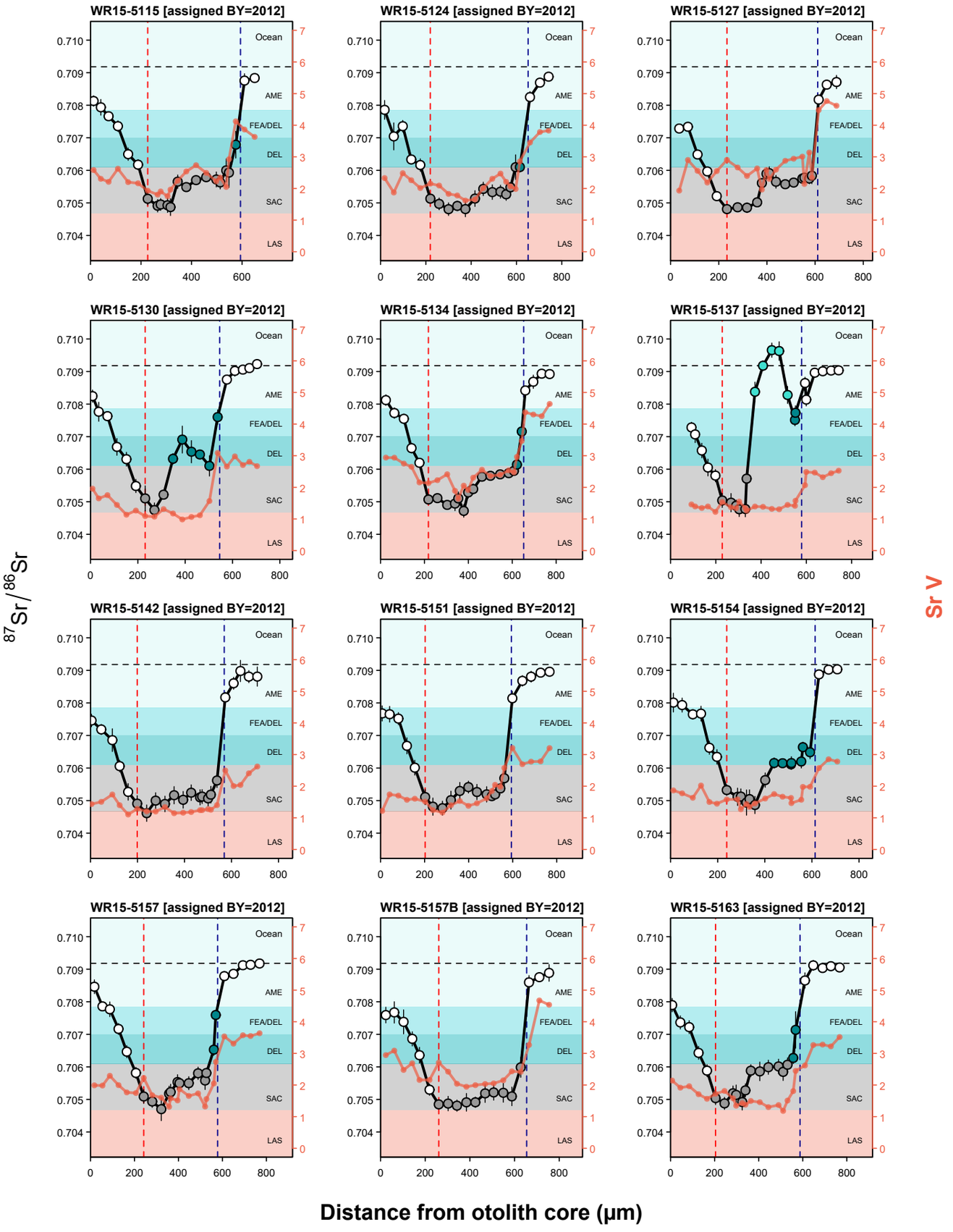

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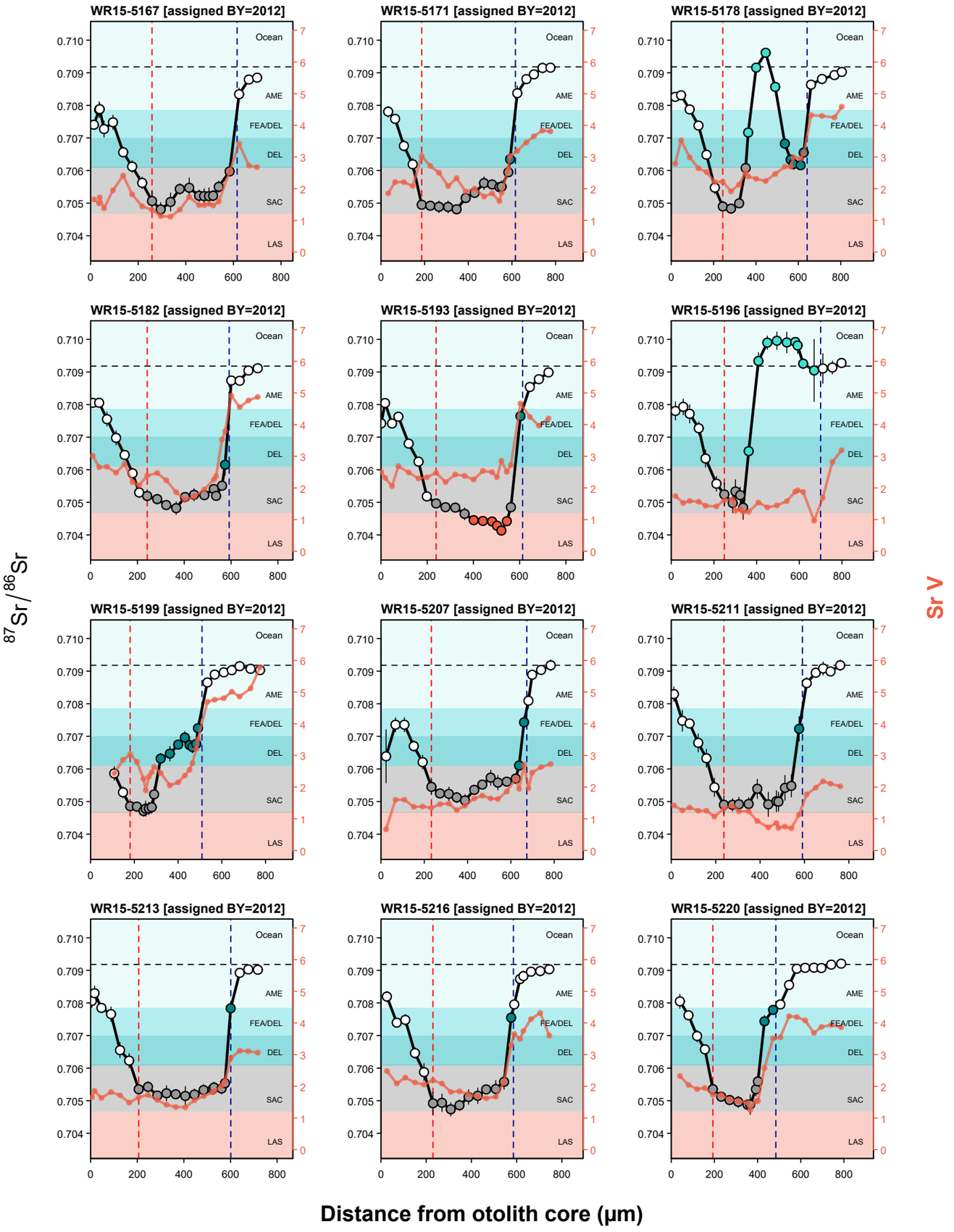

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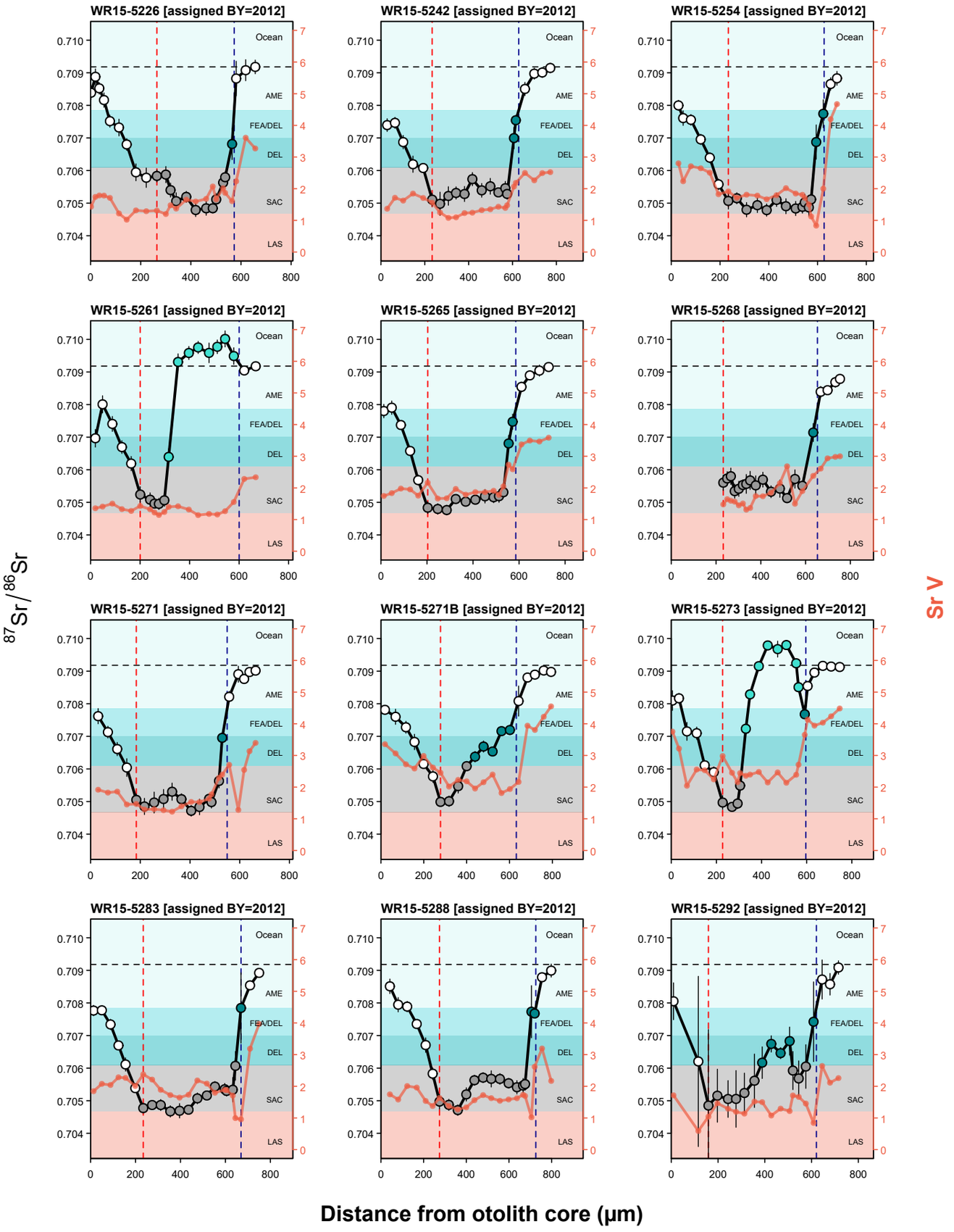

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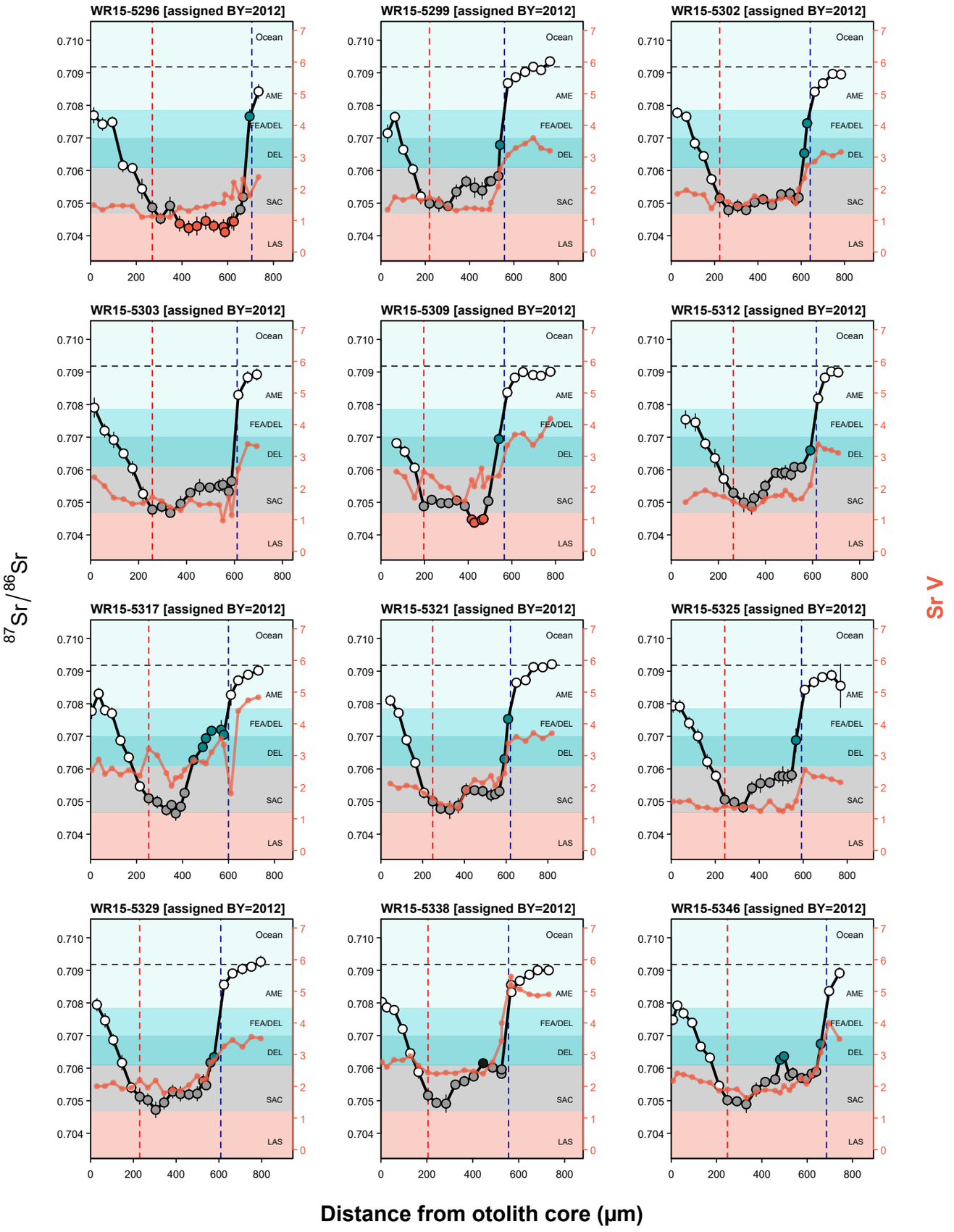

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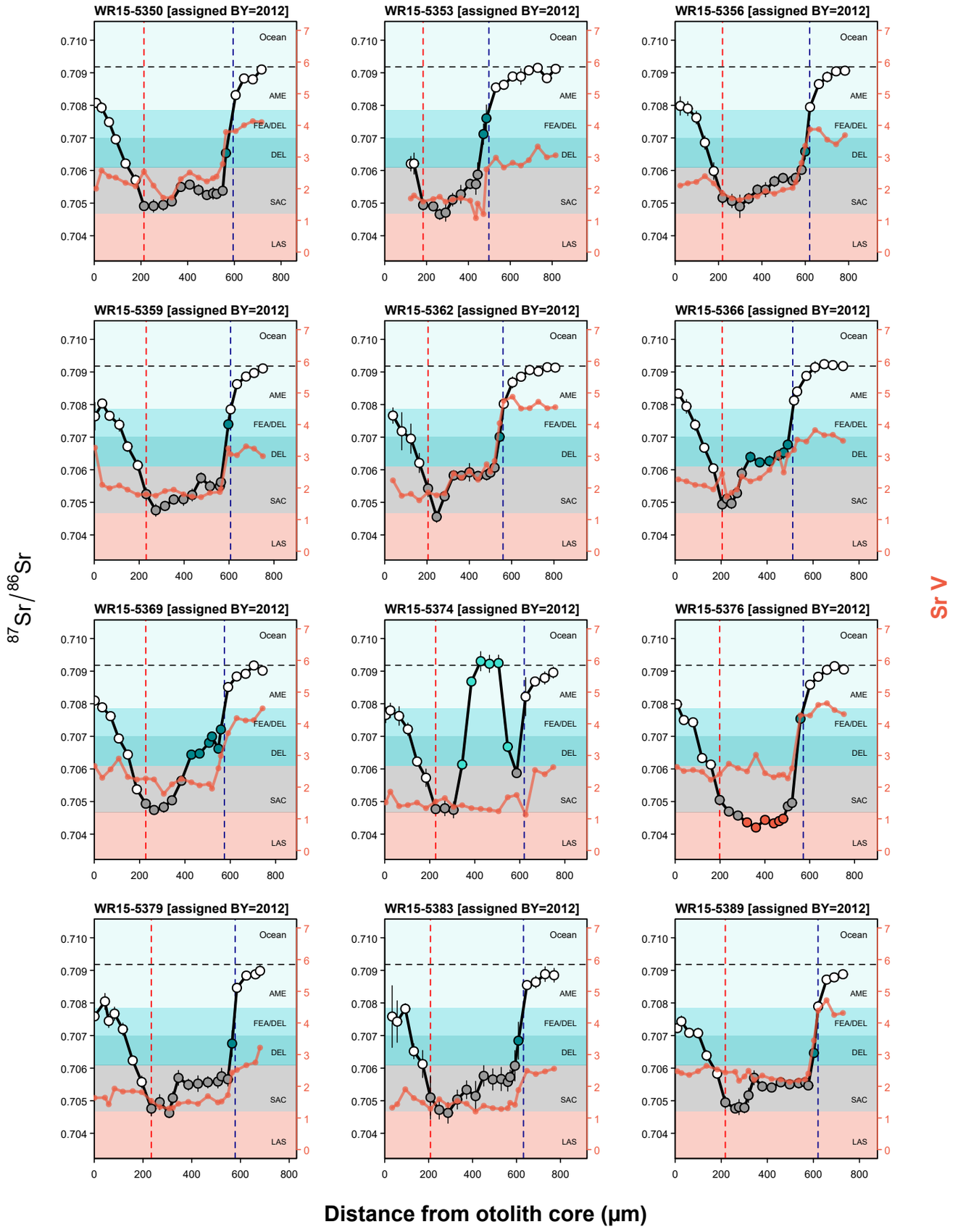

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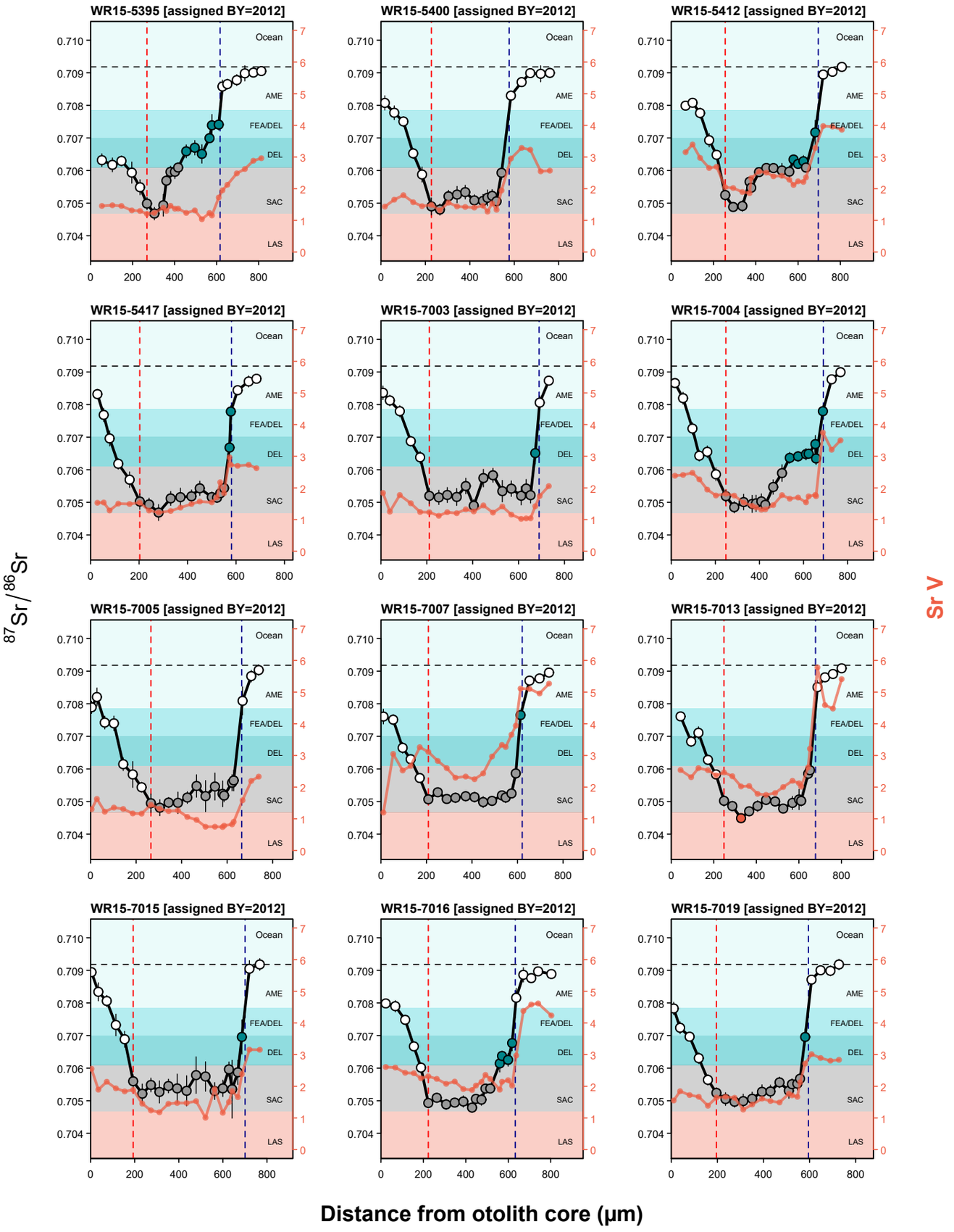

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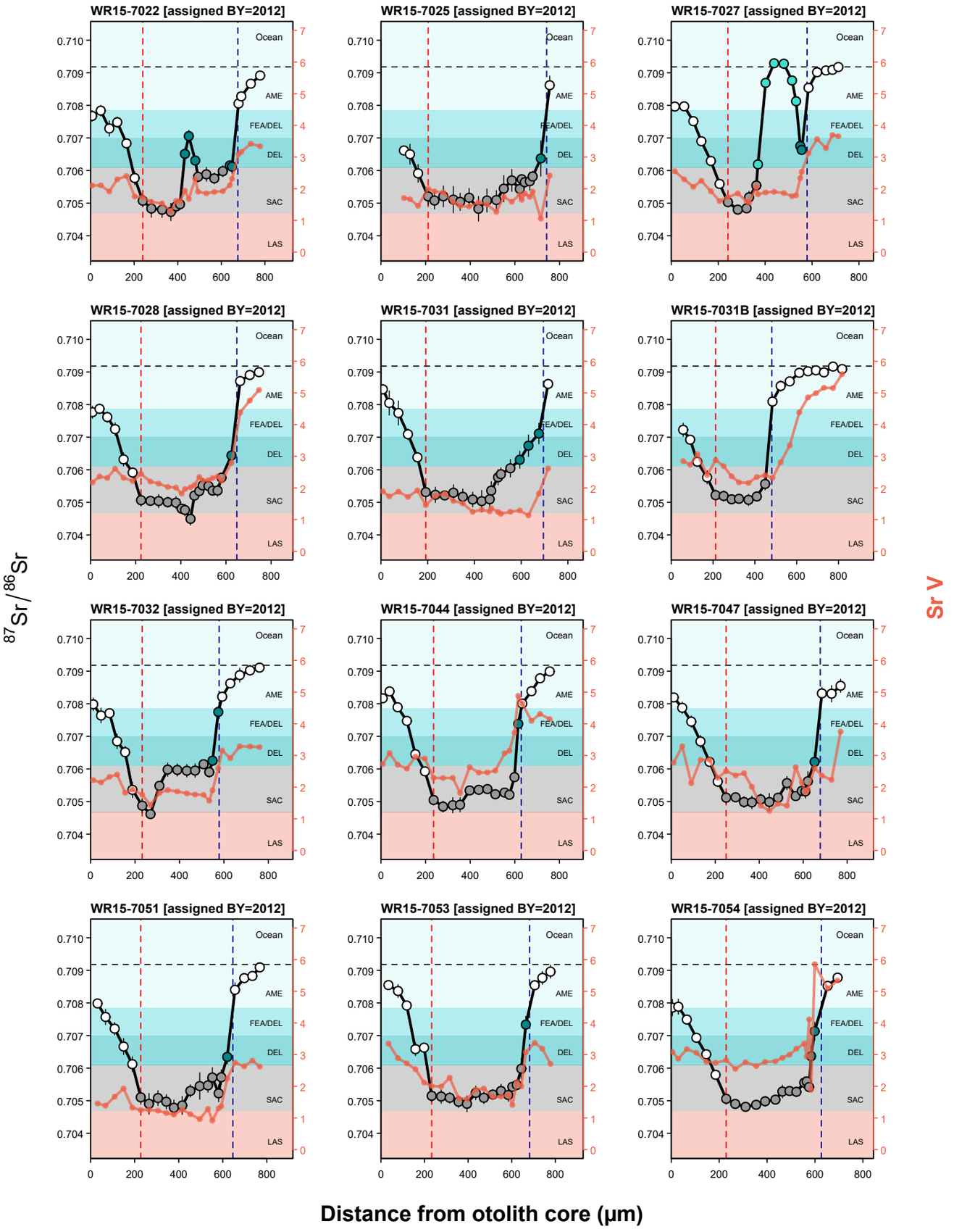

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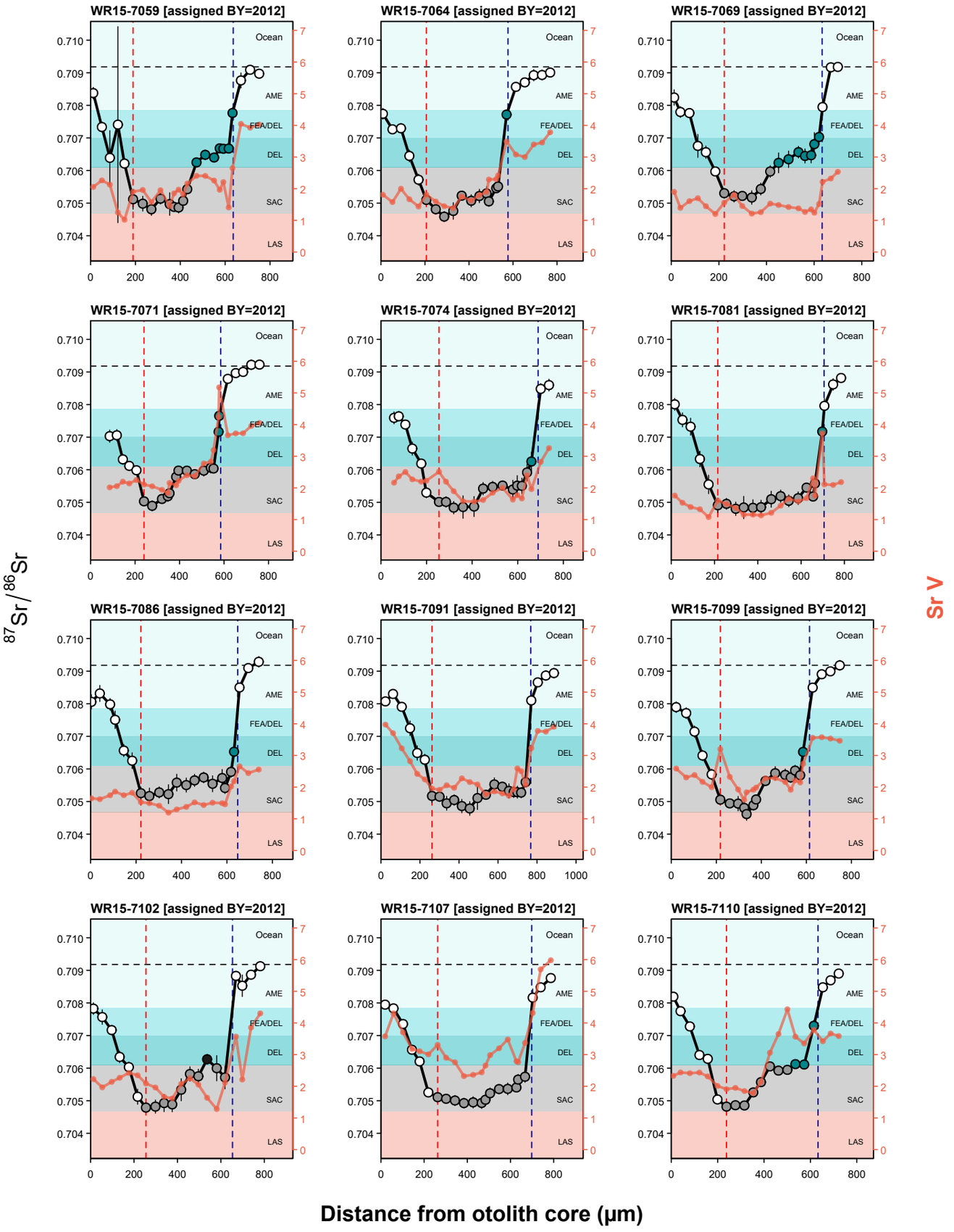

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Ocean
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FEA/DEL
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SAC
LAS
87Sr/86Sr
Sr V
Distance from otolith core (µm)

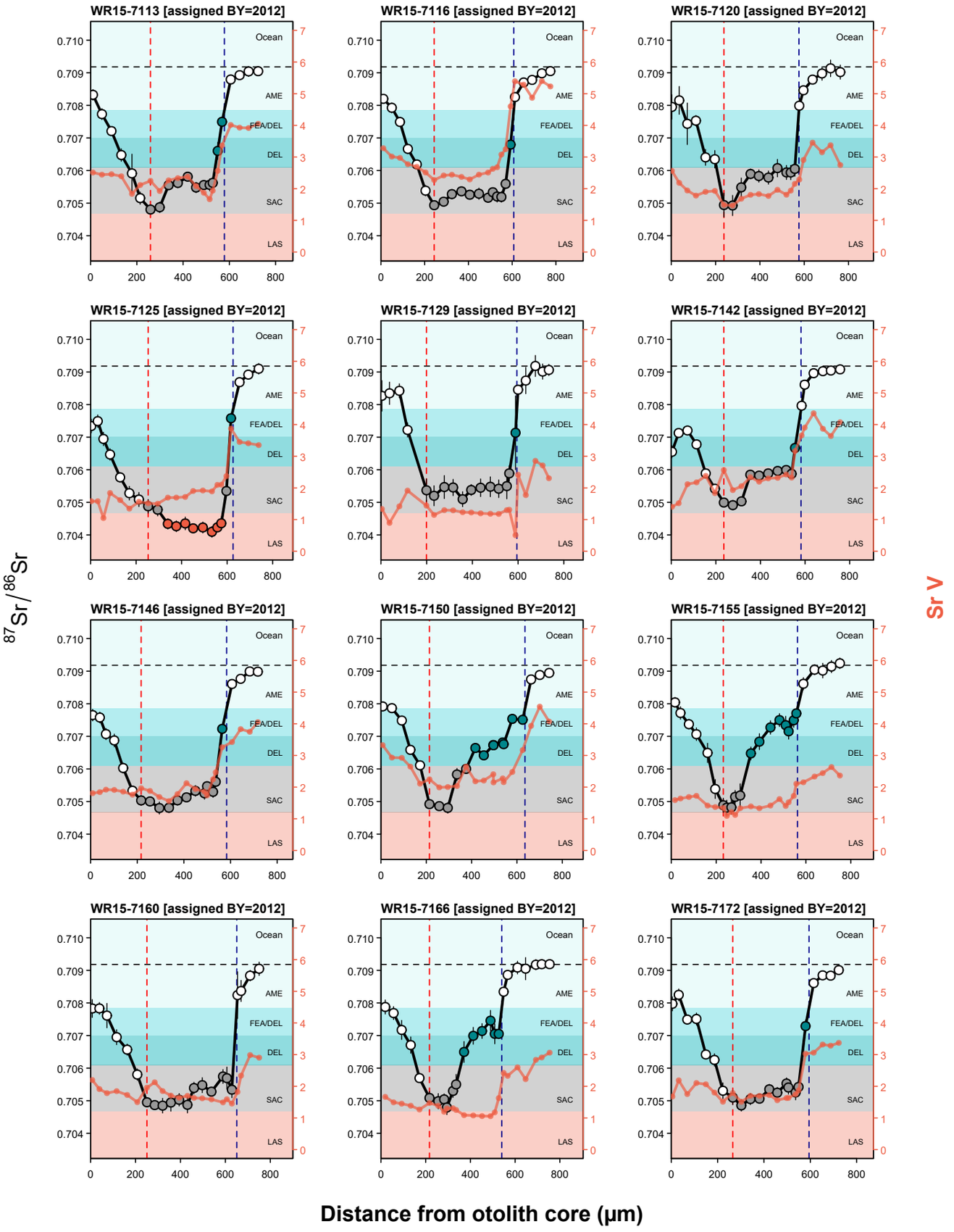

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AME
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Sr V
Distance from otolith core (µm)

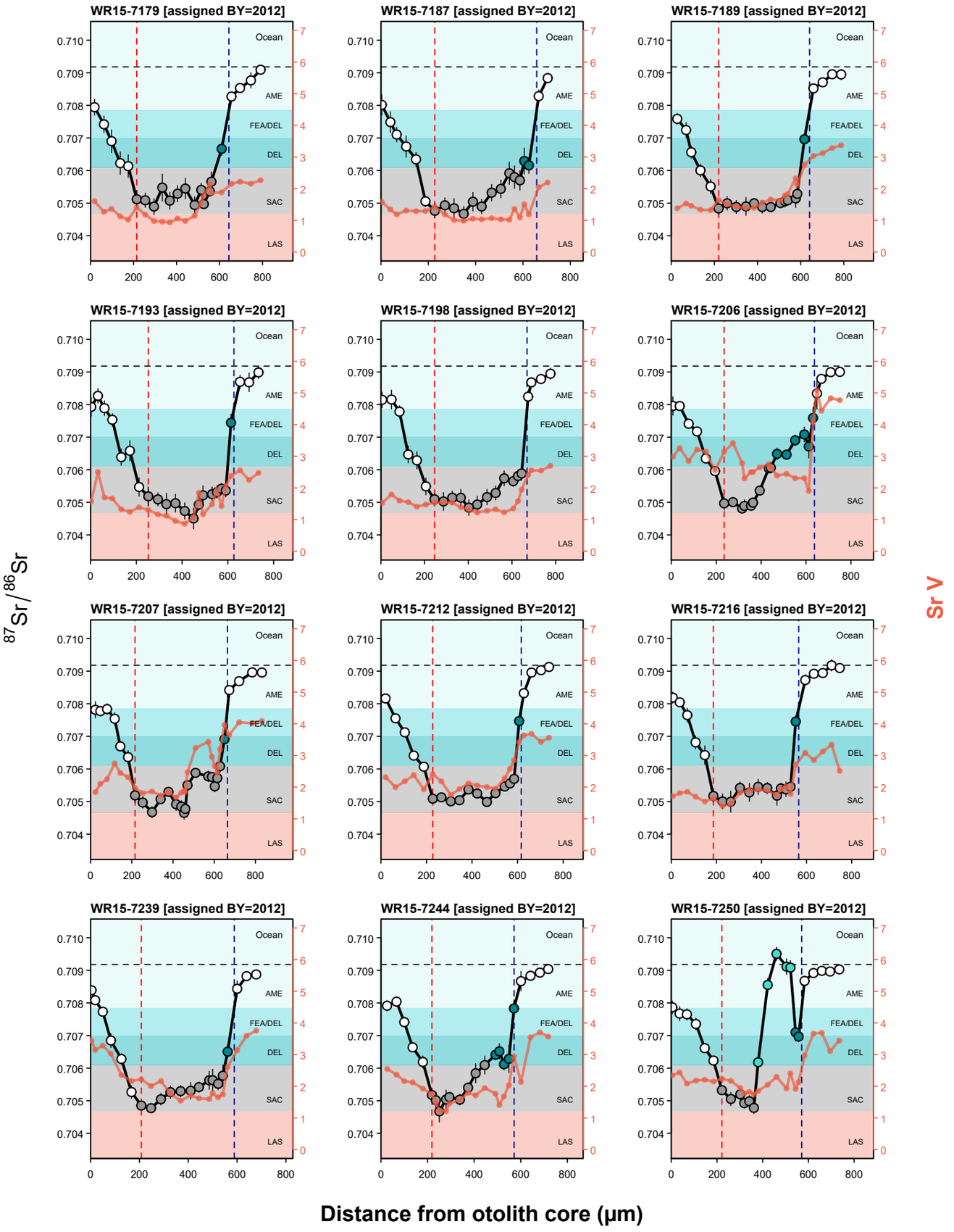

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Ocean
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$^{87}Sr/^{86}Sr$
Sr V
Distance from otolith core (µm)

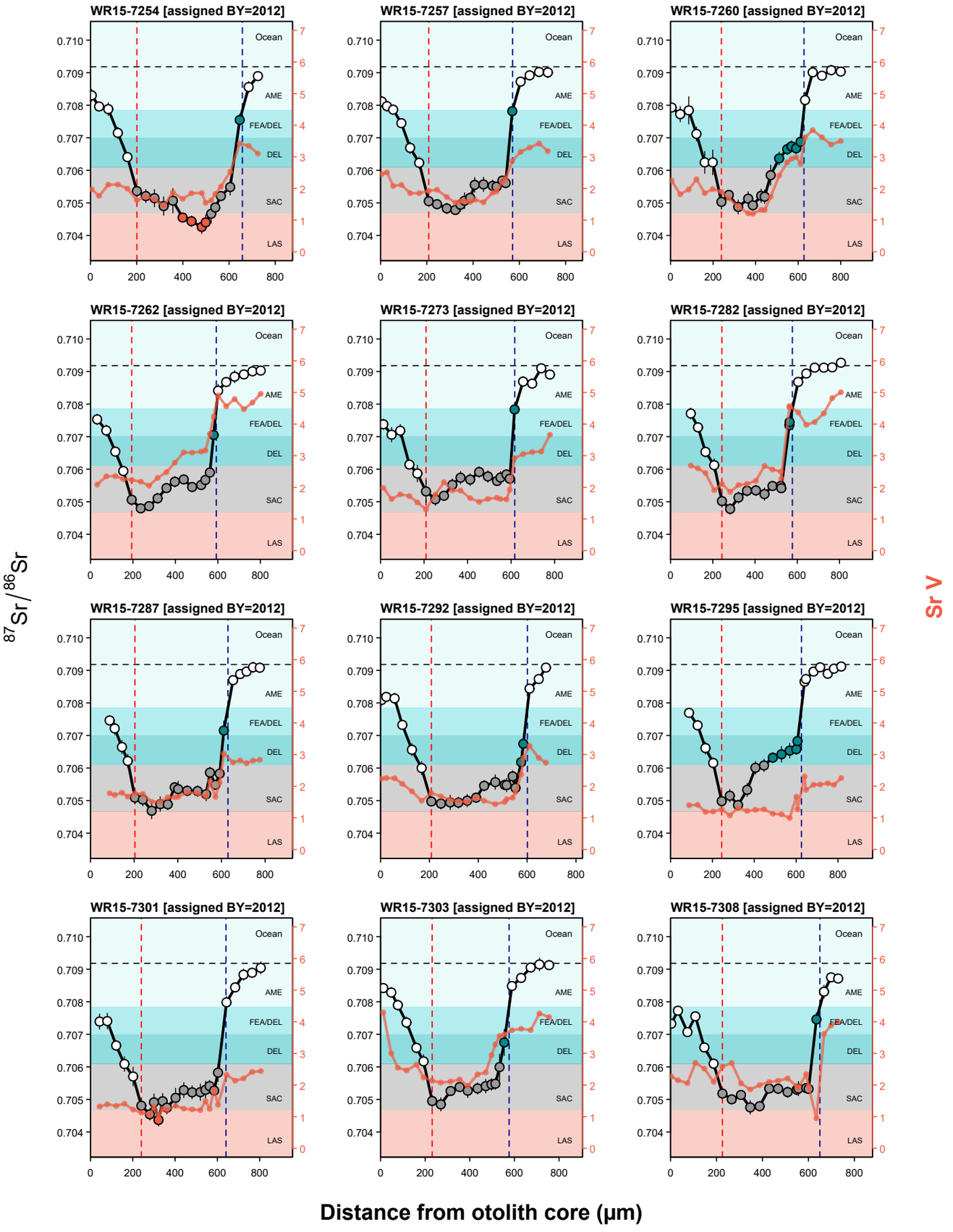

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87Sr/86Sr
Sr V
Distance from otolith core (µm)

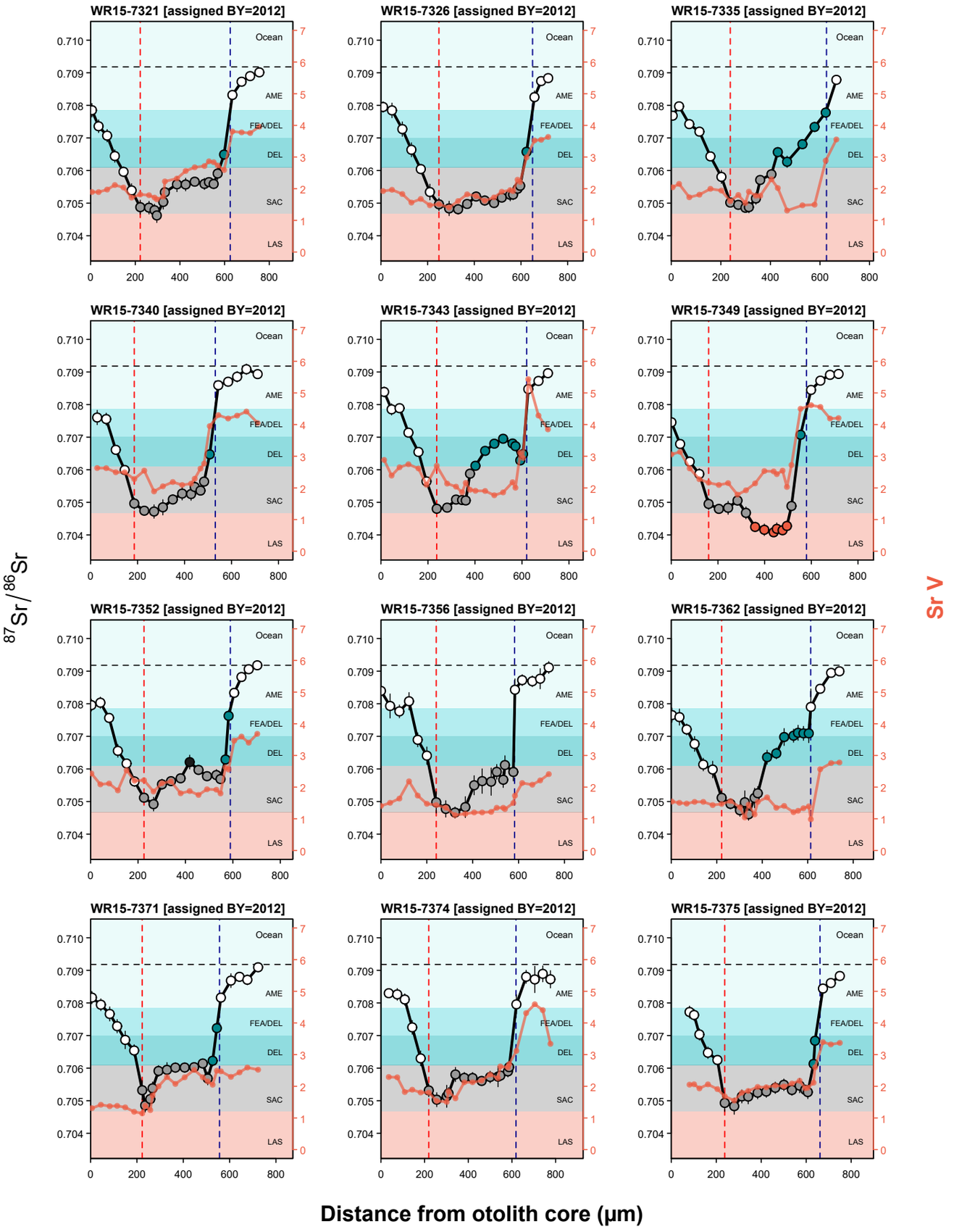

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Ocean
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FEA/DEL
DEL
SAC
LAS
$^{87}Sr/^{86}Sr$
Sr V
Distance from otolith core (µm)

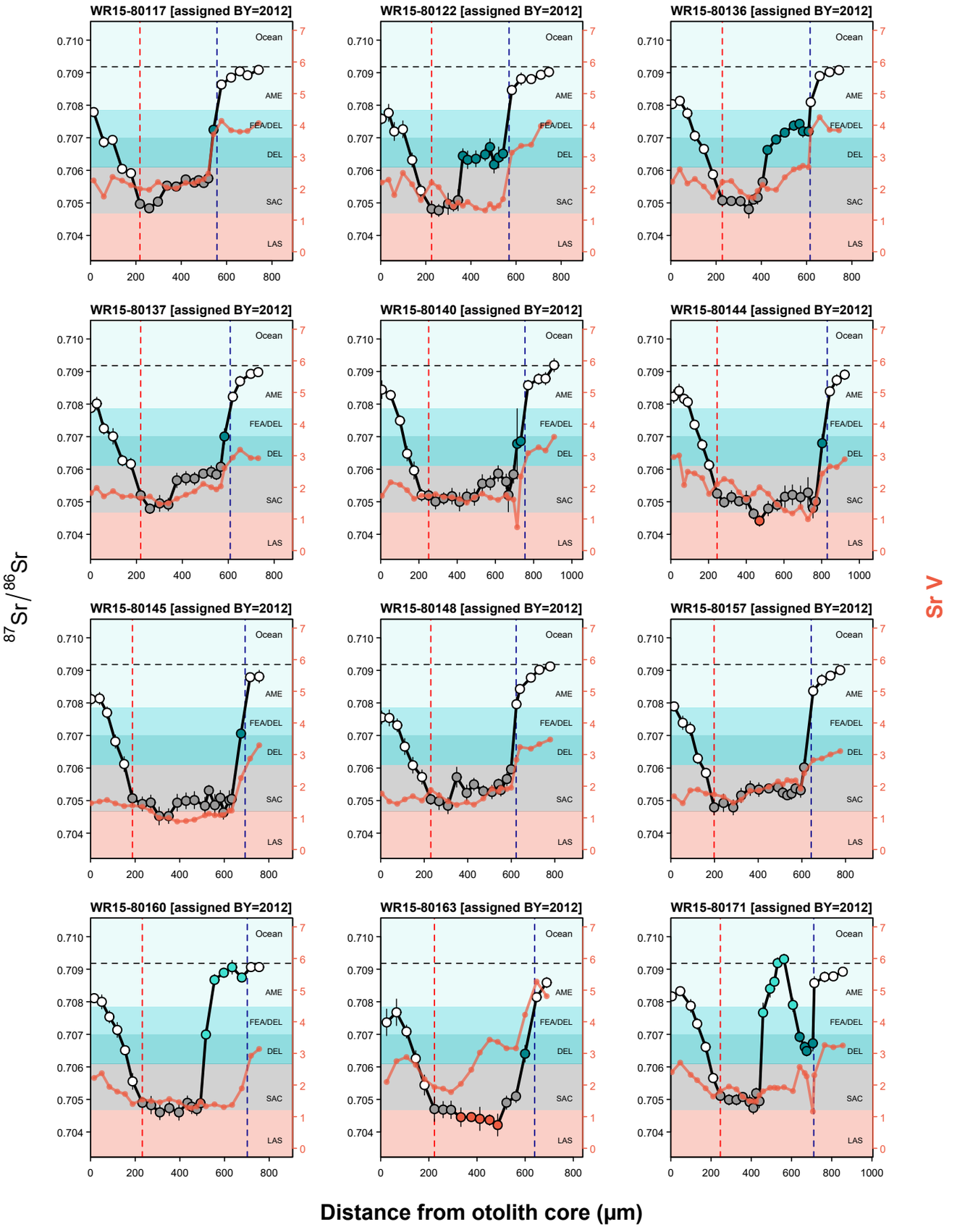

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FEA/DEL
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SAC
LAS
87Sr/86Sr
Sr V
Distance from otolith core (µm)

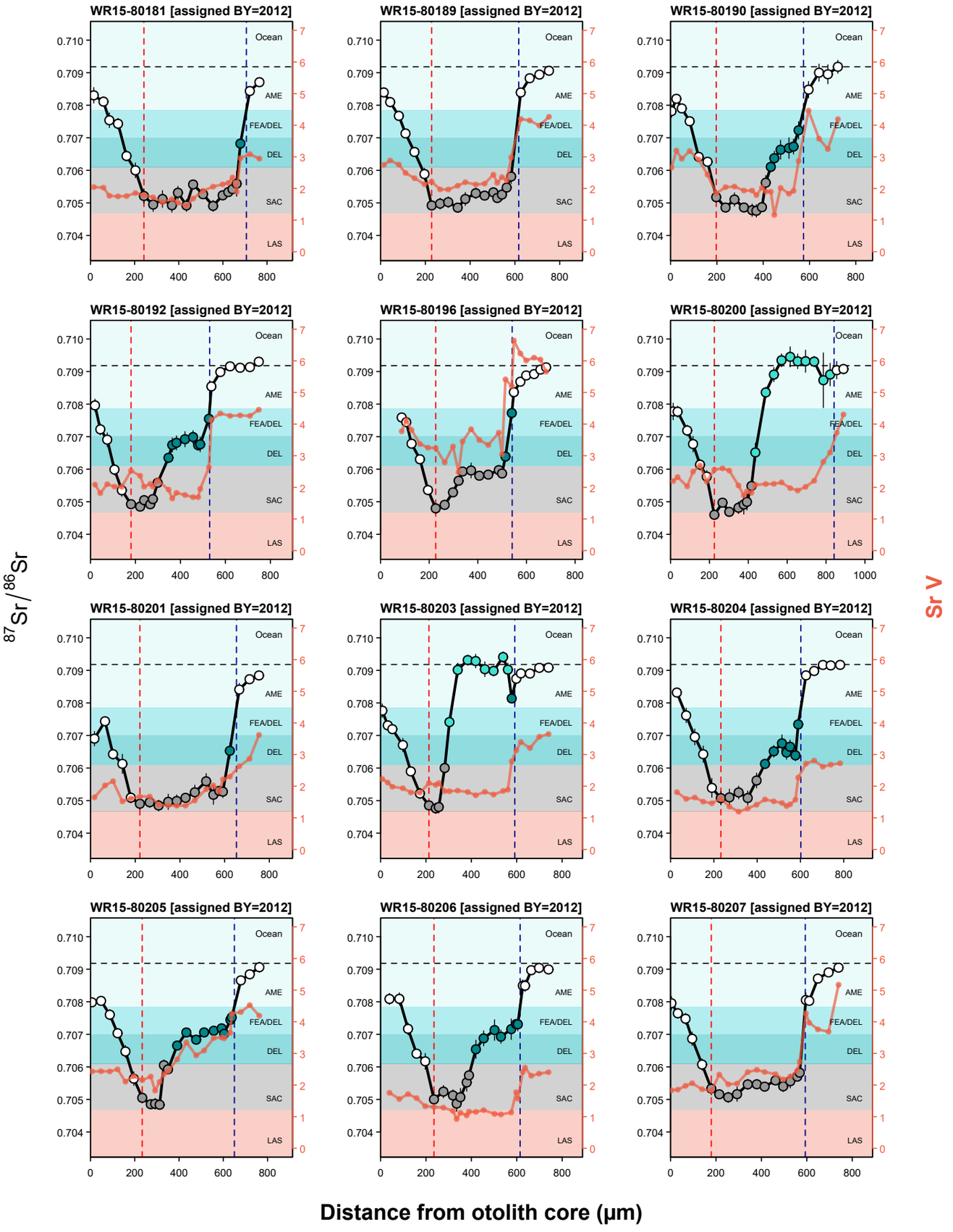

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FEA/DEL
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SAC
LAS
$^{87}Sr/^{86}Sr$
Sr V
Distance from otolith core (µm)

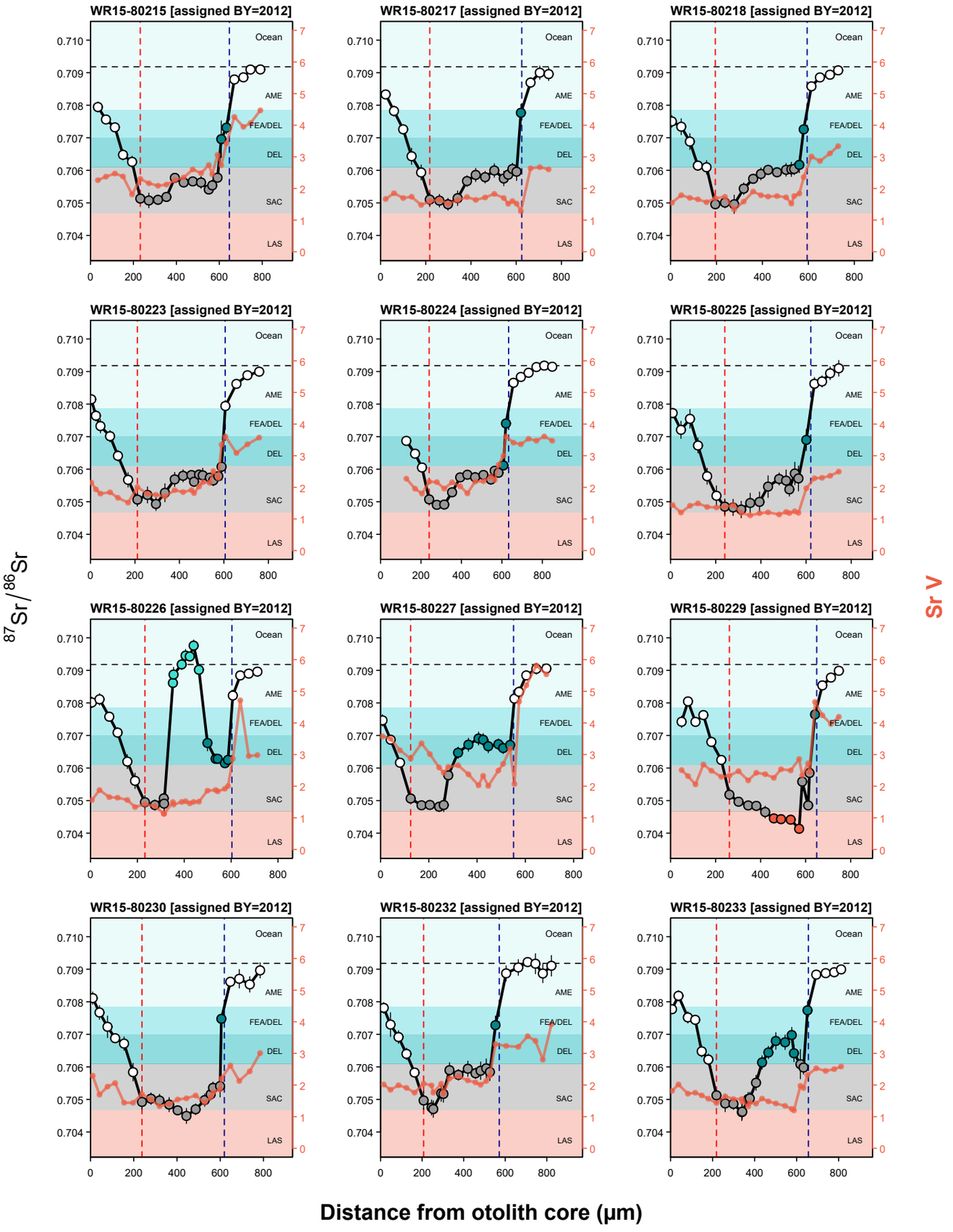

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WR15-7244 [assigned BY=2012]
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AME
FEA/DEL
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SAC
LAS
$^{87}Sr/^{86}Sr$
Sr V
Distance from otolith core (µm)

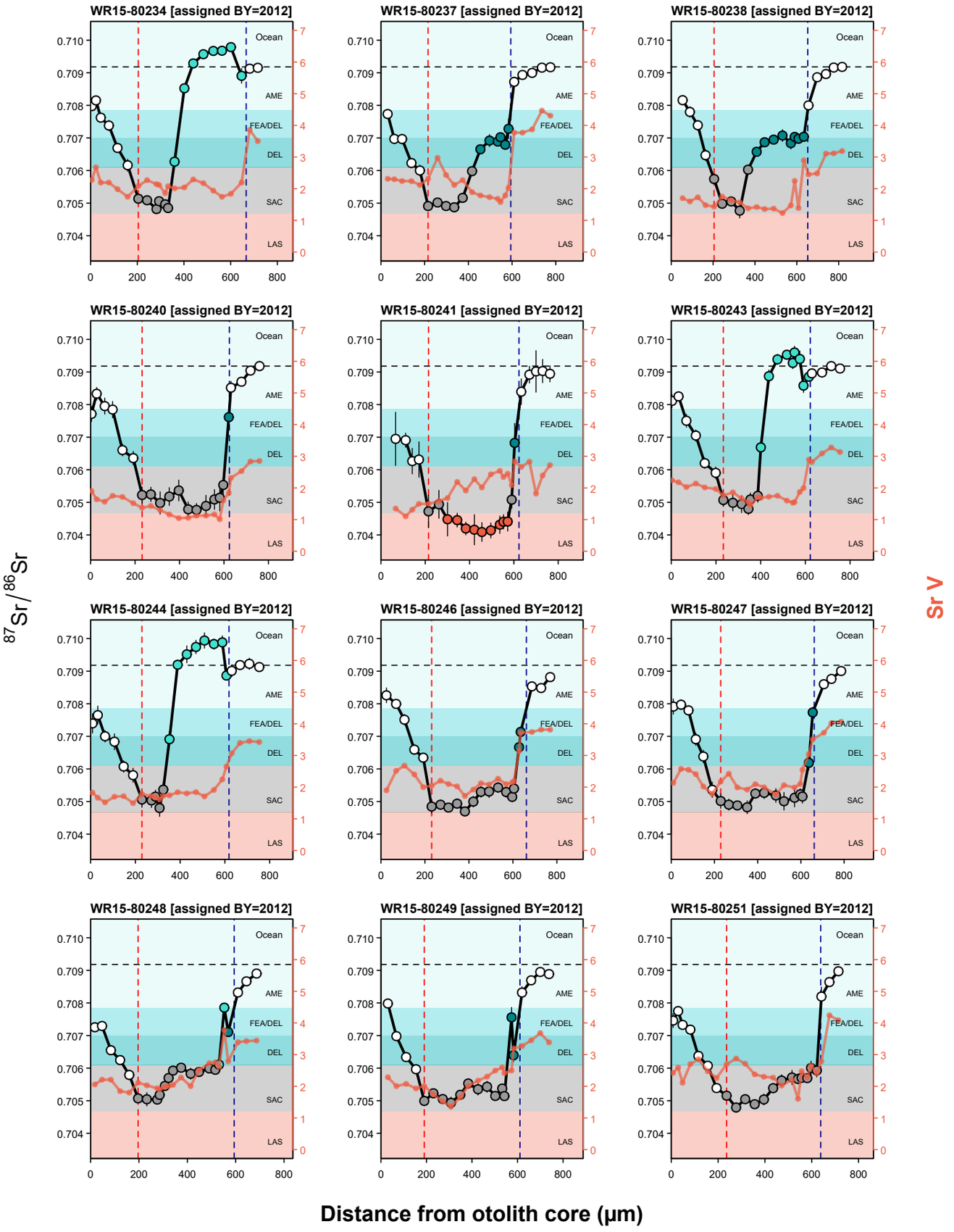

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Ocean
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FEA/DEL
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SAC
LAS
$^{87}Sr/^{86}Sr$
Sr V
Distance from otolith core (µm)

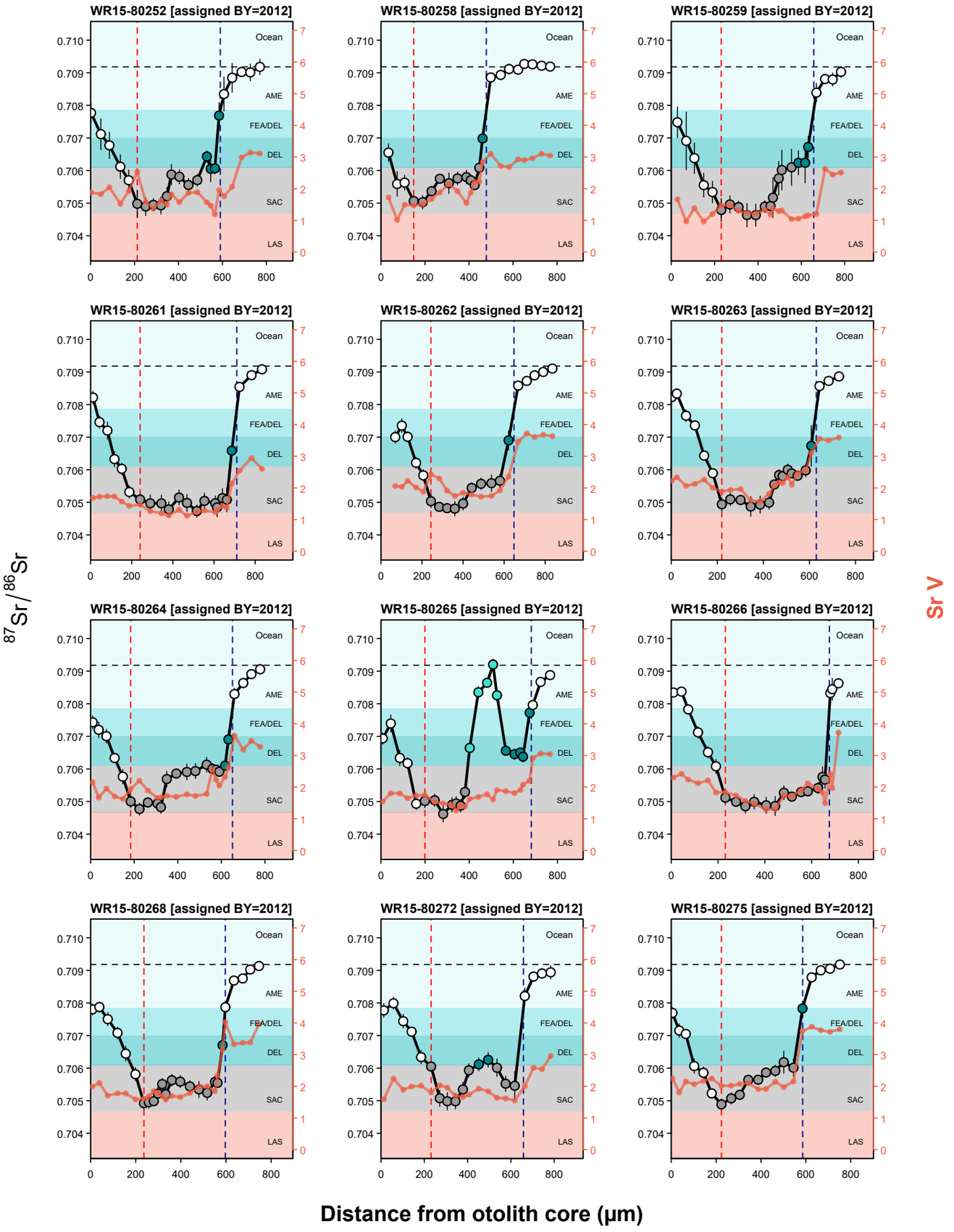

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AME
FEA/DEL
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LAS
$^{87}Sr/^{86}Sr$
Sr V
Distance from otolith core (µm)

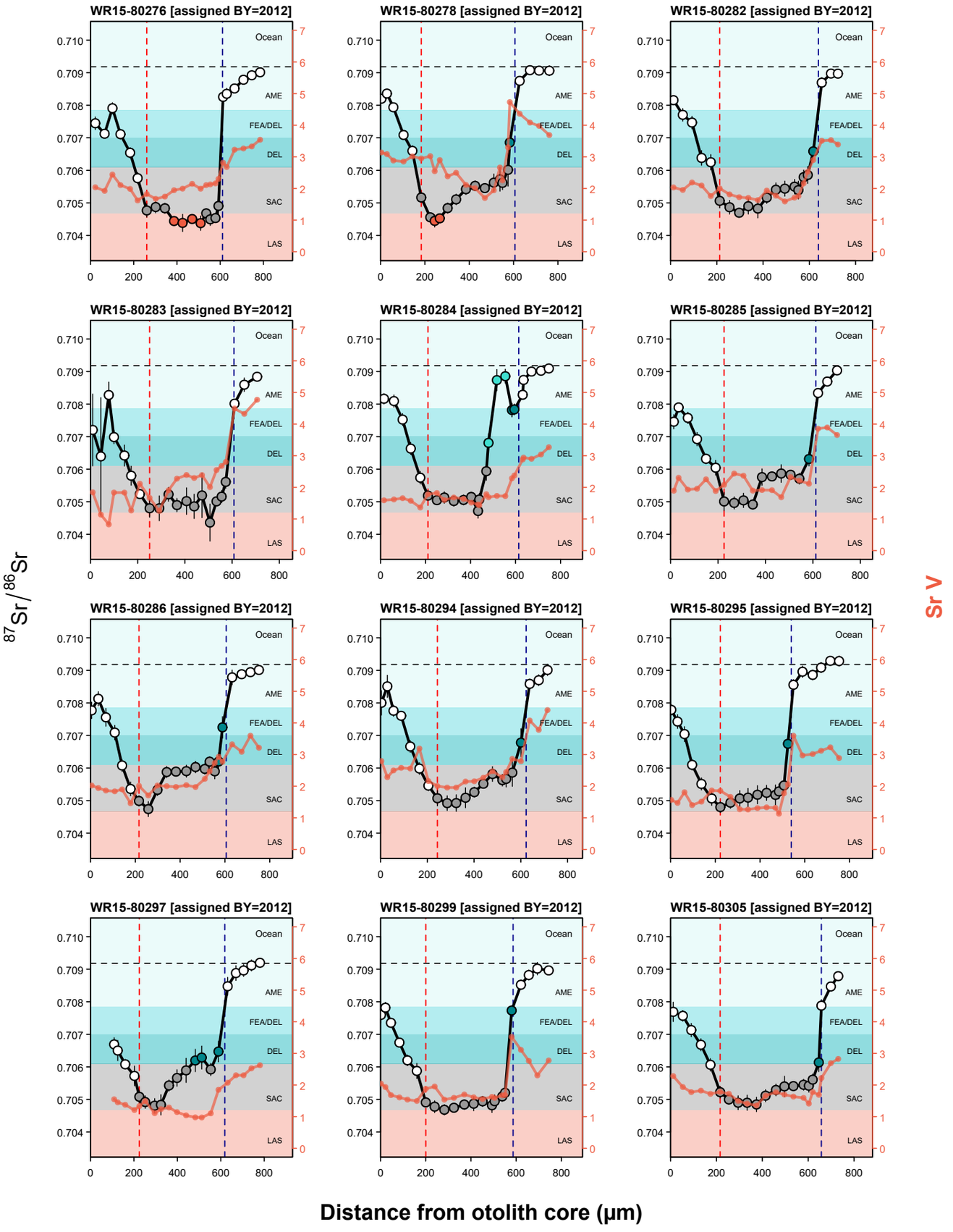

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FEA/DEL
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$^{87}Sr/^{86}Sr$
Sr V
Distance from otolith core (µm)

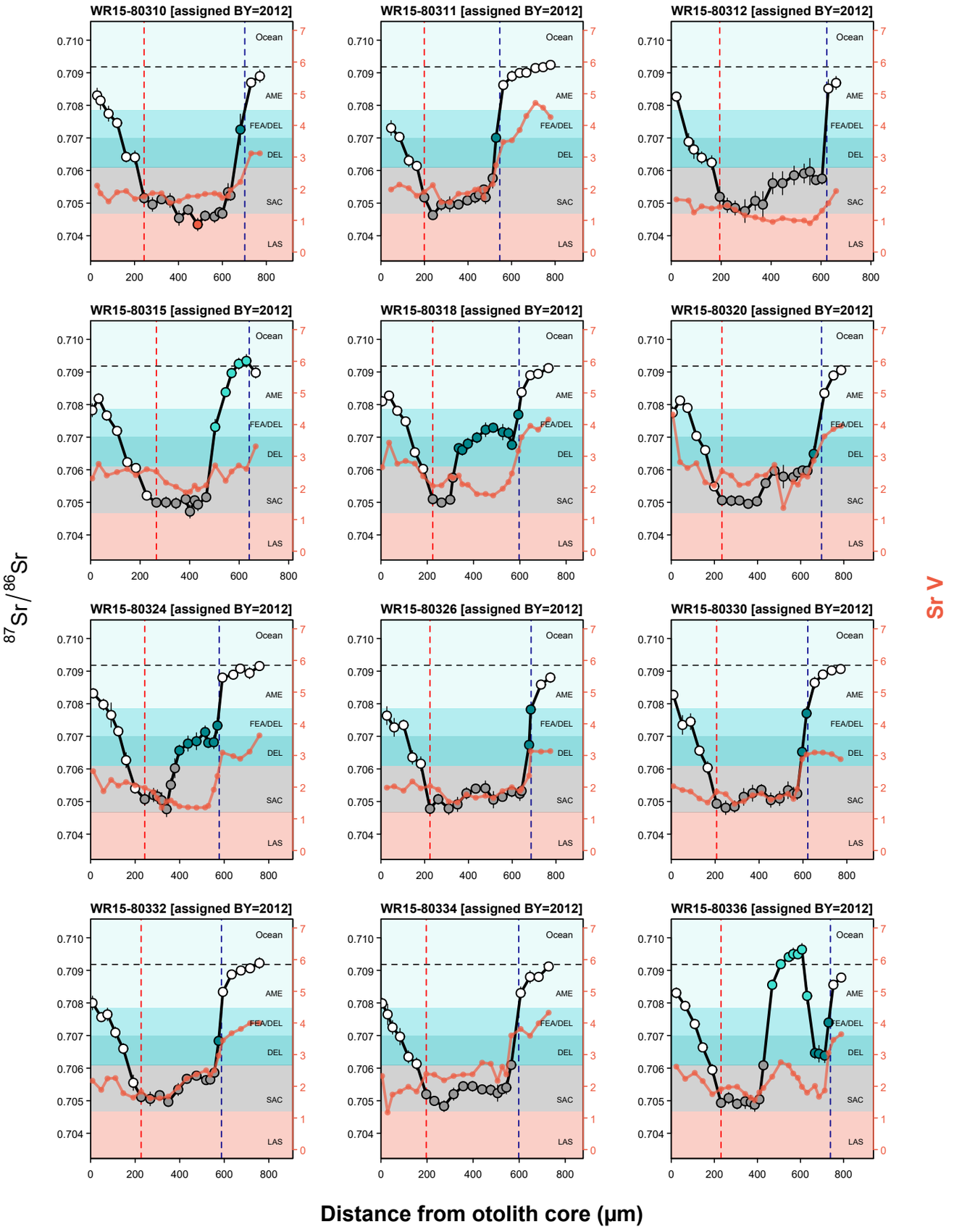

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AME
FEA/DEL
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LAS
$^{87}Sr/^{86}Sr$
Sr V
Distance from otolith core (µm)

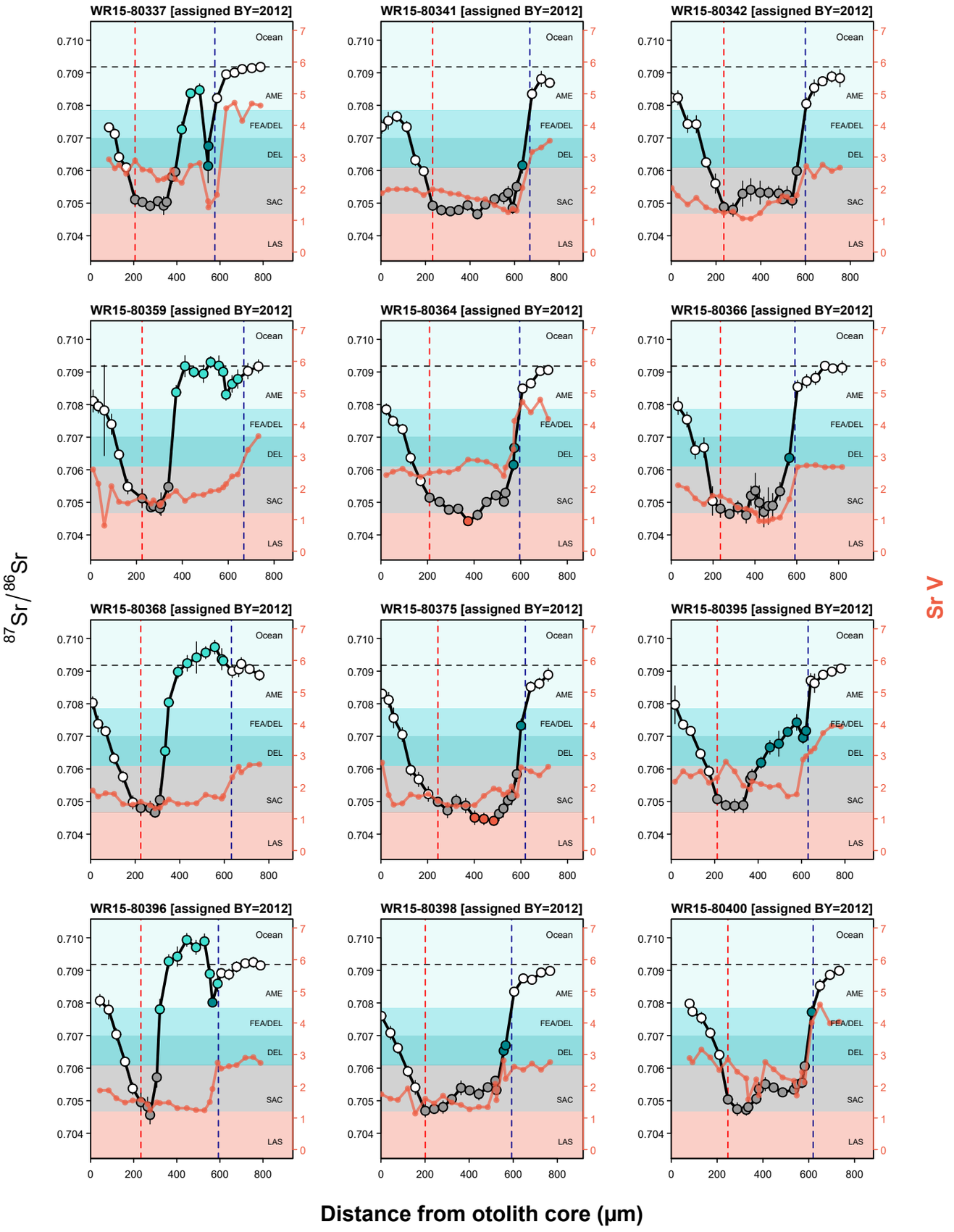

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SAC
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$^{87}Sr/^{86}Sr$
Sr V
Distance from otolith core (µm)

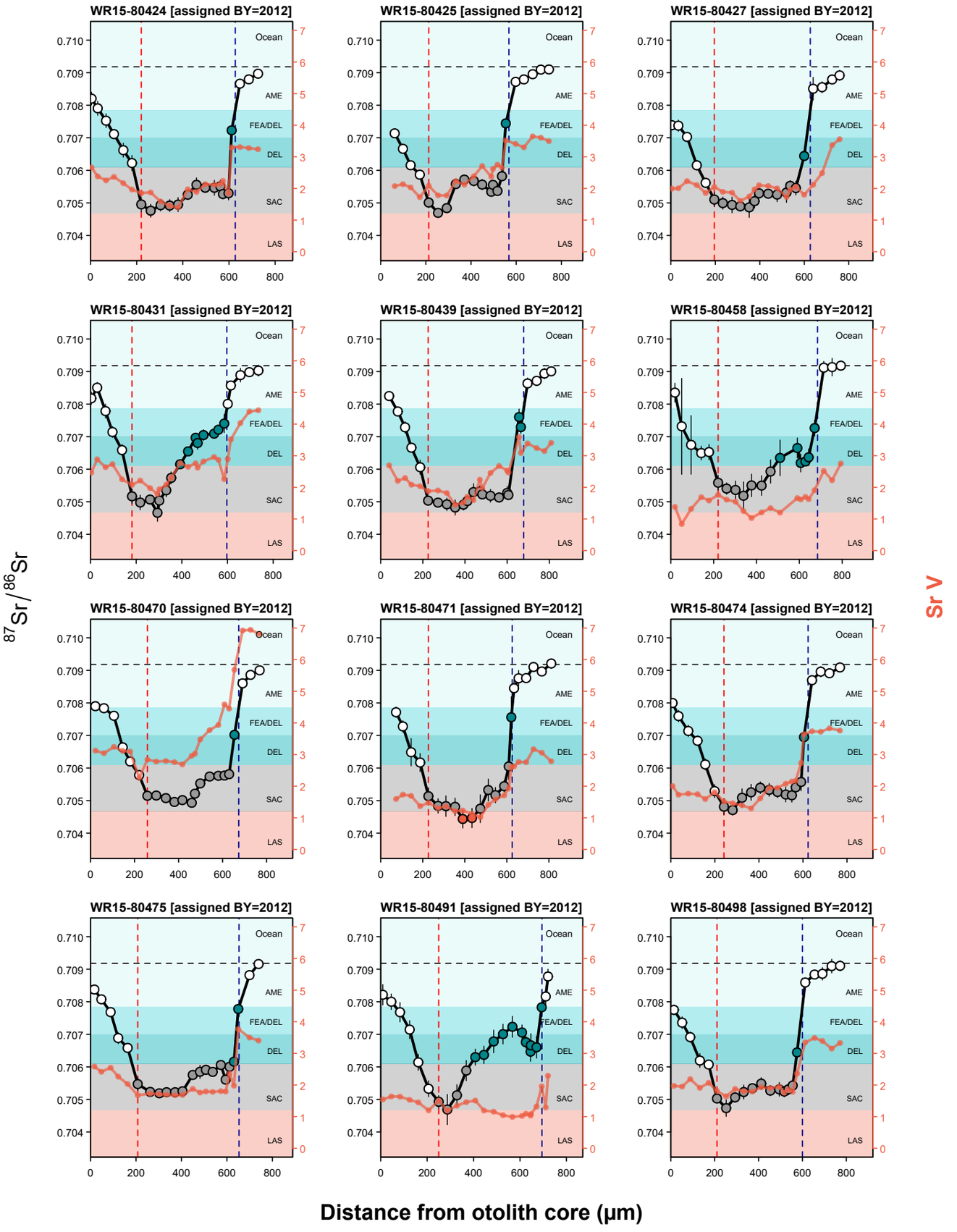

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SAC
LAS
$^{87}Sr/^{86}Sr$
Sr V
Distance from otolith core (µm)

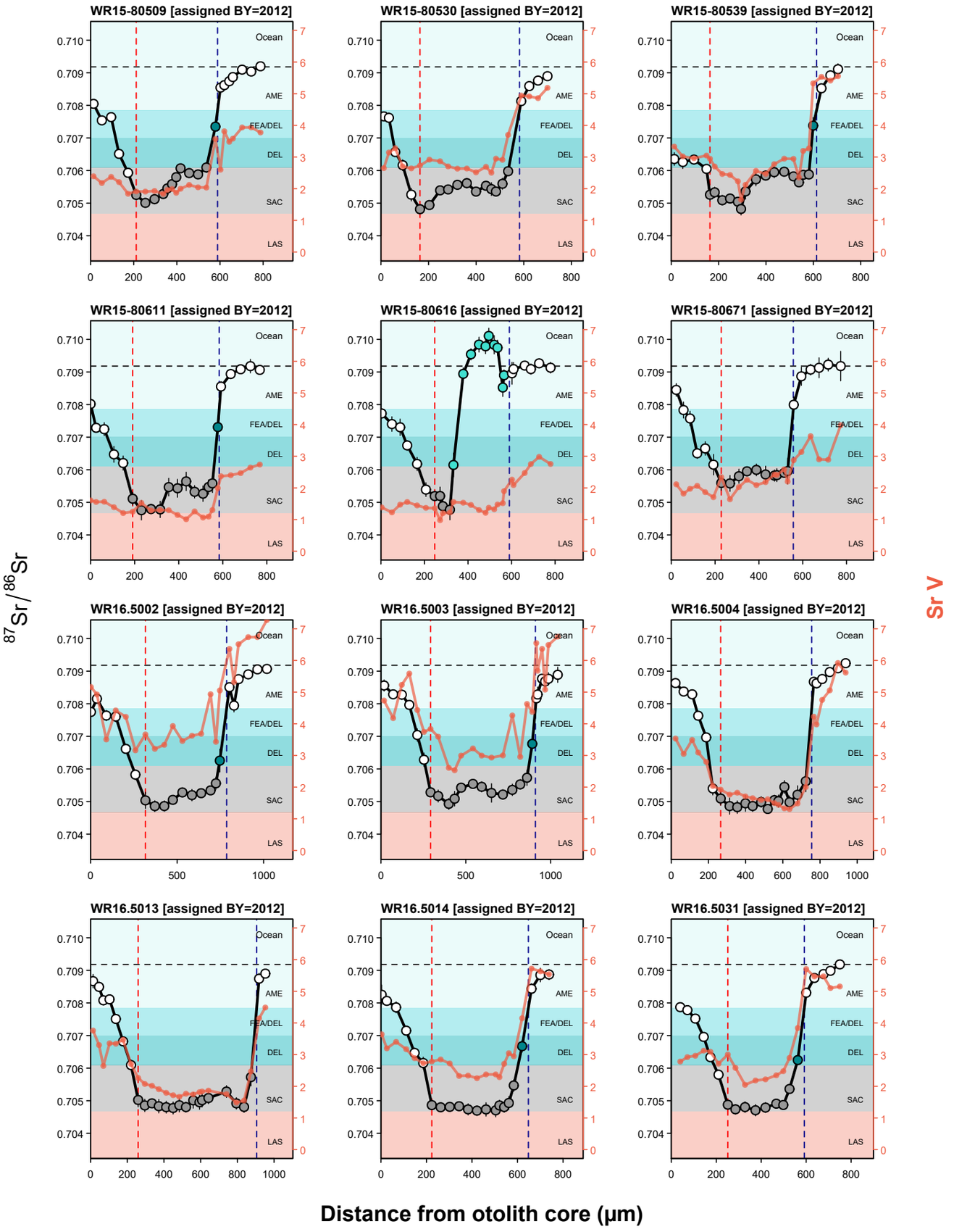

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FEA/DEL
DEL
SAC
LAS
87Sr/86Sr
Sr V
Distance from otolith core (µm)

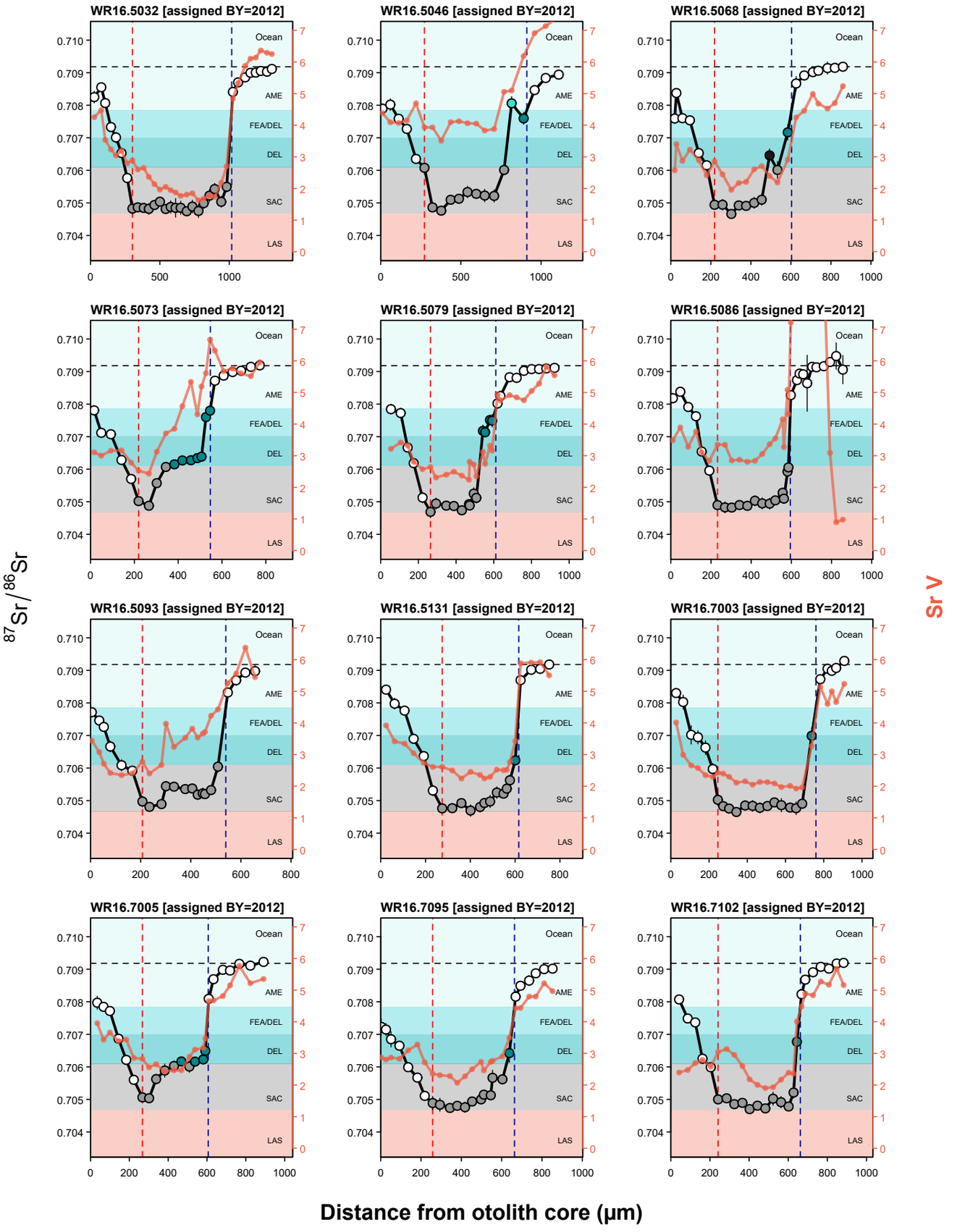

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Sr V
Distance from otolith core (µm)

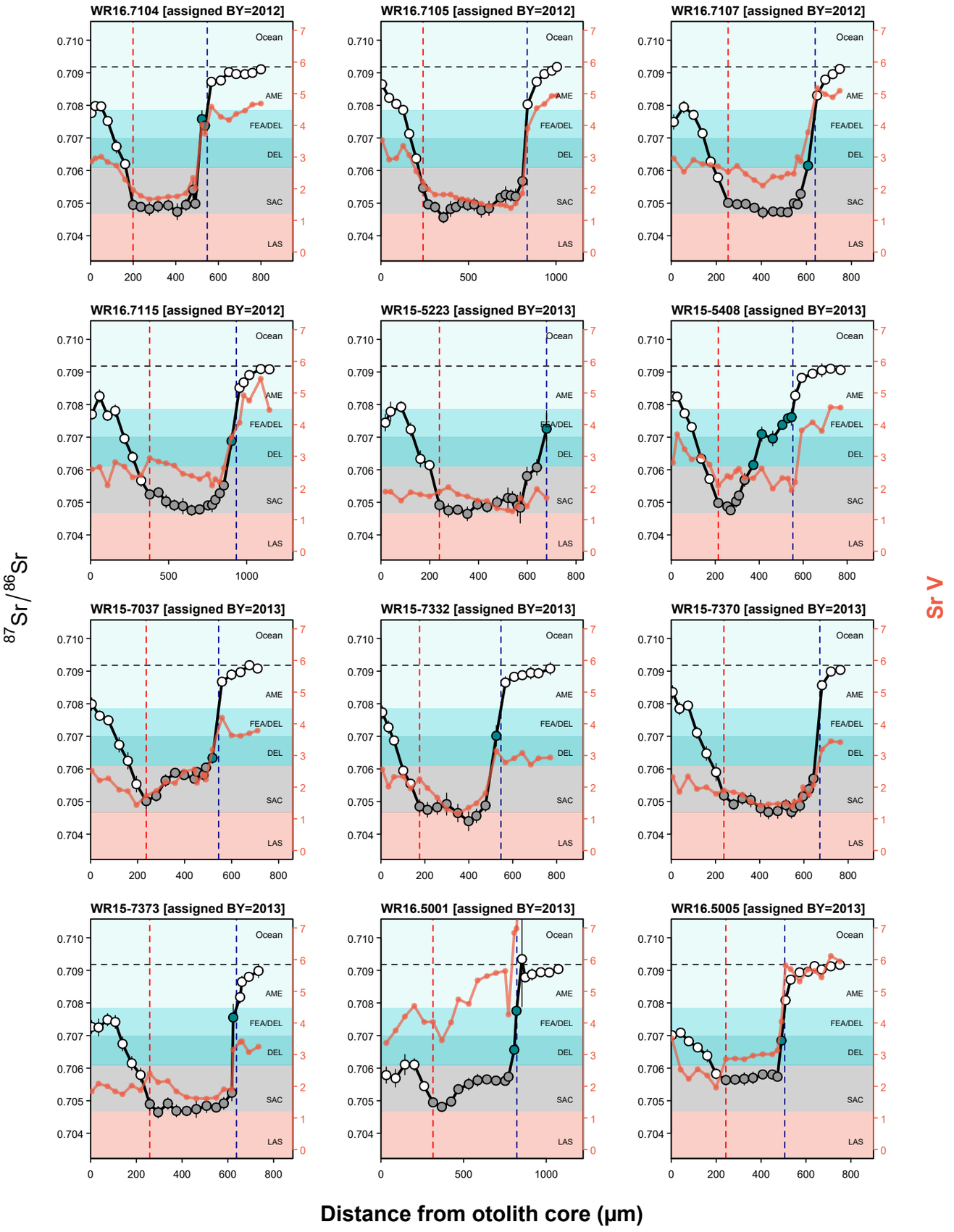

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LAS
$^{87}Sr/^{86}Sr$
Sr V
Distance from otolith core (µm)

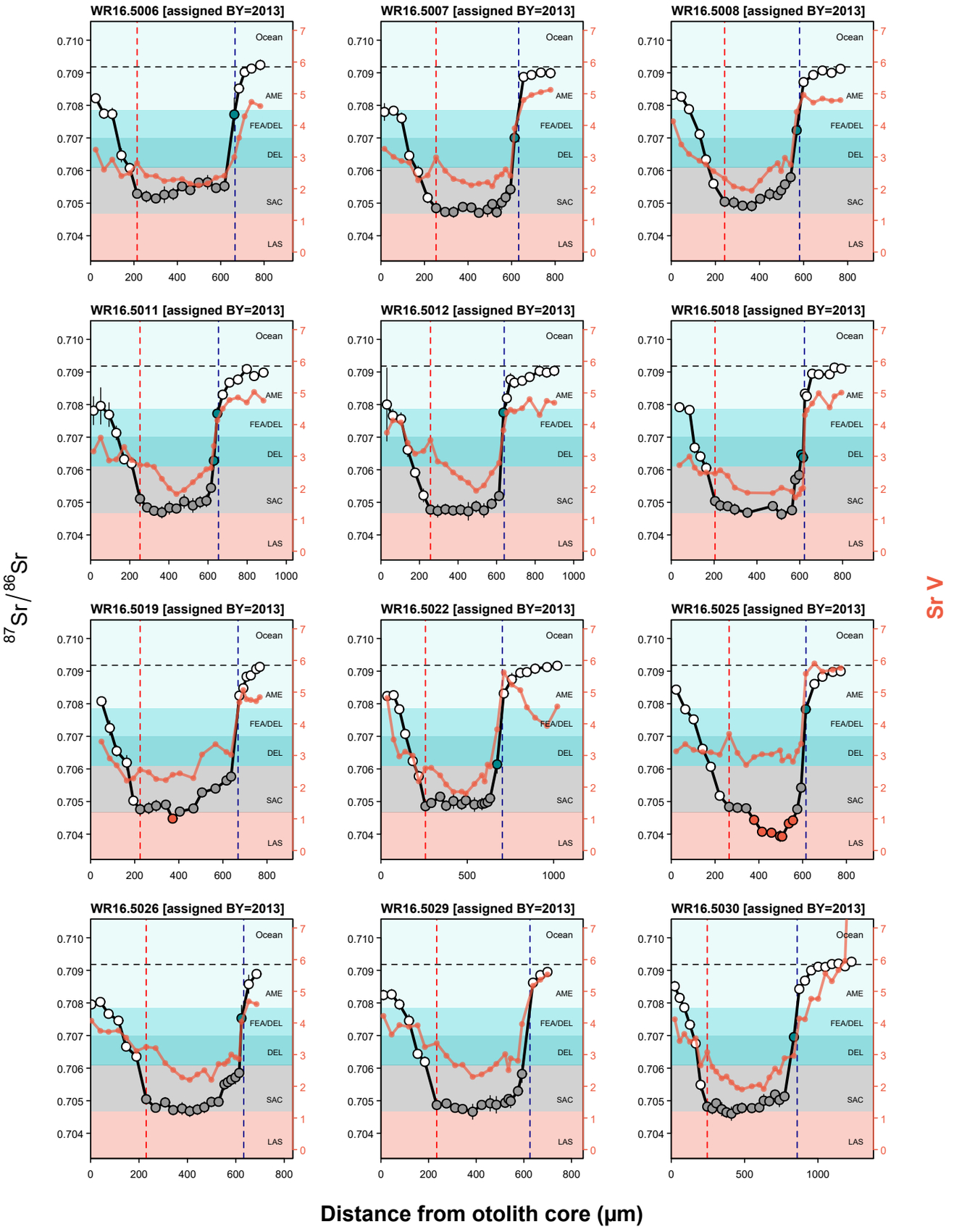

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Sr V
Distance from otolith core (µm)

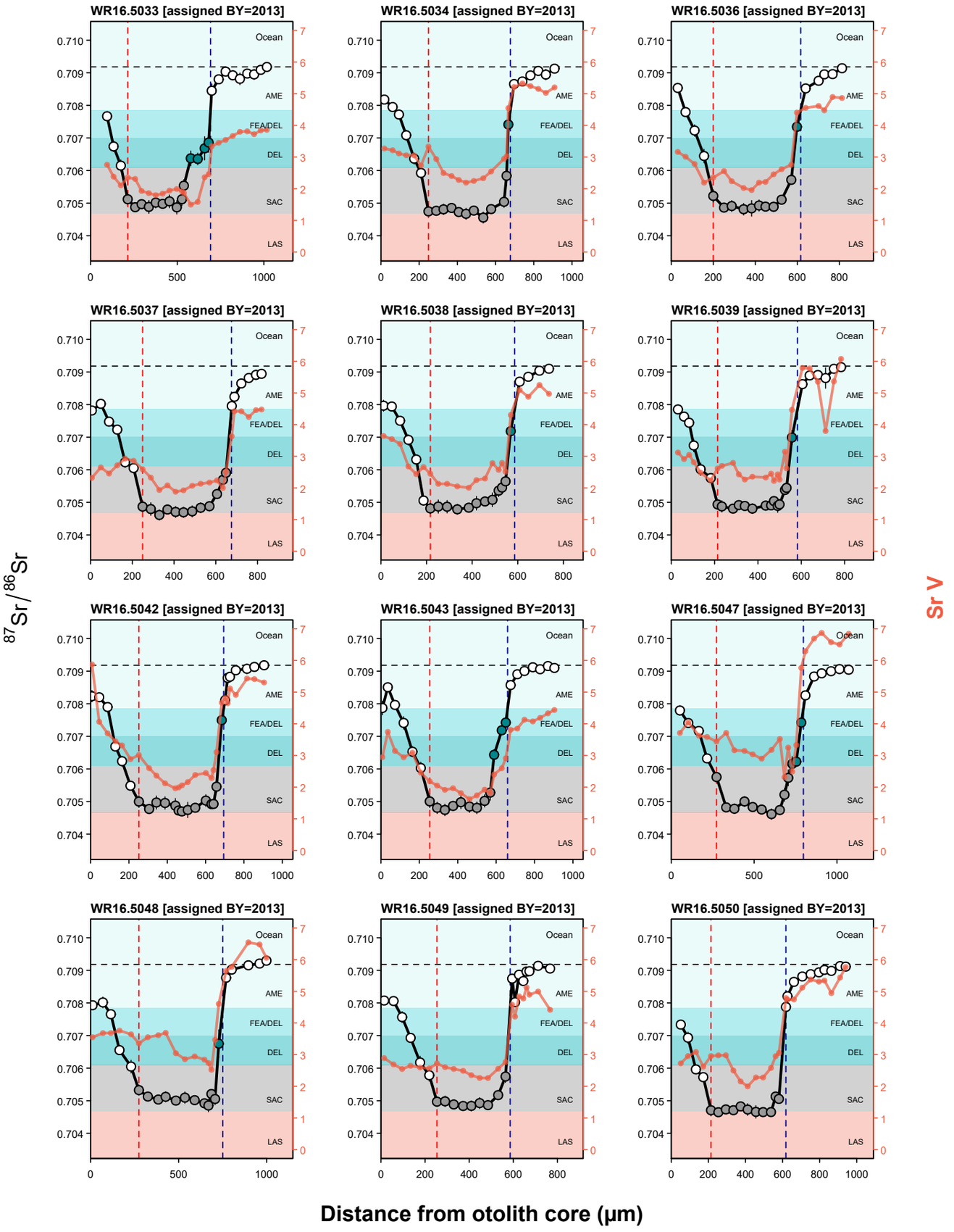

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87Sr/86Sr
Sr V
Distance from otolith core (µm)

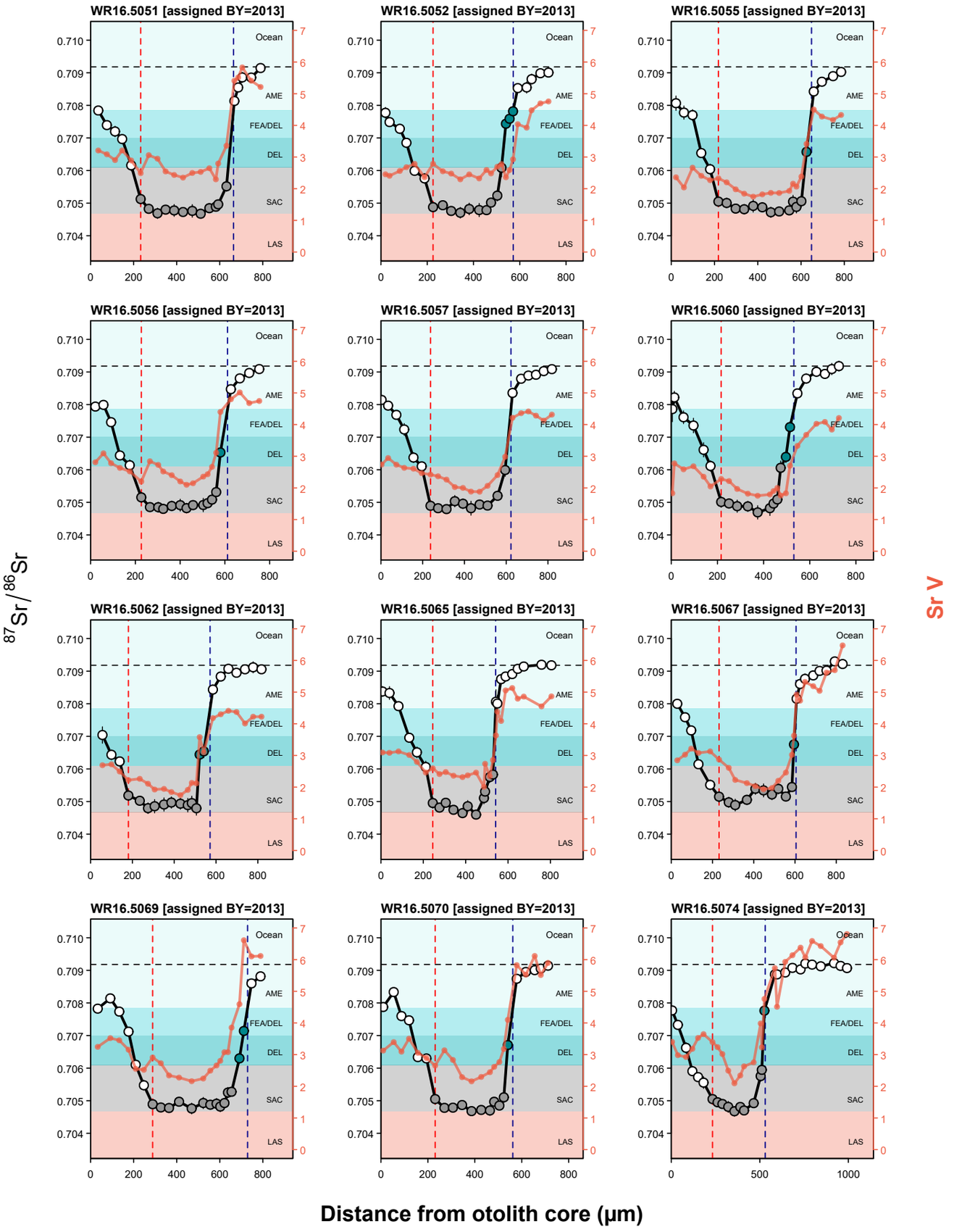

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SAC
LAS
87Sr/86Sr
Sr V
Distance from otolith core (µm)

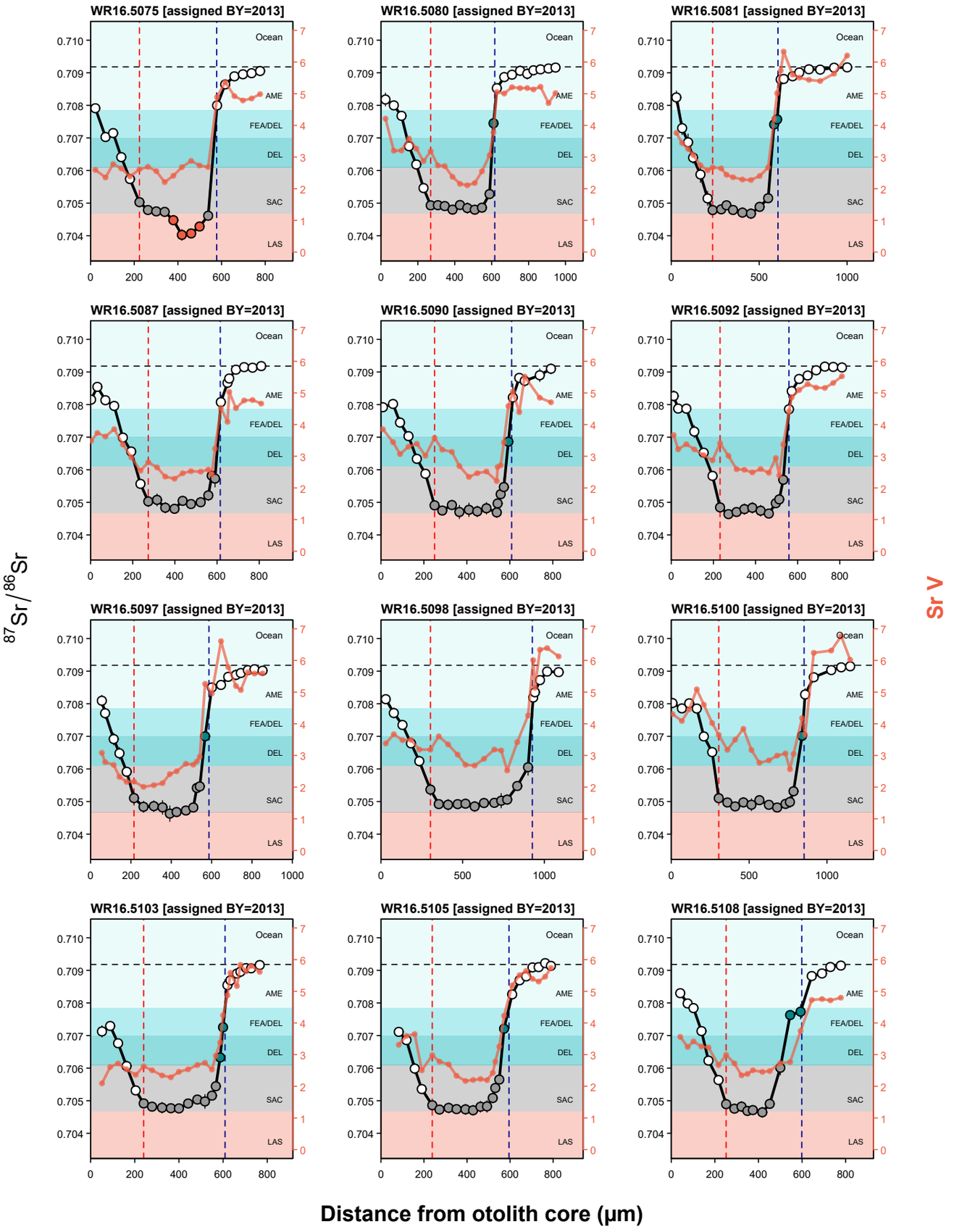

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$^{87}Sr/^{86}Sr$
Sr V
Distance from otolith core (µm)

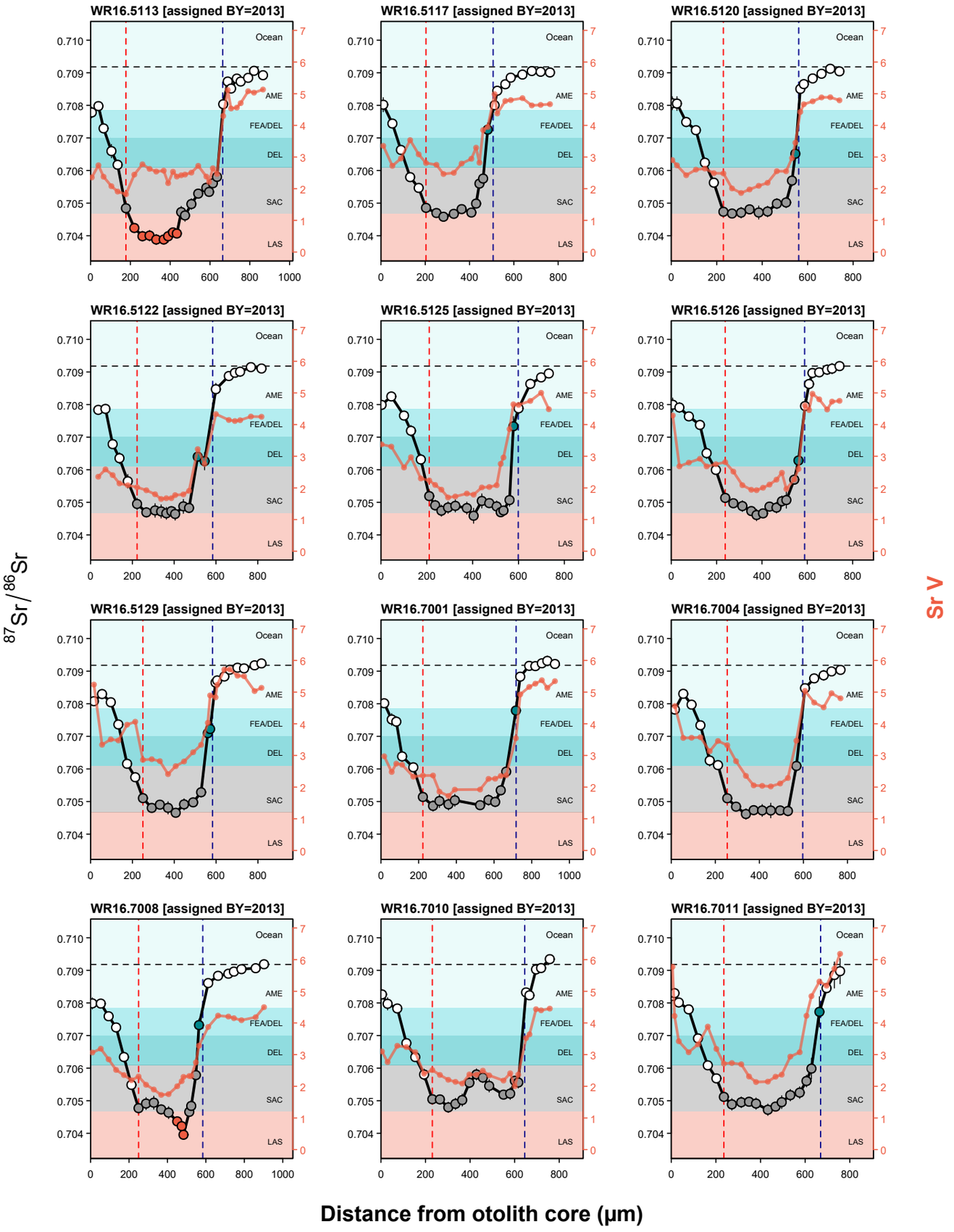

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DEL
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LAS
$^{87}Sr/^{86}Sr$
Sr V
Distance from otolith core (µm)

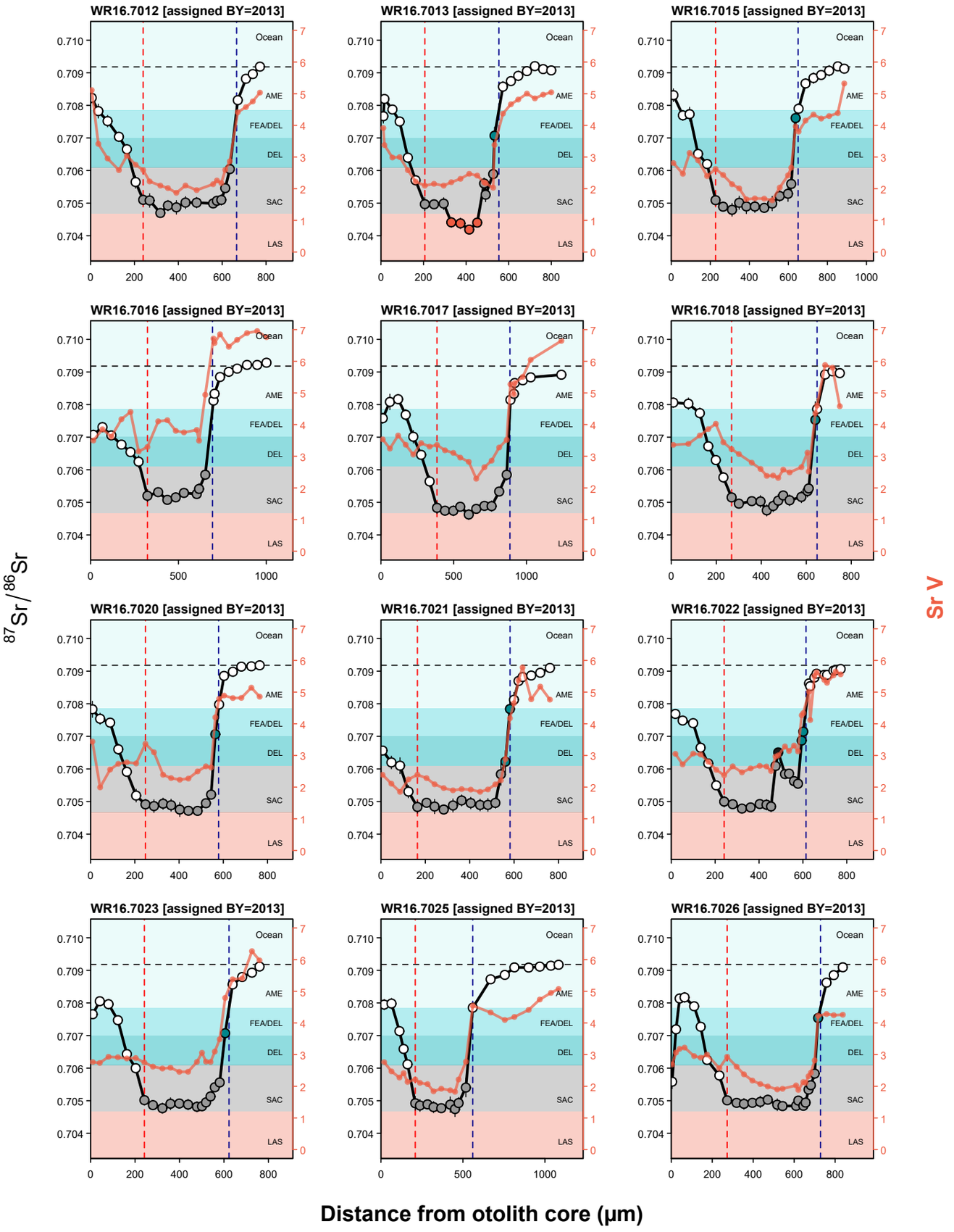

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FEA/DEL
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LAS
87Sr/86Sr
Sr V
Distance from otolith core (µm)

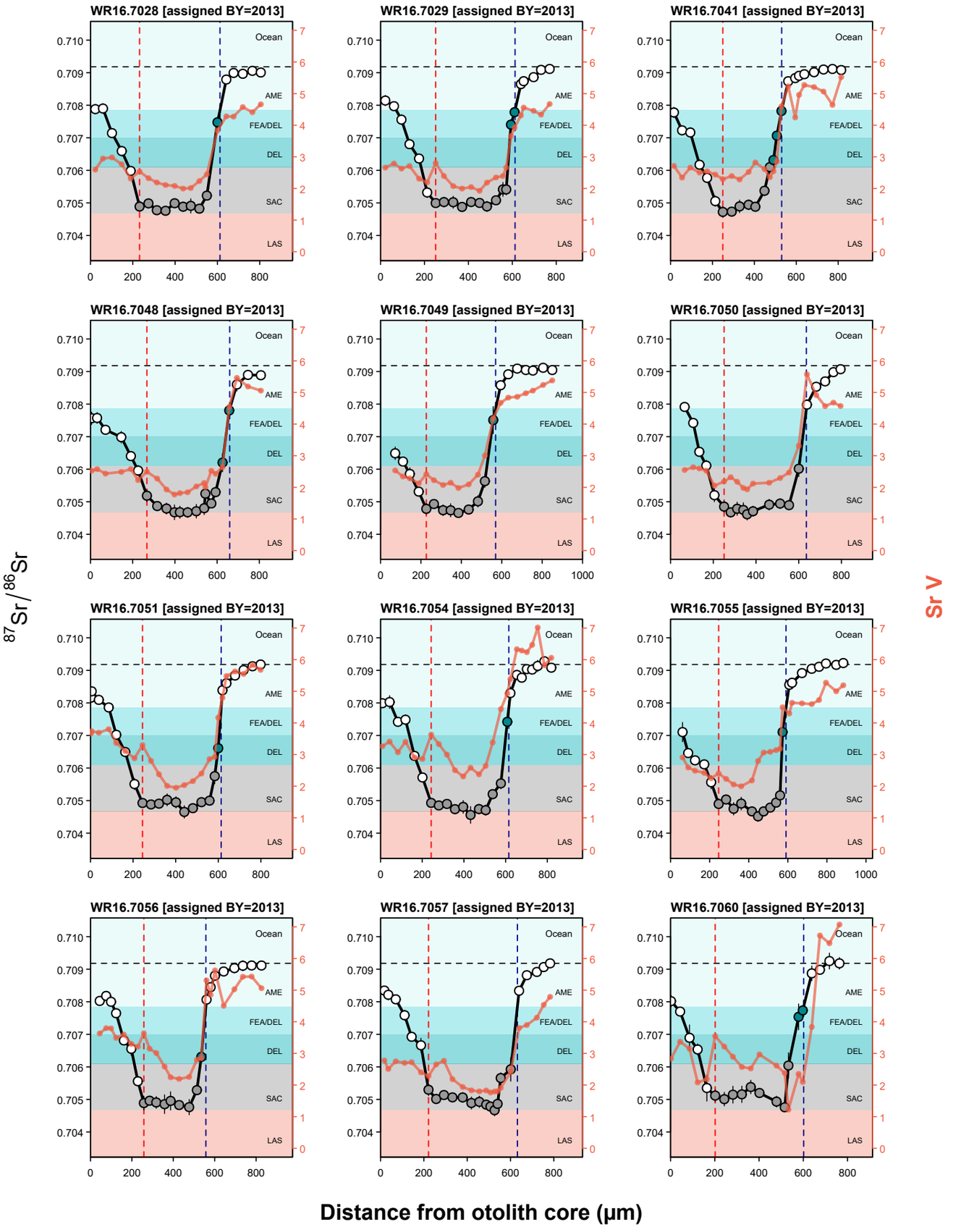

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FEA/DEL
DEL
SAC
LAS
$^{87}Sr/^{86}Sr$
Sr V
Distance from otolith core (µm)

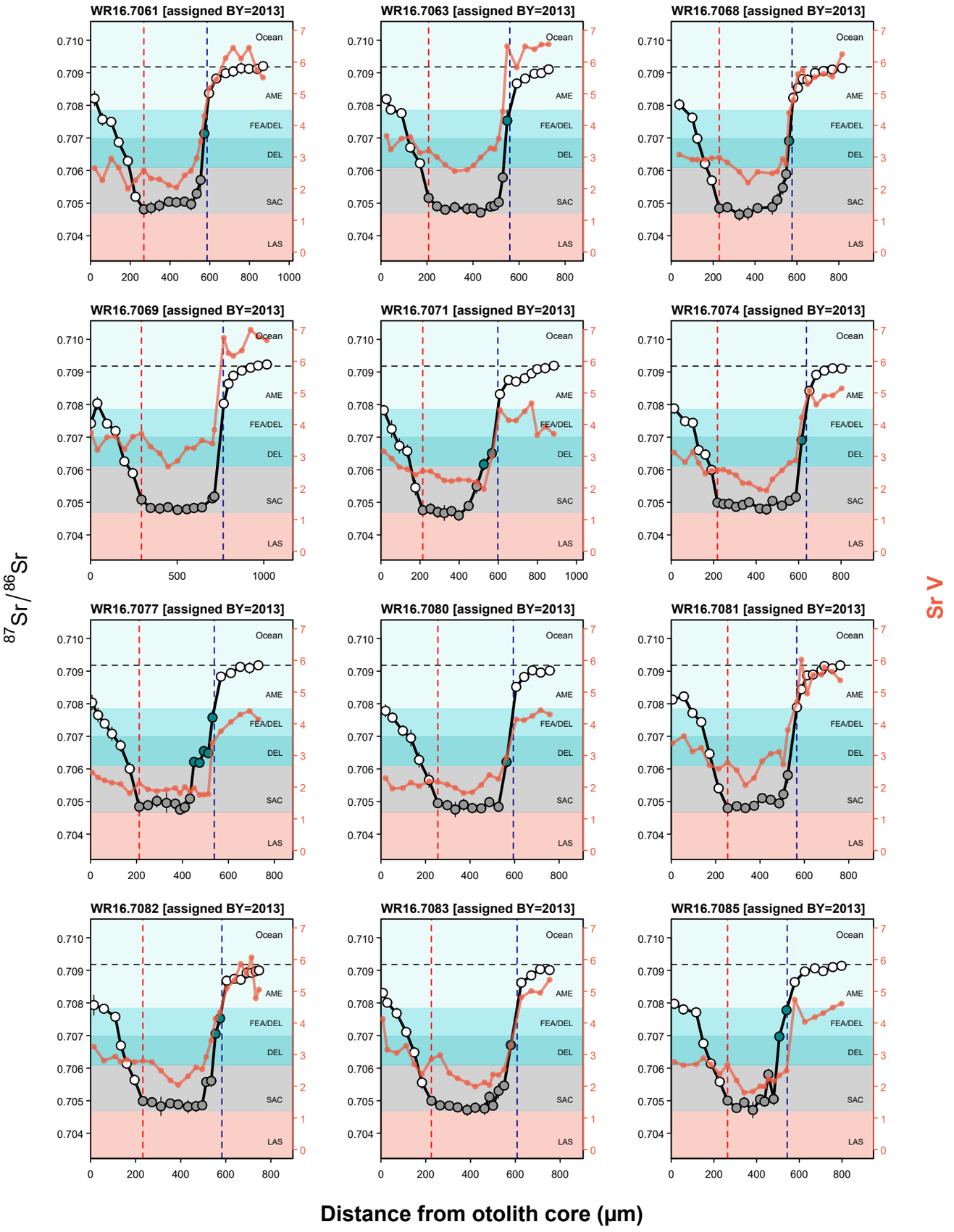

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Ocean
AME
FEA/DEL
DEL
SAC
LAS
$^{87}Sr/^{86}Sr$
Sr V
Distance from otolith core (µm)

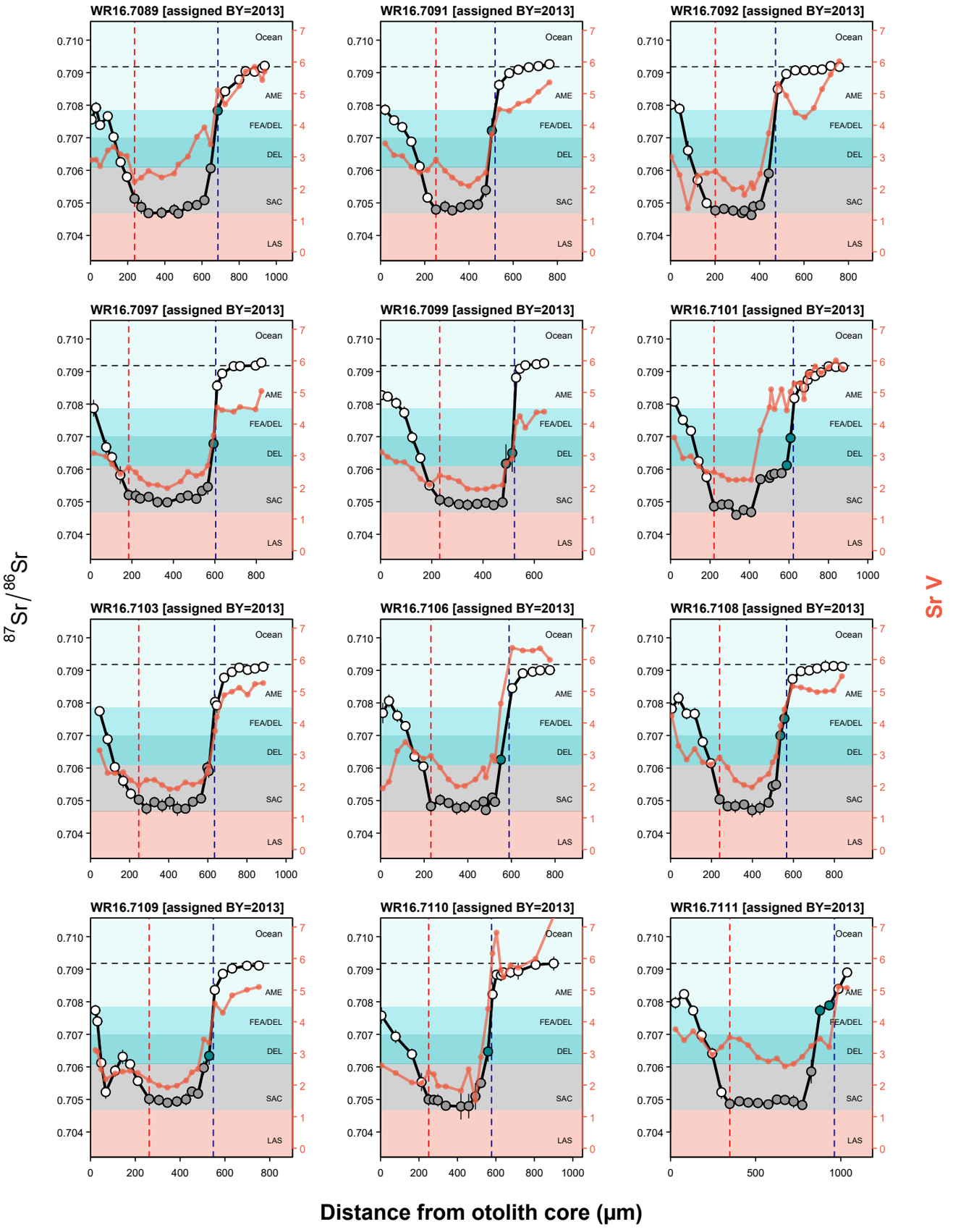

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Ocean
AME
FEA/DEL
DEL
SAC
LAS
87Sr/86Sr
Sr V
Distance from otolith core (µm)

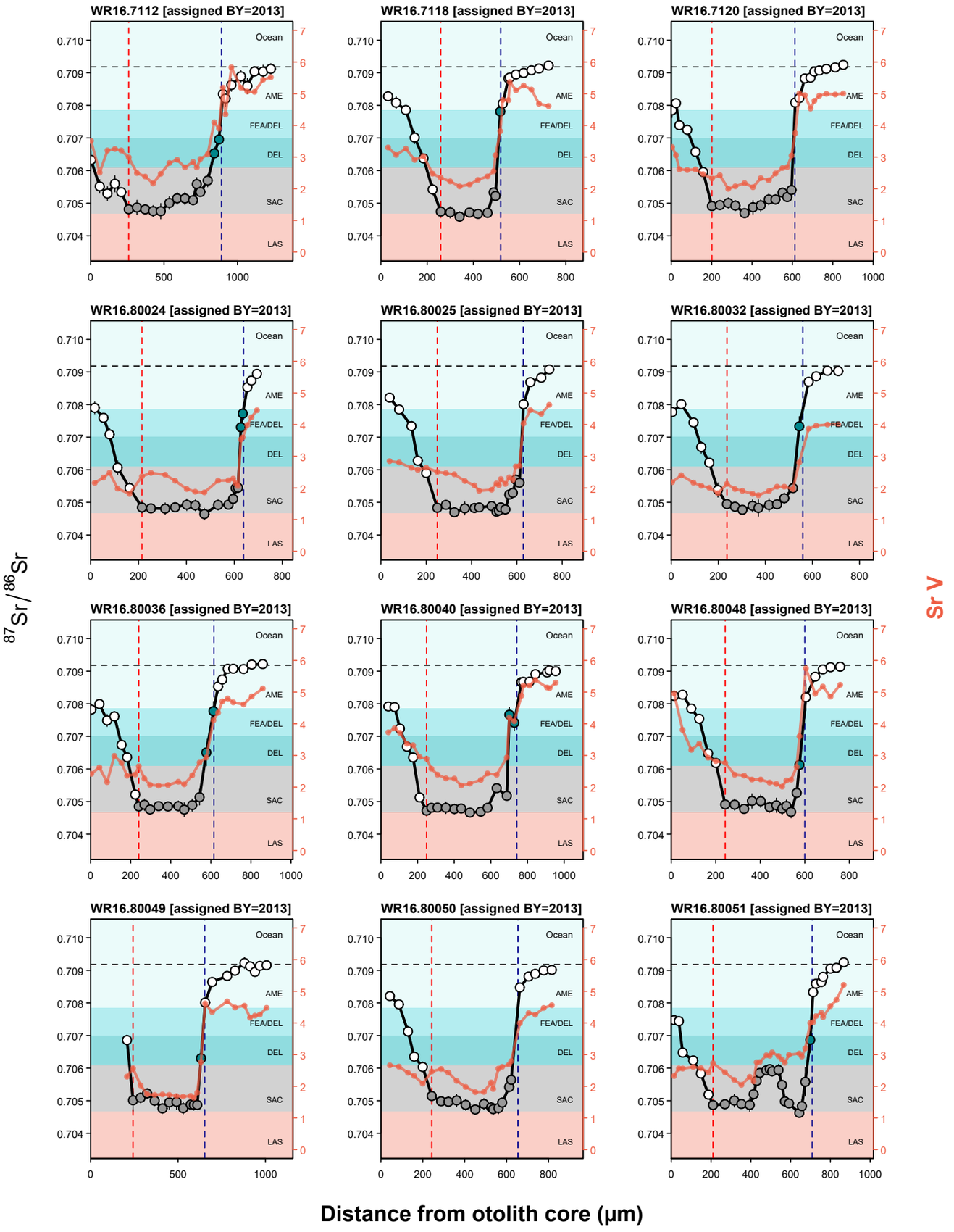

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Ocean
AME
FEA/DEL
DEL
SAC
LAS
87Sr/86Sr
Sr V
Distance from otolith core (µm)

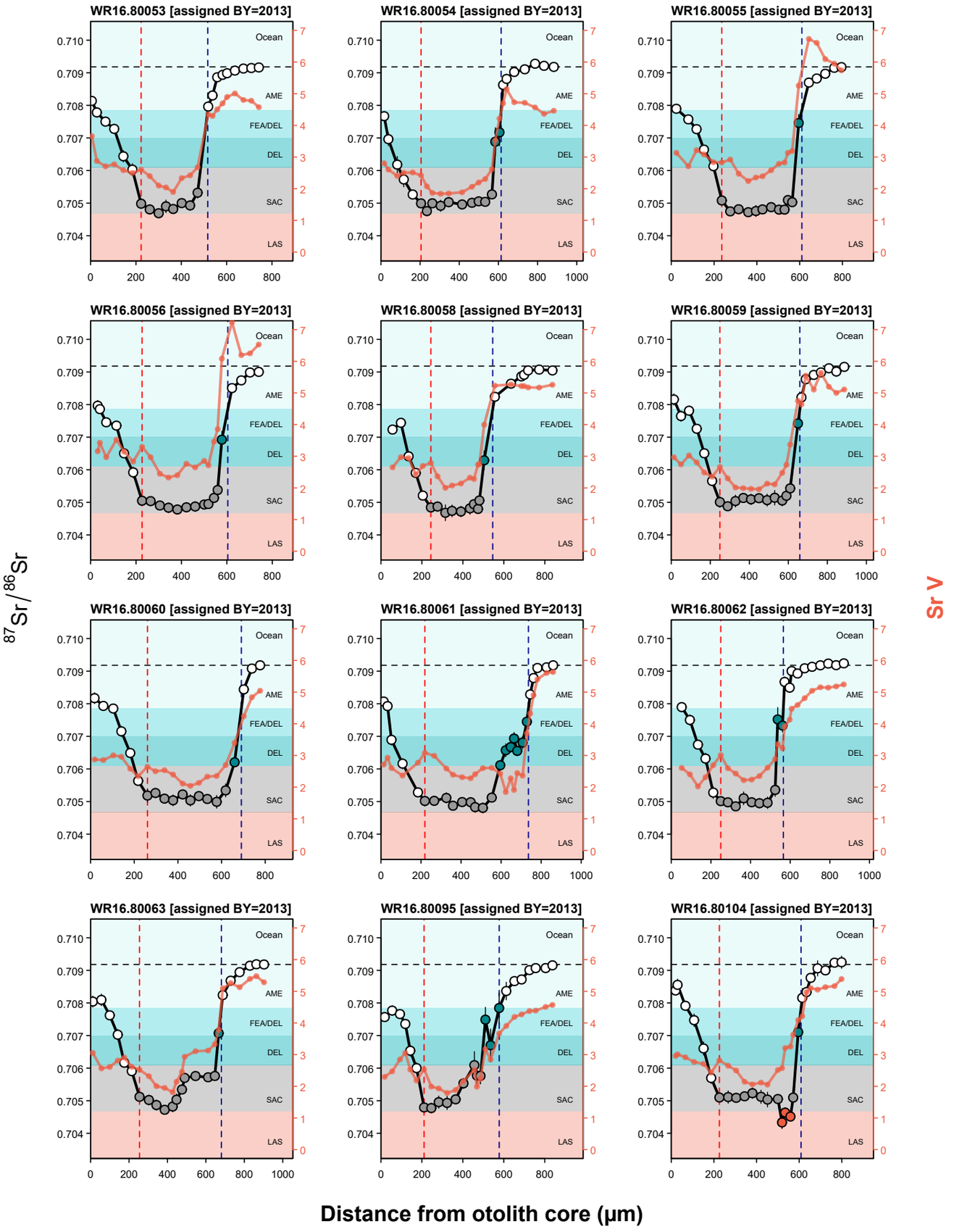

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AME
FEA/DEL
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SAC
LAS
$^{87}Sr/^{86}Sr$
Sr V
Distance from otolith core (µm)

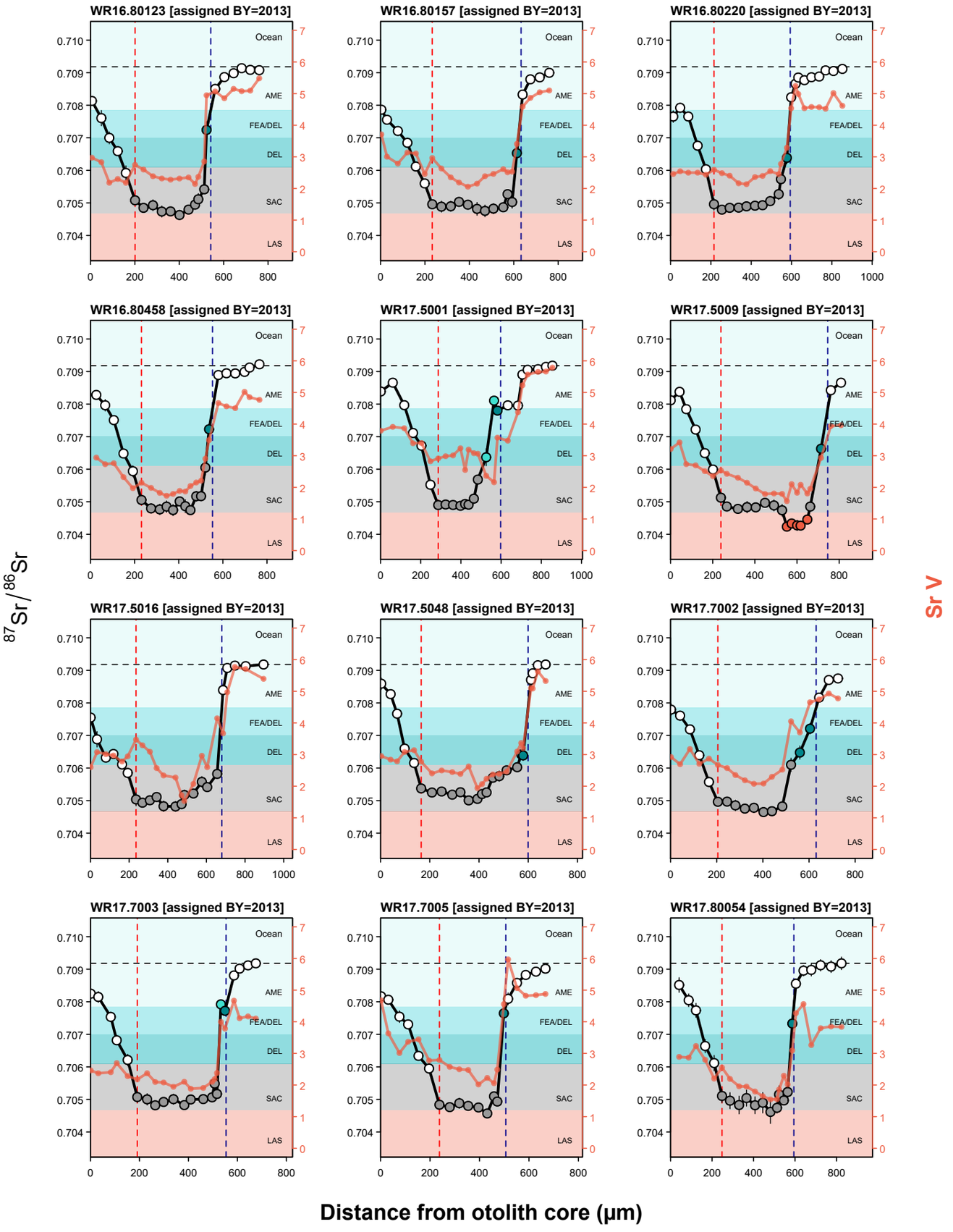

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AME
FEA/DEL
DEL
SAC
LAS
$^{87}Sr/^{86}Sr$
Sr V
Distance from otolith core (µm)

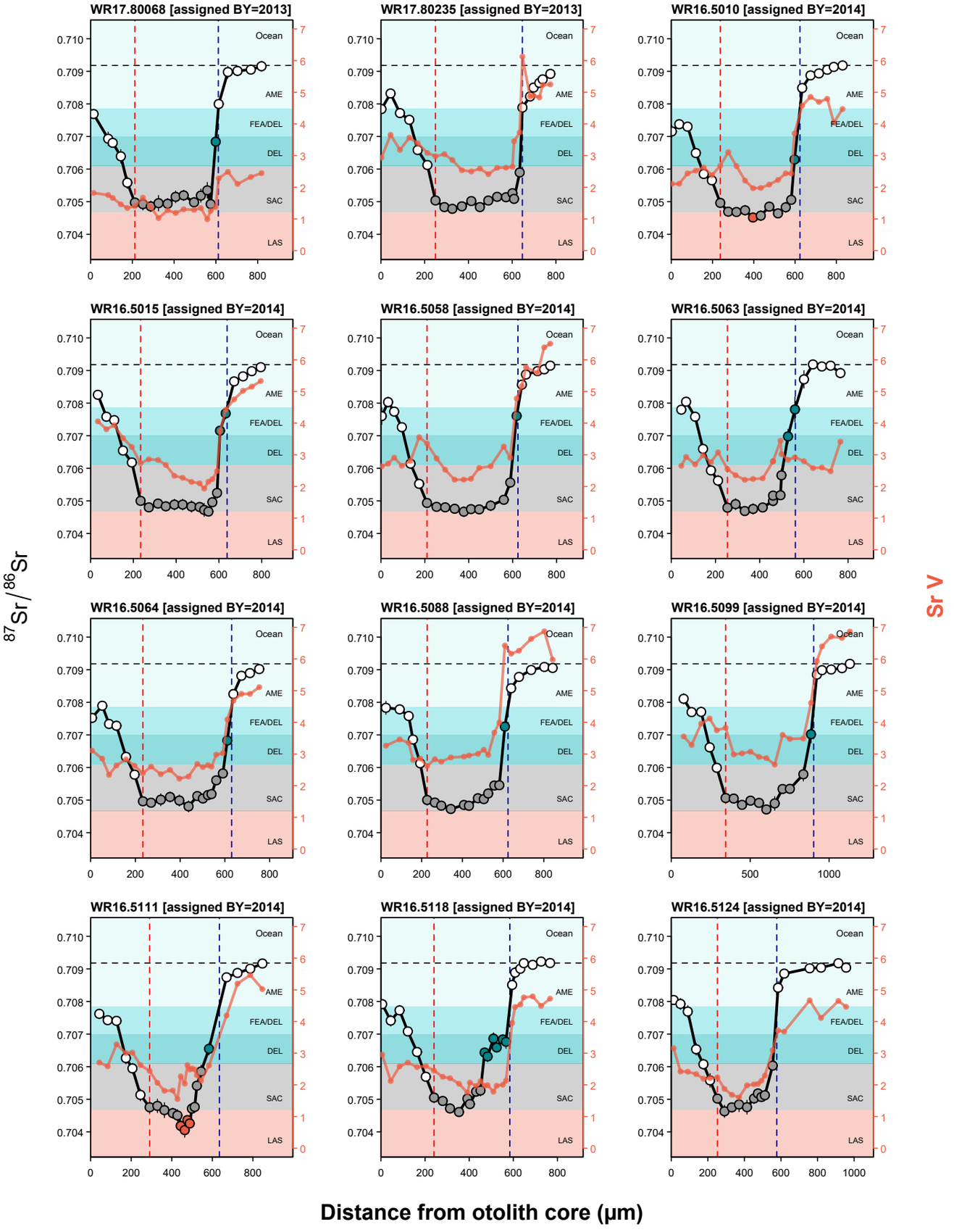

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AME
FEA/DEL
DEL
SAC
LAS
$^{87}Sr/^{86}Sr$
Sr V
Distance from otolith core (µm)

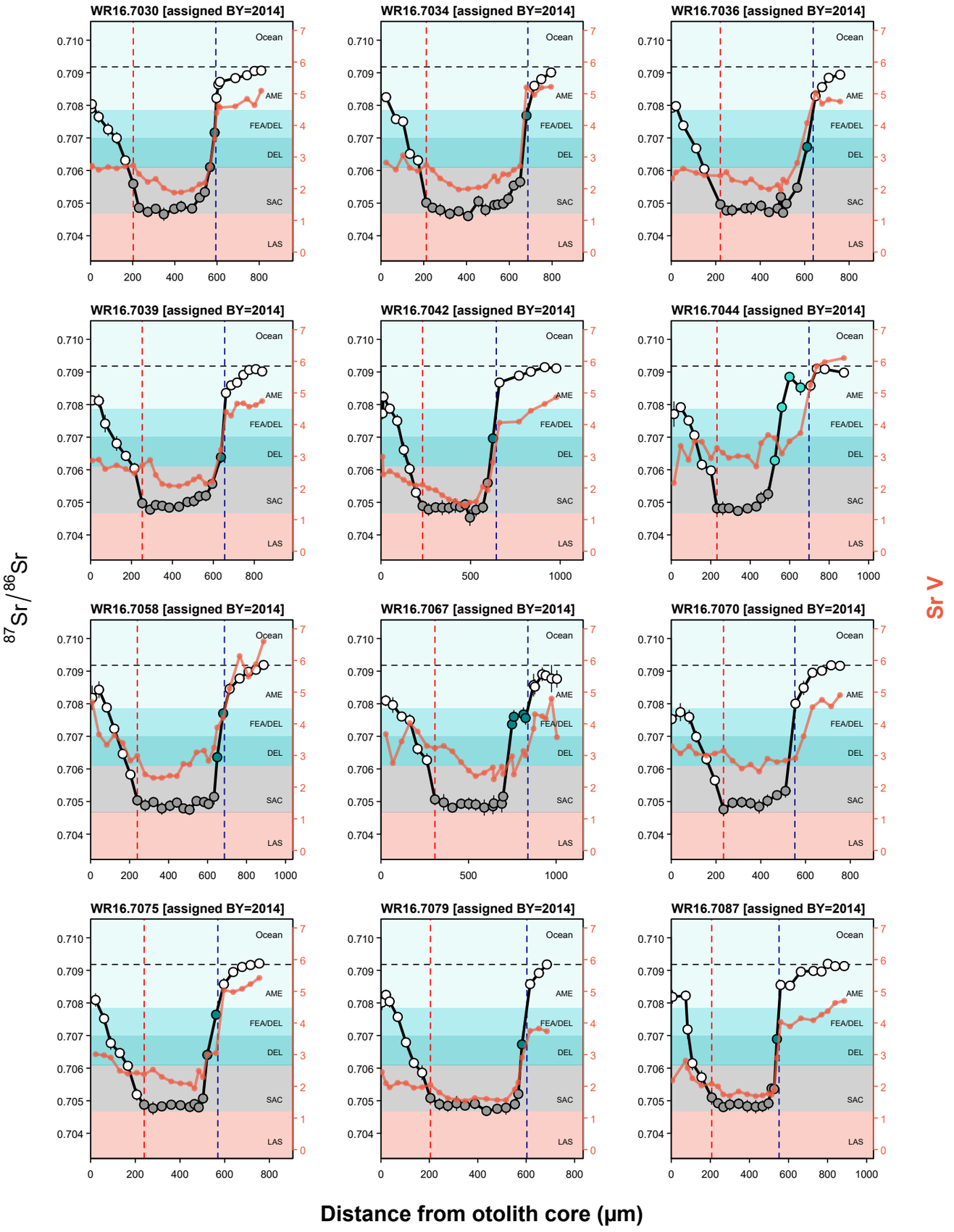

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WR16.7109 [assigned BY=2013]
WR16.7110 [assigned BY=2013]
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Ocean
AME
FEA/DEL
DEL
SAC
LAS
87Sr/86Sr
Sr V
Distance from otolith core (µm)

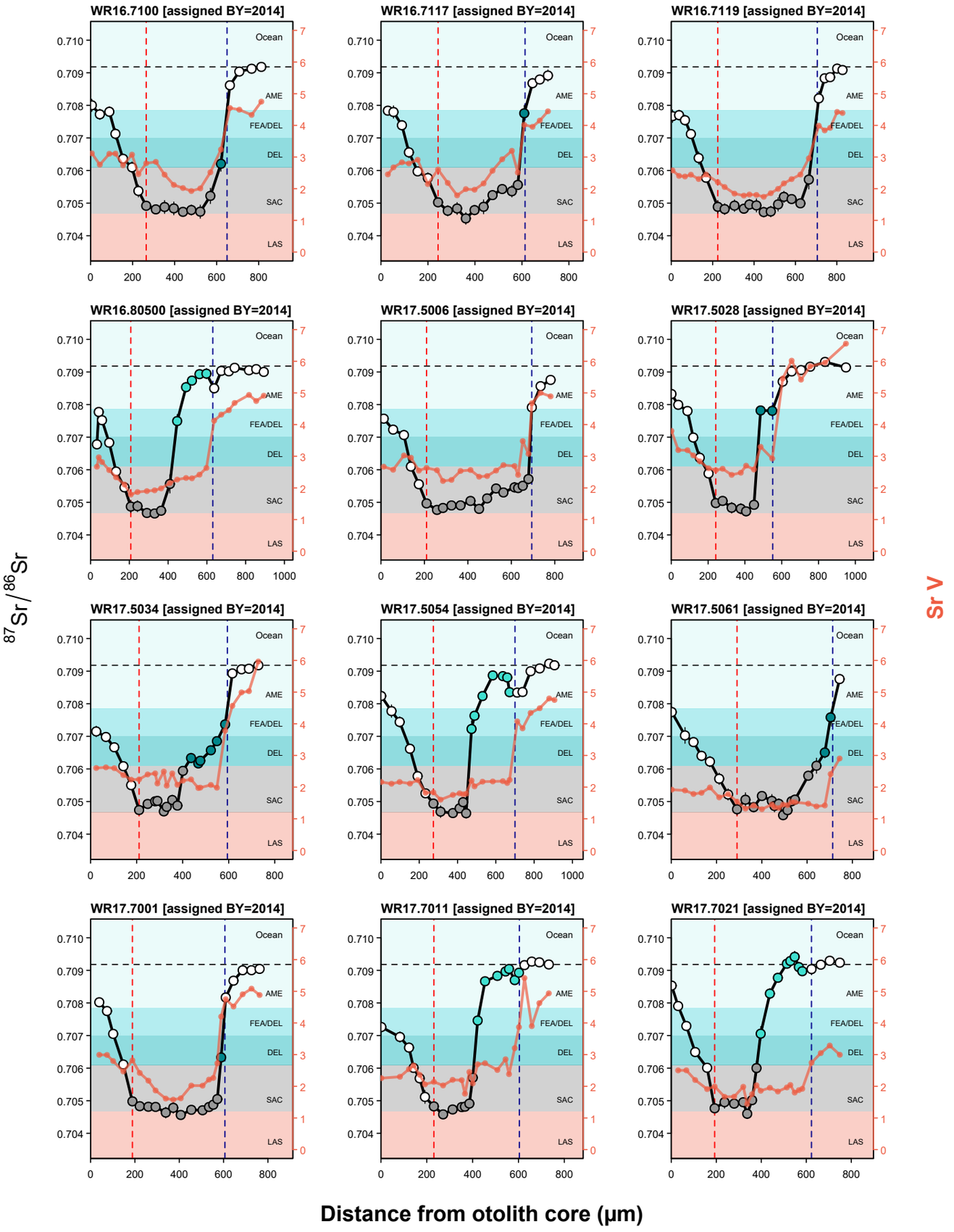

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WR16.80049 [assigned BY=2013]
WR16.80050 [assigned BY=2013]
WR16.80051 [assigned BY=2013]
Ocean
AME
FEA/DEL
DEL
SAC
LAS
$^{87}Sr/^{86}Sr$
Sr V
Distance from otolith core (µm)


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WR16.80063 [assigned BY=2013]
WR16.80095 [assigned BY=2013]
WR16.80104 [assigned BY=2013]
Ocean
AME
FEA/DEL
DEL
SAC
LAS
$^{87}Sr/^{86}Sr$
Sr V
Distance from otolith core (µm)


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WR17.7002 [assigned BY=2013]
WR17.7003 [assigned BY=2013]
WR17.7005 [assigned BY=2013]
WR17.80054 [assigned BY=2013]
Ocean
AME
FEA/DEL
DEL
SAC
LAS
$^{87}Sr/^{86}Sr$
Sr V
Distance from otolith core (µm)


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WR16.5111 [assigned BY=2014]
WR16.5118 [assigned BY=2014]
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Ocean
AME
FEA/DEL
DEL
SAC
LAS
87Sr/86Sr
Sr V
Distance from otolith core (µm)


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WR16.7079 [assigned BY=2014]
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Ocean
AME
FEA/DEL
DEL
SAC
LAS
87Sr/86Sr
Sr V
Distance from otolith core (µm)


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Ocean
AME
FEA/DEL
DEL
SAC
LAS
$^{87}Sr/^{86}Sr$
Sr V
Distance from otolith core (µm)

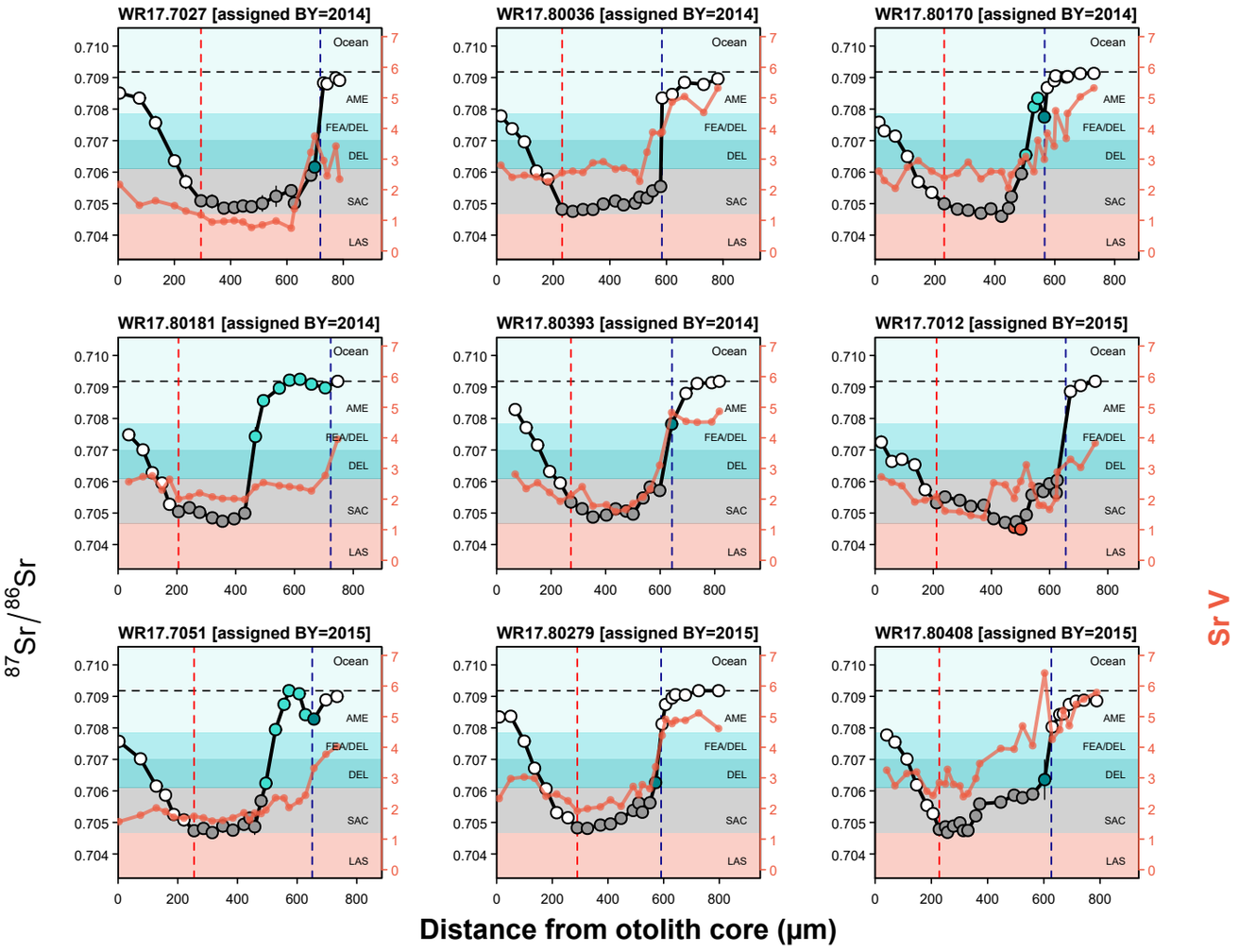


Figure S1. Individual otolith strontium-isotope profiles of juvenile winter-run Chinook salmon. Each panel presents the profile of an individual fish, identified by sample code and assigned brood year. The normalized otolith $^{87}Sr/^{86}Sr$ ratio (black line and points; left y-axis) and Sr signal in volts (light red line and points; right y-axis) are plotted against distance from the otolith core (µm). Vertical error bars around $^{87}Sr/^{86}Sr$ measurements represent ±2 SE (standard error). Point colors indicate habitat assignments: Lassen Tributaries (LAS), Sacramento River (SAC), Feather River or Delta (FEA/DEL), American River (AME), Delta (DEL), and unassigned. Shaded horizontal bands represent the reference $^{87}Sr/^{86}Sr$ ranges associated with LAS, SAC, FEA/DEL, AME, and DEL habitats. The horizontal dashed line marks the ocean reference threshold. The vertical lines delimite the distances in the otolith between emergence (red dashed line) and freshwater exit (blue dashed line).